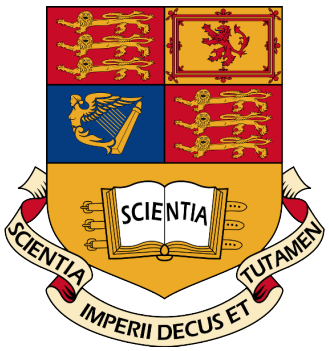
Modelling Wind Energy Potentials and Energy Pricing Between the North and Baltic Seas

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Abstract

Europe's next wave of offshore wind moves from the mature North Sea into the enclosed Baltic Sea, where wind farms generate at the same time and push down the price they all sell at. The share of the average electricity price that a wind farm actually receives is called its capture value, and it falls as more wind is built. This thesis measures capture value for every electricity market around the two seas from 2017 to 2024 and projects it to 2030 and 2040, using open data throughout. It asks three questions: how the two seas compare today and how they will compare once the planned wind farms are built, what causes the loss of value and how much of it can be avoided by building the surrounding system differently, and whether a model trained on the past can predict a future this different, or whether a model that rebuilds the market from its parts is required.

Two models are built and compared with each other. The first is a statistical model, trained on eight years of observed prices at 31 offshore wind farms. The second is a power-system model, PyPSA-Eur, which reconstructs the market from its parts, the power plants, the transmission grid and the demand in every hour, under the official European expansion scenarios for 2030 and 2040. Both models are checked against observed prices before either is trusted.

Today the two regions are equal in terms of capture value: wind farms in both capture roughly 88 per cent of the average electricity price. By 2040, with the planned buildout in place and today's grid, Baltic farms fall to 47 per cent while North Sea farms hold 57, so the Baltic loses almost half of its present value. About half of that loss is caused by congestion in the grid, and it can be removed: with transmission expanded, the Baltic recovers to 66 per cent, level with the North Sea again. The two models prove equally accurate where they can be checked, and their estimates of capture value agree to within half a percentage point, but beyond the observed record the statistical model returns almost the same answer whichever future it is given, so the long-term projection must come from the power-system model. Even with an expanded grid the Baltic captures a quarter less than it does today, so what the planned wind farms will be worth depends on the grid built alongside them.

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Ai Statement

This thesis is my own work. The research questions, the design of the study, the data collection, the modelling, the analysis, and the conclusions are my own, and every result reported was produced, checked, and validated by me.

During the project I used an AI assistant (Claude, Anthropic) as a tool: to discuss and stress-test ideas, to help structure chapters and arguments, to suggest edits and draft wording that I reviewed, revised, and verified before any of it entered the text, and to check the consistency of numbers, figures, and references. No content was included without my review and validation, no AI-generated material was used in place of my own research, and I take full responsibility for the accuracy and integrity of the entire document. All sources used are cited. Everything written on this document is my own and written by me.

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1 Introduction

In 2024, German day-ahead electricity prices fell below zero for 459 hours, roughly three times the count of two years earlier [1]. That is the structural signature of wind at scale: the more of it a power system absorbs, the less each additional unit earns on the wholesale market. That tension, between the ambition to build offshore wind and the economics that erode its value as it is built, is the subject of this thesis.

A wind farm sells its electricity at the price prevailing in the hours it happens to generate, and because wind output is correlated across a region, those are precisely the hours in which farms produce together and depress the price they all receive. What a farm earns per unit of electricity is therefore the average price of its own generating hours, which sits below the market's overall average. The fraction of the average price a farm captures in this way is its capture value, and it declines as regional wind capacity grows, a process called cannibalisation [2]. Because revenue is output multiplied by the price that output sells at, capture value is increasingly the term that decides whether an offshore project is viable.

Europe ended 2024 with roughly 36 GW of offshore wind [3] and plans on the order of 360 GW by 2050 [4], a tenfold expansion within the lifetime of a single project. Most of today's capacity sits in the mature North Sea, where Germany alone operates 9.2 GW [5]; the next wave moves east into the enclosed Baltic, where the Marienborg Declaration targets 19.6 GW by 2030 against a technical potential of up to 93 GW [6]. The question is no longer whether offshore wind expands, but where it expands and what value survives once it does.

Policy makes that question more urgent: REPowerEU lifted the EU's 2030 renewable target to 42.5% [7], the IEA and IRENA project a tripling of global renewable capacity by 2030 [8, 9], the Draghi report names European electricity prices a threat to industrial survival [10], and the 2024 market reform made the two-sided contract for difference the default for supported capacity [11]. Each of these forces more zero-marginal-cost generation into the market, and so deepens the erosion this thesis measures.

Yet the basin where the next wave will land has never been systematically compared with the North Sea on the metric that decides its viability. Cannibalisation has been quantified nationally [2] and pan-European in aggregate [12], but the statistical models used absorb each market's character into a single fitted constant. They establish that markets differ while leaving the reason inside that constant, and they cannot be carried into a future in which the characteristics they absorb have changed. And because investment horizons force projection, every projected capture value in the literature rests on a single model family never validated against another. The enclosed Baltic, earlier on its cannibalisation curve and structurally less able to export its surplus, is the case these gaps leave unanswered.

This thesis measures and projects the market value of offshore wind across the open North Sea and the enclosed Baltic, entirely on open data. It asks how the two basins compare today and how they diverge as the committed buildout is delivered, what sets the rate at which value is lost and how much of that loss system design can remove, and whether a model fitted to historical observation can describe a future lying outside it or whether a model of the power system is required. Section 2.6 states these three questions formally, after setting out the gaps they follow from.

Answering them requires two models, because no single one can both measure the present credibly and describe a future that has never occurred. A regression model fitted to eight years of observed prices measures the present. A system model, PyPSA-Eur (Python for Power System Analysis), rebuilds the market from generation costs and the physical network and projects beyond the observed record. Both are compared against observed capture value wherever it exists, and that comparison decides which model is used for what (Section 3.1).

On this design the thesis makes four contributions, the first of which is a systematic North-Sea-versus-Baltic capture-value comparison at the bidding-zone resolution where prices form, carried from today's

measurement to the 2040 buildout. The second sets the two models against observation and against each other on the same quantity, and determines from measurement, rather than by assumption, at which point the projection should pass from one model to the other: the regression model is shown to stop answering where it cannot extrapolate, the remaining gap is decomposed into the construction steps that produce it, and the system model's congestion bias is corrected against observation. The third decomposes the erosion twice, into the structural part no grid can remove and the congestion part an optimal grid does, and into the shares of the loss borne by the farm and by the state under the contracts that now dominate support. The fourth folds capture value into levelised cost to give a value-adjusted cost of energy per location, and carries it to the decision it changes, which is where to build. The whole chain rests on open data and open code, so every number in the thesis can be reproduced.

The study covers both basins through 31 offshore wind farms, sixteen in the North Sea and fifteen in the Baltic, measured hourly over 2017 to 2024 and projected to 2030 and 2040 under the official European transmission-planning scenarios. The fourteen countries bordering the two seas hold 21.1 GW of Europe's offshore capacity today, the remainder sitting mostly in Great Britain, which lies outside the modelled area. Figure 1.1 maps the farms.

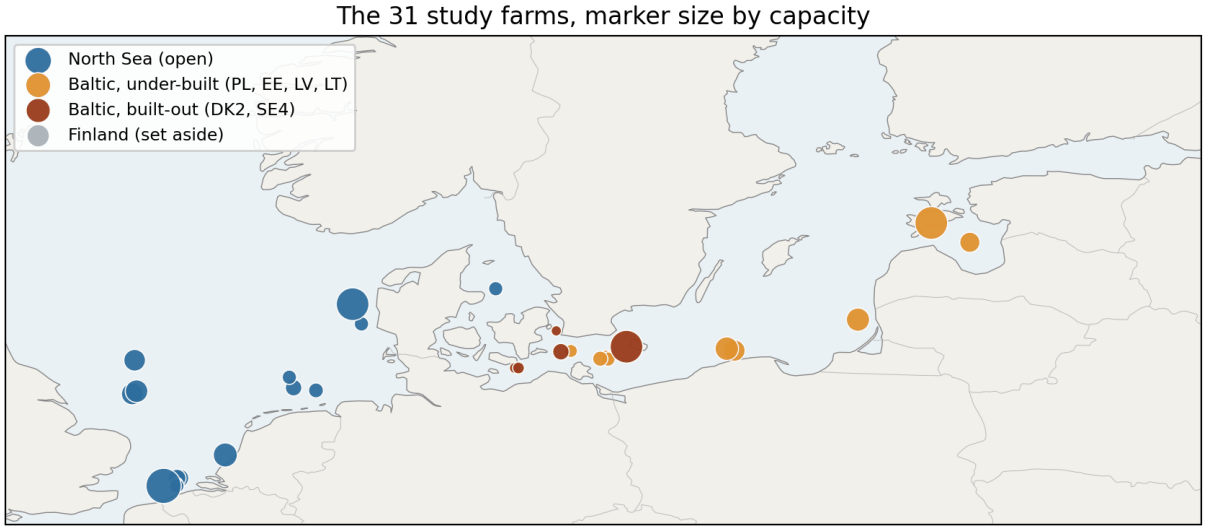


Figure 1.1: The 31 study farms in the four categories used throughout the thesis; marker size is proportional to capacity.

Chapter 2 builds the concepts from first principles and locates the gaps the questions above fill. Chapter 3 develops the two models and how they are compared. Chapter 4 measures today's value, tests the two models against observation and against each other, projects the buildout, explains what drives the erosion, and turns the result into decisions: a break-even price per site and a ranking of where to build. The accounting of who bears the loss is carried in Appendix F. Chapter 5 answers the three research questions against that evidence, sets the answers against the literature, and states the limits of the work. Chapter 6 concludes. The finding that carries through every chapter is that cannibalisation in the Baltic is a consequence of the system built around the basin.

2 Background and Literature Review

Offshore wind is usually measured in gigawatts installed and terawatt-hours generated. Neither quantity determines whether a wind farm earns money. A generator is paid the price prevailing in the hours it produces, and because wind output across a sea basin arrives together, those are precisely the hours in which the price is lowest. This chapter builds the argument from that single observation. Section 2.1 sets out how European wholesale prices form. Section 2.2 defines the resulting value erosion and reviews its measured magnitude. Section 2.3 asks the question on which this thesis turns, namely whether that erosion is an inherent property of the resource or a consequence of system design that can be reversed. Section 2.4 contrasts the two basins under study, and Section 2.5 reviews the model families available for answering the question. Section 2.6 states the gaps that remain and the questions this thesis puts to them.

2.1 Price formation in electricity markets

The day-ahead market, which carries 80 to 90 % of merchant wind revenue [13], clears pay-as-clear. Generators offer at short-run marginal cost, the market operator stacks those offers into an ascending supply curve, and every dispatched unit receives the price of the most expensive unit required to meet demand. Wind turbines, whose fuel is free, enter at the bottom of that stack. Additional wind shifts the supply curve to the right, displaces the marginal thermal unit, and lowers the clearing price. This is the merit order effect, illustrated in Figure 2.1.

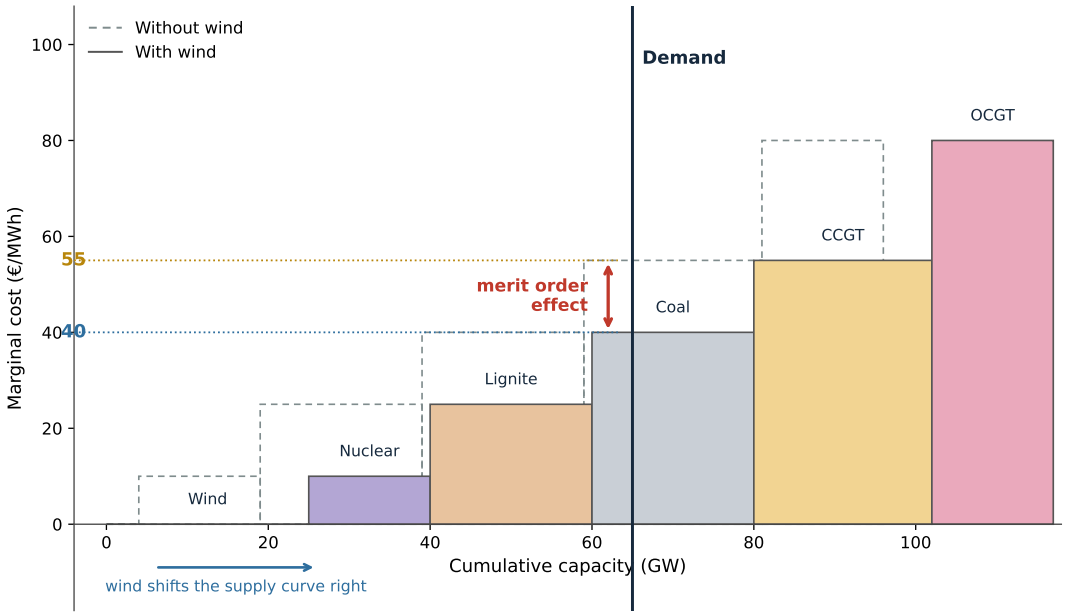


Figure 2.1: The merit order effect. Wind enters at the bottom of the supply curve and shifts it rightward, displacing higher-cost thermal plant and reducing the clearing price from 55 to 40 €/MWh in this stylised example. The depression is deepest in exactly the hours wind generates most.

Prices are not formed zone by zone in isolation: the Single Day-Ahead Coupling mechanism matches orders across participating zones simultaneously, subject to available cross-zonal transmission [14]. Where transmission is adequate prices converge and the merit order effect is shared across borders; where it binds, prices separate and the surplus is trapped [15]. A basin cannot be analysed as a closed system.

Geographic resolution matters as much as mechanism. Europe is divided into bidding zones, each clearing at one price: Germany prices a single zone despite substantial internal congestion, a configuration under review whose resolution would change North Sea wind revenues [16, 17], while the Nordic countries split into eleven zones that let transmission constraints surface as genuine price areas [18]. The three Baltic states each form their own zone and, since February 2025, are synchronised with the continental grid rather than the former Soviet system [19, 20]. Figure 2.2 sets out the zones around both basins with the interconnectors between them, and groups them by the kind of generation that usually sets their price, which is the distinction Section 2.4 develops. A zonal structure localises price signals, so cannibalisation propagates to neighbours only as far as interconnection carries it.

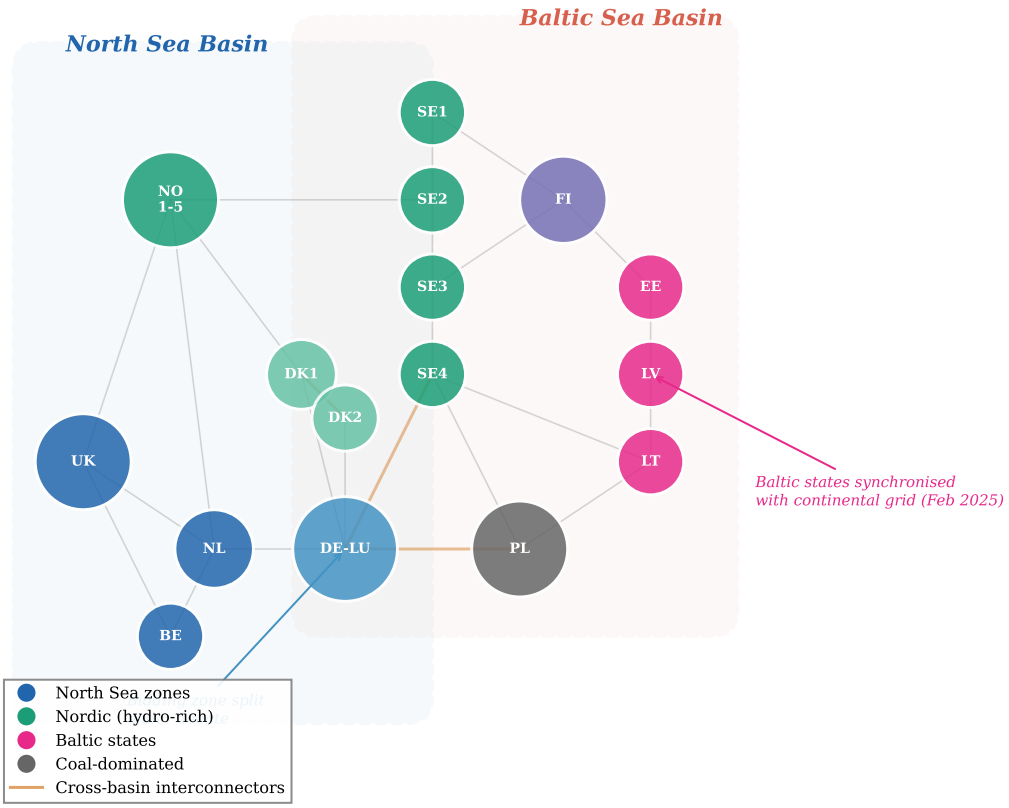


Figure 2.2: Bidding zones and the main interconnectors around the North Sea and the Baltic, grouped by the kind of generation that usually sets each zone's price.

2.2 Capture price, cannibalisation and value erosion

2.2.1 Capture price and capture value

A baseload plant running continuously earns approximately the time-averaged wholesale price; a wind farm earns the average price weighted by its own output. Two measures established by Hirth [2] formalise the difference. The *capture price* is the generation-weighted average spot price received. Dividing it by the baseload price, the simple time-average of the wholesale price over the same period, gives the *capture*

value:

$$\text{Capture value} = \frac{\text{Capture price}}{\text{Baseload price}} = \frac{\sum_t p_t\, g_t}{\bar{p} \sum_t g_t} \tag{2.1}$$

where p_t is the spot price in hour t, g_t the generation, and $\bar{p}$ the time-averaged price. At negligible penetration the ratio sits near unity; as wind's share grows its own output moves the price in the hours it generates and the ratio falls. That shortfall is what Hirth [21] and Ueckerdt et al. [22] term profile cost, and it is precisely the quantity a cost-only assessment of where to build omits.

Hirth's early estimates placed the decline at roughly 1.10 at negligible penetration falling to between 0.50 and 0.80 at a 30% wind share, the range separating thermal-dominated systems from hydro-dominated ones (Figure 2.3) [2]. That spread is not incidental here: most North Sea markets are thermal-heavy, the Baltic rim borders Scandinavian hydro. Later work at higher penetration has confirmed the pattern and established that the decline steepens as penetration rises [12, 23, 24, 25].

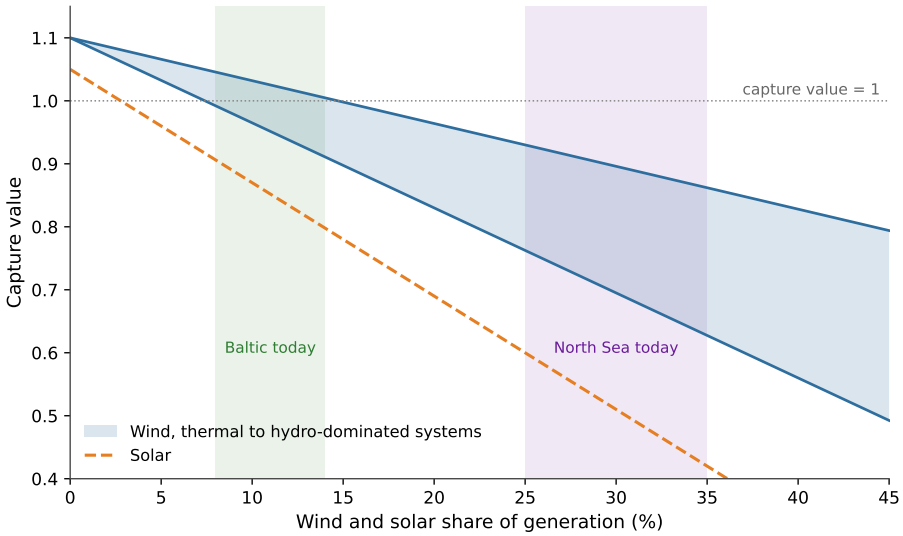


Figure 2.3: Capture value against wind and solar market share, synthesised from Hirth [2]. The shaded band for wind spans thermal-dominated systems (lower bound) to hydro-dominated systems (upper bound); the approximate current penetration ranges of the North Sea and Baltic markets are indicated.

Figure 2.4 shows the divergence directly. Germany's capture value fell from 0.93 to 0.76 as wind penetration rose from roughly 20% to 35%, while Poland, where wind remained below 18% of generation and coal sets the marginal price in most hours, stayed near unity. The Scandinavian markets sit between the two, buffered by hydro, and the 2021 to 2022 uptick is the gas crisis lifting wholesale prices faster than capture prices could follow. The spread between the highest and lowest trajectories widened from five percentage points in 2015 to twenty by 2024, which is both the severity of the problem and the reason a basin-level comparison is worth making.

2.2.2 Magnitude and measurement

Estimates of the merit order effect span more than an order of magnitude across markets, periods and methods [28, 29, 30], partly genuine variation and partly the fragility of the identification assumptions behind them [31, 32, 33]. For Baltic markets published estimates are effectively absent, the pan-European panel of Cevik and Ninomiya [25] being the only work that includes Baltic zones, and that absence is part of the motivation for this thesis.

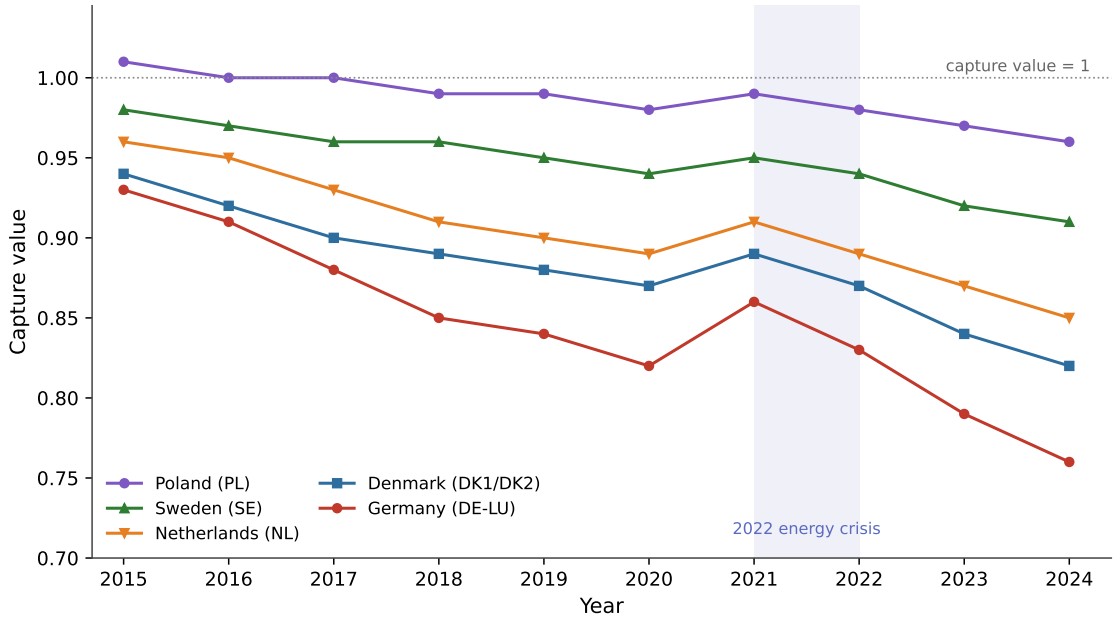


Figure 2.4: Observed wind capture value for selected European bidding zones, 2015 to 2024, synthesised from Stiewe et al. [12], Ember [24, 26] and ACER market monitoring [27, 20].

Cevik and Ninomiya find that each percentage point of additional renewable share reduces average wholesale prices by 0.6%, accelerating at higher penetration [25]. Since wind sells precisely in the hours its own output has depressed [2], erosion becomes severe past roughly a 20% wind share, beyond which Hirth estimates additional wind stops adding system welfare [34]. Offshore is more exposed than onshore: higher and steadier capacity factors concentrate output in time and correlate it across a basin, so a sustained blow depresses prices across several coupled zones at once [35].

2.2.3 From price depression to cannibalisation

The merit order effect describes an instantaneous price depression; cannibalisation is its cumulative consequence, the revenue earned per megawatt-hour falling for every existing and prospective asset as capacity accumulates. Its severity depends on penetration, the spatial correlation of output, fuel and carbon prices, and the flexibility of the surrounding system [2, 12]. Its most visible symptom is the growth of negative prices, abundant wind meeting inflexible plant that will not curtail and generators that keep producing because support rewards volume; negative hours have risen sharply where penetration is highest [24, 26], masked briefly by the 2022 crisis and reasserted as gas normalised [36]. The German fall of Figure 2.4 compounds higher penetration, greater temporal concentration of offshore output, and the inflexibility of lignite [37, 12, 24].

Whether this is permanent is contested: theory predicts convergence as marginal generators exit [38, 39], but European support decouples investment from market signals and delays that exit [2]. The gap is starkest in small, fast-expanding Baltic markets, where projected revenue losses reach 15 to 20 €/MWh by 2029 against near-zero today [40], too fast for the fleet adjustments equilibrium arguments assume.

2.2.4 Cross-border cannibalisation effects

Most analyses treat each bidding zone as an island. Stiewe et al. [12] tested that directly, regressing the domestic capture value on both domestic wind penetration and that of directly interconnected neighbours across more than 30 zones from 2015 to 2023. A one point rise in domestic penetration removes 0.61 points of capture value; a simultaneous one point rise across all neighbours removes a further 0.33, so coordinated European expansion costs nearly a full point for every point added. The profitability of a Danish wind farm depends on Dutch, German, Norwegian and Swedish deployment as well as Danish.

Two features of that identification, the fixed effects absorbing each market's character and the linear penetration term, are structural limits of statistical models, and Section 2.5 returns to what they imply for projection.

2.3 Moderators of cannibalisation

The literature so far establishes that capture value falls with penetration and that the fall propagates across borders, but it largely treats the surrounding system as fixed. Every mechanism above depends on system configuration: whether surplus can leave the zone, whether anything absorbs it, where the fleet is placed, and who is contractually exposed. This section reviews those four levers.

2.3.1 Interconnection

Transmission plays a dual role: capacity exports surplus, relieving domestic price depression, but imports the depression of high wind next door. Stiewe et al. find the mitigating effect slightly outweighs the exacerbating one [12], and at system scale the benefit is larger, transmission expansion cutting the total cost of a highly renewable European grid by roughly 25% [41].

The scale of planned investment reflects this: more than €400 billion of offshore transmission by 2050 under the ENTSO-E Offshore Network Development Plan [42], with the 70% cross-zonal trading requirement pushing markets toward convergence [27, 20]. Convergence is double-edged for a generator: surplus exports more easily, while the importing zone absorbs the corresponding depression.

Congestion is also a revenue stream: whenever a link binds the price differs across it, and the rent accrues to transmission operators. Joskow and Tirole [43] formalised the awkward consequence, that an investor relieving the constraint destroys the income justifying the investment, which is why regulated planning dominates. Article 19 of Regulation (EU) 2019/943 ring-fences those revenues for maintaining or increasing cross-zonal capacity [44], so a congested basin is already earmarking the money that would relieve it. Neither that accounting nor the question of how much of a zone's capture value decline is transmission scarcity rather than penetration has been brought together with capture value analysis, and this thesis undertakes the decomposition.

2.3.2 Flexibility and sector coupling

Storage and conversion attack the same problem in time rather than space. Hybrid configurations pairing offshore wind with electrolysis absorb surplus locally [45], sector coupling converts it into demand that would not otherwise exist [39], and demand-side flexibility remains largely unquantified [21]. The important observation is negative: none of these pathways has been integrated into spatial capture value assessment, so published projections implicitly assume that today's level of flexibility persists into the future, an assumption that has not been tested at the scale of a whole basin.

2.3.3 Spreading the fleet

The third lever is spatial: if output arrives everywhere at once, placing capacity where it blows at different times should raise the portfolio's value. The idea is Markowitz's mean-variance argument [46] applied to European wind by Roques et al. [47]. Two things follow that the literature has not pursued: portfolio treatments optimise the variance of *generation* while an investor is exposed to the variance of *revenue*, which market coupling can correlate even where wind is not; and the mean effect has been neglected, since less severe cannibalisation arrives as higher average capture value rather than lower volatility.

2.3.4 Support mechanisms and risk allocation

The fourth lever is contractual, and it determines not whether value is lost but who loses it. European governments have moved from guaranteed feed-in tariffs toward instruments that expose generators to price risk while structuring how it is shared (Table 2.1).

Table 2.1: Principal revenue support mechanisms for offshore wind, ordered by market exposure. Basin prevalence is as of 2024.

Mechanism	Description	Market exposure	Basin prevalence
Feed-in tariff	Fixed payment per MWh, independent of market price	None	Legacy, phased out
Two-sided CfD	Strike price agreed; the state pays any shortfall and the generator returns any surplus	Medium	North Sea (UK, NL, DE, PL)
Electricity certificates	Tradable credits per MWh, valued by a separate certificate market	High	Baltic (SE)
Premium schemes	Fixed or sliding premium added to the wholesale price	Medium to high	Baltic states (LT, LV, EE)
Merchant	No direct support; revenue follows wholesale prices entirely	Full	Emerging (NL, DE)

The two-sided contract for difference has become dominant: the state pays the difference when the wholesale price falls below the strike and recovers it above. The stabilisation lengthens debt tenors and lowers risk margins [48, 49], an effect Kitzing et al. argue is structural, since without it the cost of capital rises enough to negate a decade of levelised-cost reductions [50]. Premium and certificate schemes leave operators exposed in exactly the hours cannibalisation is most severe [49], and the 2024 EU market reform mandated two-sided contracts for all new supported capacity [11]. The geography differs: the North Sea has converged on mature contract-for-difference frameworks [51] while the Baltic rim remains a patchwork [52]. What this literature does not do is connect contract design back to the physical decomposition, asking which component of a capture value decline a given instrument absorbs. That question is taken up here.

2.4 The North Sea and Baltic Sea as contrasting market environments

2.4.1 Wind resource and correlation

European wind is governed by the westerly circulation carrying Atlantic systems eastward, modulated by weather regimes that swing European mean wind generation by a factor of two [53]. Both basins respond to the same patterns, but partial shielding by the Scandinavian land mass gives the Baltic some decorrelation from the North Sea.

The resource is unambiguously stronger in the west: mean speeds near 9.75 $\mathrm{m\,s^{-1}}$ against 7.5 to 8.0 [54], capacity factors of 40 to 50% against 35 to 40% [55]. The gap narrows once wake losses are counted, German Bight clusters losing wind speed for up to 70 km downstream [56] where the sparser Baltic is less affected.

Correlation structure matters as much as resource strength, because simultaneous output depresses the same hours. Wind power correlation decays exponentially with distance [57, 58] but neither isotropically nor uniformly: east-west pairs lag as Atlantic fronts propagate eastward, so a North Sea event precedes its Baltic counterpart by several hours [59], and cross-basin correlations sit consistently below within-basin ones [58]. Mehrens et al. [60] estimate a decorrelation length of about 425 km (Figure 2.5); set against the North Sea's span of roughly 580 by 960 km, farms within a basin are strongly correlated hourly, so high-generation events are basin-wide. Decay is slower offshore than onshore [61] and persists over far greater distances at weekly aggregation [62].

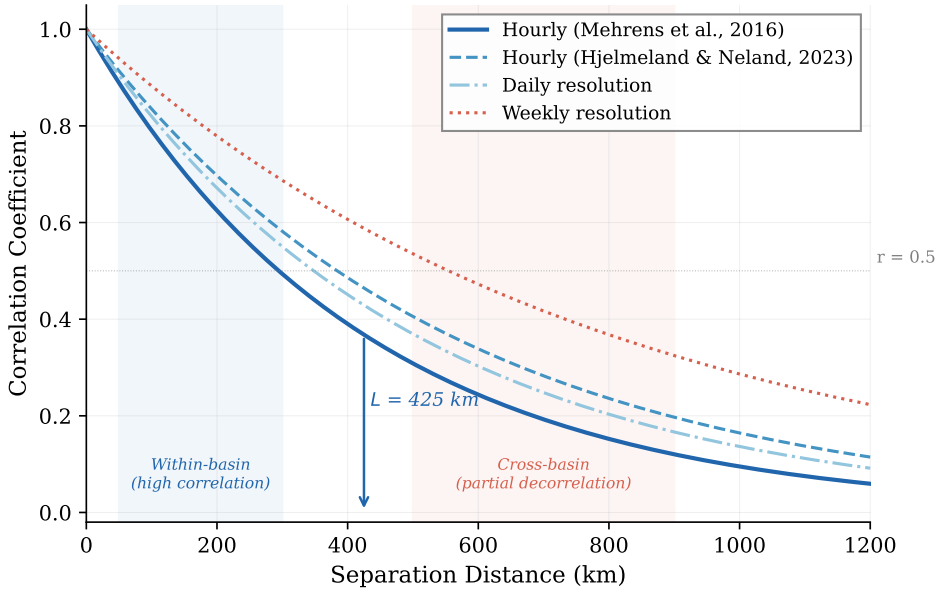


Figure 2.5: Wind power correlation against separation distance at different temporal resolutions, after Mehrens et al. [60] and Hjelmeland and Neland [62]. Typical within-basin separations (50 to 300 km) sit on the high-correlation shoulder; cross-basin separations (500 to 900 km) fall past the decorrelation length of about 425 km.

Market coupling transmits price effects across borders, so zones that are physically distant but well interconnected can show more correlated prices than their wind resources would suggest. Whether the partial decorrelation of output between the basins survives into a corresponding decorrelation of capture prices, or whether coupling erodes that diversification benefit, has not been established, and it bears directly on whether spreading investment across both basins is a genuine hedge.

2.4.2 Market structure and the penetration gradient

The composition of the surrounding fleet sets the slope of the merit order curve and with it the price response to wind (Figure 2.6), and four archetypes span the range. Germany, its dispatchable fleet still dominated by coal, lignite and gas, presents a steep curve in which each increment of wind produces a large price swing, and its size propagates that volatility to its neighbours [24, 63]. Denmark, despite

Europe's highest wind penetration at roughly 55% of generation, has not seen capture values collapse as theory predicts, because exceptional interconnection to Nordic hydro flattens the curve it faces [64, 2]. Sweden, with hydro near 45% of production, offers the region's most favourable environment: at 30% penetration Hirth estimates wind capture value about 18% higher in hydro-dominated than thermal-dominated systems [2], wide enough to move a marginal project from loss to viability. Poland sits at the opposite pole, coal above 60% of generation [26] and 5.9 GW of offshore planned by 2030 [65], so early projects earn high capture prices by displacing expensive coal while the programme itself erodes the conditions that made them attractive. Finland and Estonia are just entering the same cycle [66, 67]. Table 2.2 consolidates the archetypes.

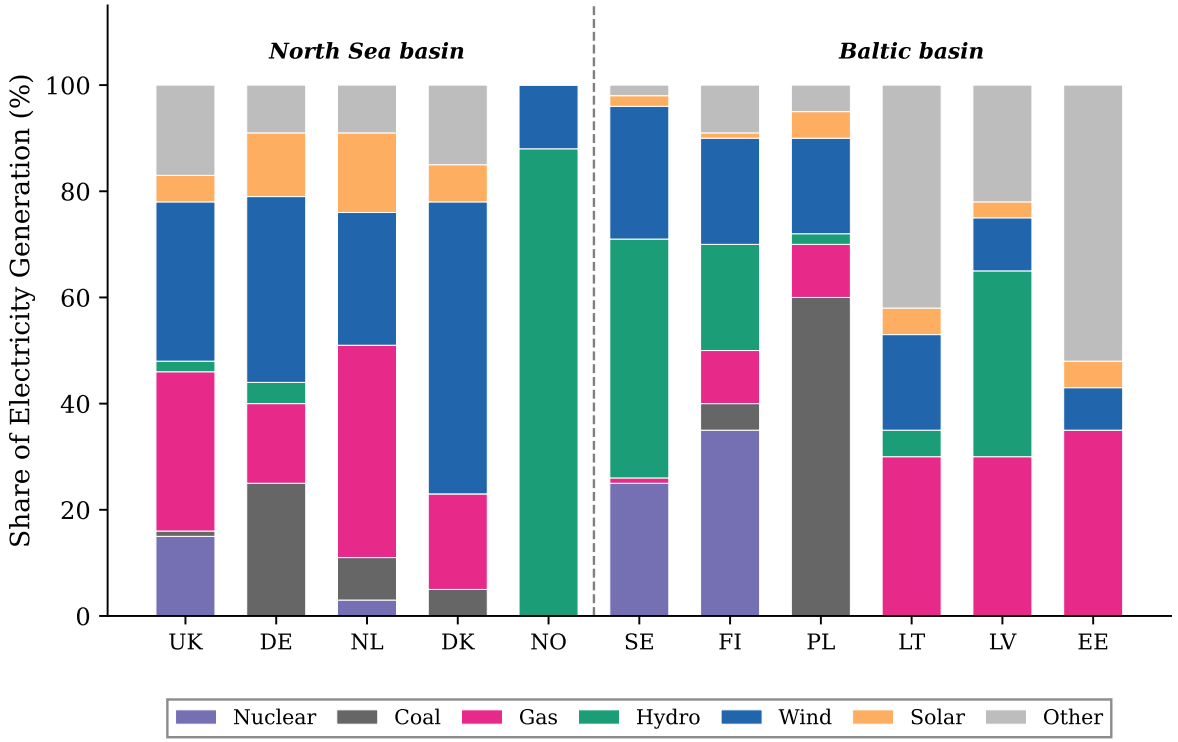


Figure 2.6: Electricity generation mix of the countries bordering the two basins, approximate 2024 shares, from Ember [24] and national system operators.

The pattern is striking and, on the conventional reading, paradoxical: every Baltic market currently records a higher capture value than every North Sea market, despite uniformly lower interconnection and, in Poland's case, minimal system flexibility. Penetration is the dominant predictor of capture value today. Whether that ordering survives as Baltic capacity scales from single to double-digit gigawatts is the question a cross-basin comparison exists to answer.

Both basins have committed to expansion on a scale that will settle it: at least 65 GW of North Sea offshore by 2030 under the Esbjerg Declaration [68] and 19.6 GW in the Baltic under Marienborg, a sevenfold increase [69], consolidated by ENTSO-E into 146 and 43 GW by 2050 [42] against a Baltic technical potential of 93 GW [70]. North Sea markets, already at high penetration, face the steepest marginal declines; Baltic markets begin further back on the curve. The timing and severity of what follows is a forecasting problem, and it is to the tools for that problem that the chapter now turns.

Table 2.2: Market archetypes around the two basins. Wind penetration and value factor estimates reflect approximate 2024 values; the final column gives the dominant revenue support instrument for offshore wind.

Country	Basin	Wind (%)	Dominant gen.	Flexibility	Intercon.	Capture value	Support
Germany	NS	∼35	Gas, coal, wind	Limited; gas peakers	Medium	0.76–0.85	Two-sided CfD
Denmark	NS/BS	∼55	Wind, gas	Nordic hydro (import)	Very high	0.80–0.90	CfD (from 2025)
Netherlands	NS	∼25	Gas, wind	Gas peakers; import	High	0.85–0.90	SDE++ tender
UK	NS	∼30	Gas, wind, nuclear	Gas CCGT; storage	Medium	0.80–0.85	CfD (AR rounds)
Sweden	BS	∼25	Hydro, nuclear, wind	Large hydro reservoirs	High	0.90–0.95	Certificates
Finland	BS	∼20	Nuclear, hydro, wind	Hydro; nuclear baseload	High	0.90–0.95	Premium / CfD
Poland	BS	∼18	Coal, gas	Limited; coal inflexible	Low	0.95–1.00	CfD auction
Baltic states	BS	<15	Gas, import, wind	Limited; import-dependent	Medium	∼1.00	Premium schemes

2.5 Models for estimating capture value

The question posed in Section 2.3, how much of the value erosion is removable by design, cannot be answered by any single tool. It demands four things at once: an hourly generation profile at any point in either basin, including places where no turbine yet stands; a price for that generation in the local market, both as it has behaved and as it would behave under a fleet that does not yet exist; a way to establish that those prices are credible; and a spatial frame in which locations can be compared on revenue rather than on energy. Each requirement has its own literature, and each of those literatures stops short of the others. This section takes them in turn.

2.5.1 Modelling generation from meteorological data

A comparison spanning both basins requires one reanalysis product applied identically to each. Higher-resolution regional products represent coastal gradients better [55, 71], but none spans both basins at the hourly resolution capture value requires. ERA5, at 0.25° and hourly from 1940, is the only product that does, and it has been validated against tall towers [72, 73] and North Sea platforms at $r = 0.85$ to 0.93 [74]. Its known weakness is that it underestimates strong offshore winds, by roughly 15 % at 15 $\mathrm{m\,s^{-1}}$ [75], which the cubic relationship between speed and power amplifies into an error in capacity factor.

Converting wind fields into power follows the standard procedure of Staffell and Pfenninger [72], now implemented in open tools such as atlite [76]. A bias correction against fleet data is not optional in that procedure, since uncorrected reanalysis understates offshore capacity factors by 12 to 15 %.

2.5.2 Models of electricity prices

Electricity prices are among the hardest commodities to model, combining extreme volatility, seasonality on several timescales, and mean reversion punctuated by spikes [77]. Two of Weron's five families matter here, and the distinction organises this thesis: statistical models estimate the relationship between wind and price from what markets have done, and system models rebuild it from what the system is made of. Behind both sits a general trade-off [78, 79]: tools with fine spatial resolution simplify temporal dynamics, and tools with rich temporal detail aggregate space. Table 2.3 places the principal tools against the requirements above. Zeyringer et al. [80] show what the gap costs: designs that neglect the coincidence of high wind and low prices overestimate how much capacity is worth building.

Statistical models estimate the relationship from observed data. Hirth [2] combined meta-regression with dispatch modelling to estimate the slope of capture value against penetration. Stiewe et al. [12]

Table 2.3: Principal modelling tools against the requirements of this thesis. Each covers one dimension of the problem, resource, price, network, or space, and stops short of the others.

Tool	Type	Spatial	Temporal	Key strength	Missing dimension
atlite [76]	Resource conversion	0.05–0.25°	Hourly	Open; flexible weather cutouts	No market or revenue component
Renewables.ninja [72]	Resource conversion	Site	Hourly	Bias-corrected; validated globally	No market or revenue component
PyPSA-Eur [81]	System optimisation	Nodal	Hourly	Full European network; open	Shadow prices rather than observed prices
Balmorel [82]	System optimisation	Zonal (Nordic–Baltic)	Hourly	Regional hydro coupling; open	Regional scope; same price caveat
EuroMod [83]	Dispatch / market	Zonal (27)	Hourly	Realistic price volatility	Proprietary data; tied to calibration period
PLEXOS [84]	Unit commitment	Nodal/zonal	Sub-hourly	Industry standard; generator detail	Proprietary and costly
Stiewe et al. [12]	Panel regression	Zonal (30+)	Annual	Cross-border spillovers	Fixed effects absorb the why; no map
Bosch et al. [85, 86, 87]	GIS spatial	0.05°	32 slices/yr	Depth, distance, LCOE	No market value or revenue

extended this to a panel across more than thirty zones,

$$VF_{i,t} = \alpha_i + \beta_1 WP_{i,t} + \beta_2 \overline{WP}_{N(i),t} + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t}, \tag{2.2}$$

where $VF_{i,t}$ is the wind capture value in zone i, $WP_{i,t}$ is domestic wind penetration, $\overline{WP}_{N(i),t}$ is the capacity-weighted penetration of directly interconnected neighbours, and α_i is a zone fixed effect. That fixed effect is where the explanatory limit sits: everything that makes one market different from another is absorbed into a single intercept. Maciejowska [88] showed that wind compresses the upper tail of the price distribution more than it shifts the mean, which is why capture prices fall faster than average prices, and machine-learning variants have improved short-term forecasting accuracy over linear specifications [89, 90].

The defining constraint of this family is structural: a model fitted over a range of conditions describes that range. Tree ensembles, the strongest performers in recent comparisons [89], are piecewise constant by construction, so beyond the training data no splits exist and the prediction stops changing. A linear specification will extend beyond its data, but only by asserting that a slope estimated at penetrations of 10 to 35 % still holds at 60 %, which is precisely where the evidence says the relationship steepens [25, 23].

System models construct the price instead of fitting it. A cost-minimising dispatch and investment program returns the shadow price of the energy balance at each node, which under marginal-cost pricing is the wholesale price. Green and Vasilakos [91] applied this to the British market early, finding that large-scale offshore deployment lowers wholesale prices and with them the revenue of the deployers themselves. PyPSA-Eur [81, 92] provides an open implementation with the full European transmission network at nodal resolution, since extended globally [93]. Because such a model re-derives the merit order for any system state, it can price configurations that have never existed, which is what a 2040 scenario requires.

Its errors are equally structural and run in a known direction. The shadow price excludes mark-ups, strategic bidding and scarcity rents [94]. Without unit commitment the model cannot bid below zero, so it cannot reproduce the negative prices that are cannibalisation's sharpest symptom [36]. Perfect foresight lets storage, and Nordic hydro in particular, absorb surplus at exactly the right moment. Clustering the network hides congestion inside each node. Every one of these approximations raises capture value, and each is largest in small, hydro-adjacent, well-interconnected zones. Commercial tools relax some of them through non-convex generator constraints [84, 83] but require proprietary plant data, and the

Nordic and Baltic region has its own open lineage in Balmorel [82], which shares the same approximations. One boundary applies to both families: adding neighbouring countries' generation and prices measurably reduces day-ahead forecasting error [95], so no single-zone model of either kind sees the whole mechanism.

2.5.3 Comparison of the two model families

A statistical model is accurate where observations are dense, and beyond them it stops responding to its inputs. A system model can price any future, but it simplifies the market it rebuilds, and the error that follows is consistently in the same direction. Neither limitation can be removed by improving a model within its own family.

No study compares them directly: where both estimates of the same quantity exist, the literature offers no way to decide which to believe, no test for the point at which one should replace the other, and no comparison with observation that would reveal the consistent error of the one and the flat response of the other. Supplying that comparison is one contribution of this thesis.

The tools for making it are standard and have simply not been applied to this question. They cover the choice of error measure [96], the separation of a consistent offset from other kinds of error [97], testing whether a difference in accuracy is real [98], and holding out whole locations or whole periods rather than individual observations when data are structured in space and time [99]. Section 3.6.3 sets out how each is used here. What matters for the argument is that the two families fail differently, so a comparison reporting only an average error would conceal exactly the difference that decides which model carries a projection.

2.5.4 LCOE and its extensions

The most developed spatial framework is the progression by Bosch, Staffell and Hawkes [85, 86, 87], which moved from gridded capacity factors, to offshore depth and distance constraints, to spatially resolved levelised cost, and was later extended globally [100], all on open bathymetry and resource data [101, 102]. What these frameworks optimise is the levelised cost of energy (LCOE), the lifetime cost of a project per megawatt-hour produced, which is the standard basis for comparing technologies and sites [103, 104].

These frameworks characterise cost comprehensively and revenue not at all. A site with low levelised cost may still return poorly if it sells into a market where cannibalisation has driven capture prices below that cost.

A parallel literature has long argued that levelised cost is the wrong comparator. Ueckerdt et al. [22] decomposed integration cost into profile, balancing and grid components, of which profile cost is exactly the capture value shortfall of Section 2.2; Hirth [21] made the same argument through market value; the International Energy Agency operationalised it as the value-adjusted levelised cost of electricity [105]; and Joskow [106] had earlier objected that comparing levelised costs across technologies with different output profiles is economically meaningless.

The gap is that the two literatures have been kept apart. Value-adjusted metrics are computed for a technology within a system; spatial frameworks for a location within a map. Nobody has computed a value-adjusted cost *per location*, which is what an investor choosing between a Polish and a German site needs, and which is necessary before asking whether a ranking of sites by cost survives the addition of value.

This thesis closes that gap with one derived metric. A merchant farm breaks even when the price it captures covers its levelised cost; its captured price is $\bar{p}\,\mathrm{CV}$ (Equation 2.1), so the average wholesale price a site needs is

$$\mathrm{vLCOE} \;=\; \frac{\mathrm{LCOE}}{\mathrm{CV}}, \tag{2.3}$$

the *value-adjusted LCOE*. Where the IEA's VALCOE adjusts a technology's cost additively within one system [105], this metric is multiplicative, carries the measured capture value directly, and is computed per location, so it can be mapped. Because capture value sits in the denominator it can rise even as the raw LCOE falls, whenever cannibalisation erodes value faster than learning cuts cost.

2.6 Research gaps and questions

Four gaps follow from the review.

No systematic cross-basin comparison of capture value. Capture values have been measured within national markets and across a European panel [12], but no study compares them across the full set of bidding zones surrounding both basins under a consistent method, and merit order estimates for Baltic markets are effectively absent [25].

Cannibalisation is treated as a property of penetration rather than of system design. The decline is well documented and interconnection and flexibility are known to moderate it [12, 41, 45], but the literature does not decompose a decline into the portion a different system configuration would remove and the portion that is structural. Without that split, a capture value figure cannot distinguish a transmission problem from an oversupply problem, and the policy implications of the two are opposite.

Market value is absent from spatial assessment, and support design from both. GIS frameworks rank sites on cost while ignoring revenue [85, 86, 87], and the contract-design literature [48, 50, 49, 52] treats revenue risk as a financing question rather than asking which physical component of a value decline a given instrument absorbs.

No method both measures and projects. Statistical models are accurate in sample and cannot extrapolate beyond their support; system models project freely and carry structural bias. The literature offers no validated procedure for combining them or for establishing at which point one should replace the other.

These gaps motivate three research questions.

1. How does offshore wind capture value in the enclosed Baltic compare with the open North Sea today, and how does it evolve under the committed buildout to 2030 and 2040?

2. Which market mechanisms determine the rate of value erosion, do they differ between an enclosed and an open basin, and how much of the erosion is removable by system design rather than inherent to the resource?

3. Can a model trained on historical observation project these futures, or is a system model required, and how should the two be combined and validated against each other?

The third question is the one on which the other two depend, since every projected number in this thesis comes from one model or the other, and it is the one the literature leaves most open. This thesis builds one model of each kind, a statistical model fitted to eight years of observed prices and a system model that rebuilds the market from physics and cost, and sets them side by side. Somewhere the two will disagree, and the practical decision is which of them produces the 2040 numbers this thesis reports. Section 3.6 sets out how that decision is made from measurement rather than by inspection.

Answering these requires four objectives, running from the apparatus outward to the decisions it informs.

1. *Build and validate the two models.* Construct a virtual wind farm from reanalysis and validate it against metered generation; fit a capture value model carrying market characteristics as explicit features rather than fixed effects, scored by leaving whole farms out; configure an open system

model of both basins under official capacity trajectories; and confront both against observation where reality can arbitrate.

2. *Measure and project capture value across both basins.* Establish where each bidding zone stands today and how it evolves to 2030 and 2040 under the committed buildout.

3. *Decompose the erosion into removable and structural parts.* Separate what an unconstrained network or a flexible demand side would remove from what the buildout imposes regardless, and identify who bears each part once support contracts are accounted for.

4. *Map the value-adjusted cost of energy and test whether capture value changes where to build.* Carry the validated capture values onto the map alongside resource, water depth and distance to shore, and fold them into cost to give a *value-adjusted levelised cost of energy*, the average wholesale price a site must earn to break even, computed per location rather than per technology. Then compare the ranking of candidate sites by cost alone against the ranking once value is included. If the rankings agree a cost-only screen is sufficient; if they diverge, site selection has been made on an incomplete criterion, and the divergence grows over the period across which projects are financed.

The remainder of this thesis pursues these objectives using open data throughout, so that every step from raw download to final figure can be inspected and re-run.

3 Methodology

3.1 Rationale for a two-model approach

The rationale for building two models is a single comparison: one model is constructed from first principles and the other learns from past trends, and setting both against observed prices establishes which is better suited to projecting a future that has not been observed. Furthermore, observing if one is better suited for short term vs long term projection. More details on the logic flow of both models can be found in figure 3.1 and table 3.1.

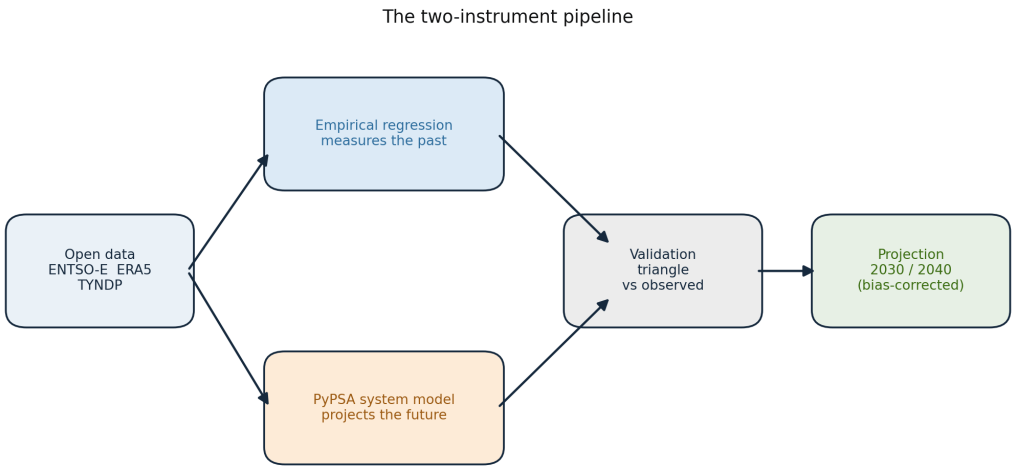


Figure 3.1: The design of the thesis. Shared data feed the two models, both are compared against observed capture values, and the comparison decides which model is used for what.

Table 3.1: How the two models are built, what they produce, and what each is used for.

	Regression model	**System model**
Algorithm	Gradient-boosted trees (XGBoost)	Linear optimisation in PyPSA-Eur (Gurobi)
Built from	Seven market conditions, measured monthly for 31 farms from 2017 to 2024, giving 2,706 observations	The transmission grid, generation fleets, ERA5 weather and TYNDP scenarios, solved hourly across thirty nodes
Produces	Monthly capture value for each farm	Hourly prices and generation in every zone, and annual capture value per zone
Compared against	Observed capture value, with whole farms held out of the fitting	Observed capture value in each zone and year
Used for	Measuring the present and explaining the mechanism	Projecting 2030 and 2040, with its measured error subtracted

3.2 The use of capture value

The regression model applies it monthly to each farm, using day-ahead prices. The system model applies it annually to each zone, using its own modelled prices. The comparison of Section 3.6.1 applies it to both against observation. In every case it is calculated only over the hours for which both a price and a generation figure exist, which is what makes the three sets of numbers comparable. Denmark is treated as two zones throughout, DK1 and DK2.

One feature of the definition matters for what follows. Capture value depends on the prices in a zone, weighted by when a farm generates. Prices are observed for every zone in the study, and generation comes from the virtual wind farm of Section 3.4.1, so a capture value can be measured for any zone with observed prices, whether or not offshore wind operates there.

The regression model uses this to train on all seven enclosed Baltic zones instead of the one zone with operating offshore wind. It weights prices that have already occurred by a wind profile that has been validated against metered output, so the only unobserved element is the farm itself. The projection in Chapter 4 asks more than this, since it prices farms in markets that do not yet exist.

3.3 Data sources

3.3.1 Market data

All market data come from the ENTSO-E Transparency Platform and cover 2017 to 2024. They comprise day-ahead prices for 22 bidding zones, hourly load, hourly generation by technology, flows across 28 borders, and the yearly capacity of each border. Everything is hourly and in UTC, and prices are resolved by bidding zone rather than by country, so DK1, DK2 and SE4 are separate.

The platform under-reports Dutch renewable generation, which is therefore rescaled to match physical capacity factors while keeping its hourly shape [107], and British day-ahead prices cease publication after Brexit, so the three British farms contribute only up to 2020. No other corrections are applied to the market data.

3.3.2 Meteorological and wind farm data

The study covers 31 offshore wind farms, sixteen in the North Sea and fifteen in the Baltic, including sites that are operating, under construction and planned. The operating farms are the ones whose modelled generation can be checked against metered output, and they are used for validation. Every site in a zone with observed prices is used to fit the regression model, whether it is operating or not (Section 3.4.3).

A farm belongs to the sea its foundations stand in. DK2 and SE4 are part of the Nordic synchronous area, but their farms stand in the western Baltic, experience Baltic weather, and sell into the same enclosed market as the Polish and Baltic-state fleets, so they are treated in this work as Baltic.

Within that geography the analysis uses four groups, kept the same throughout. The open **North Sea** is the reference: Germany, the Netherlands, Belgium, DK1 and France. The **under-built** Baltic is Poland, Estonia, Latvia and Lithuania, which have very little offshore wind built so far. The **built-out** Baltic is DK2 and SE4, which already have offshore wind in operation. **Finland** is kept separate, because its low capture value comes from nuclear generation holding prices down rather than from wind. The groups are chosen so that the built-out zones show today what the under-built ones are heading towards.

Hourly wind speeds at 10 and 100 metres come from ERA5, the fifth-generation reanalysis of the European Centre for Medium-Range Weather Forecasts, taken at each farm's nearest 0.25 degree grid cell. A reanalysis is a modern weather model re-run over the past and combined with every available observation, and it is the standard source of wind data for energy studies. ERA5 is known to underestimate the strongest offshore winds [75]. These speeds are left uncorrected, because the power-curve adjustment of Section 3.4.1 corrects the average level only, and the validation reported there tests whether the resulting output is nonetheless accurate.

3.3.3 Scenario data

Future capacities, demand, fuel and carbon prices and battery fleets come from the 2024 Ten-Year Network Development Plan, the official scenario framework of the European transmission system operators [108]. They are extracted once into a single set of per-country tables, which both models then use. Because these tables cover renewable technologies only, the thermal capacity they omit, and the corrections that omission requires, are set out in Section 3.5.4.

3.4 The regression model

3.4.1 Virtual wind farm model

Most of the farms in the study are not yet built, so generation is modelled for every site, including those already operating. The method follows Staffell and Pfenninger [72] as adapted by Jacquet [107], in three steps.

1. **Height extrapolation.** ERA5 gives wind speed at 10 and 100 metres, and the IEA 15 MW reference turbine used has a hub height of 150 metres [109]. Speed is carried up to that height by fitting a logarithmic profile through the two known levels,

$$u(z) = u_{\text{ref}} \frac{\ln(z/z_0)}{\ln(z_{\text{ref}}/z_0)}, \tag{3.1}$$

   where $u(z)$ is the speed at height z, z_{ref} is a height at which the speed is known, and the surface roughness z_0 is solved from the 10 and 100 metre pair. This is the standard description of how friction at the sea surface slows the wind near it and lets speed increase with height.

2. **Power conversion.** Speed is turned into power using the turbine's power curve, which rises steeply between the cut-in speed of 3 $\mathrm{m\,s^{-1}}$ and the rated speed near 10.6 $\mathrm{m\,s^{-1}}$, then stays flat up to the cut-out speed of 25 $\mathrm{m\,s^{-1}}$. Output is set to zero outside that range, so neither calm nor storm hours produce anything.

3. **Fleet adjustments.** Two corrections are applied, both calibrated by Hayes et al. on metered European farms [110]. The curve is smoothed ($\sigma = 1.17\ \mathrm{m\,s^{-1}}$), because the turbines across a single farm do not all meet the same wind speed, and it is shifted towards higher speeds ($w_0 = 0.71\ \mathrm{m\,s^{-1}}$) to account for wake losses and for the ERA5 bias noted above (Figure 3.2).

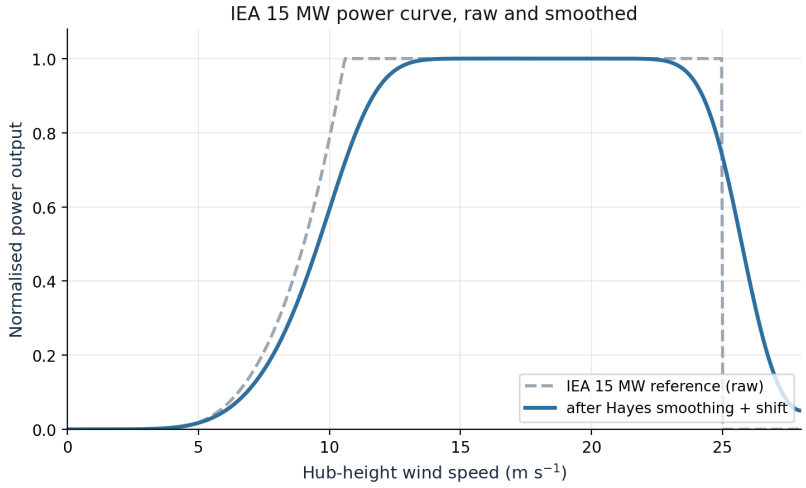


Figure 3.2: The IEA 15 MW reference power curve before and after the two adjustments, used to convert wind speed at hub height into output. Smoothing $\sigma = 1.17\ \mathrm{m\,s^{-1}}$, shift $w_0 = 0.71\ \mathrm{m\,s^{-1}}$.

The modelled generation is then checked against metered output. Aggregated to zone level, the modelled farms reach a median hourly correlation of 0.88 with the generation ENTSO-E reports, and 0.95 monthly, across five zones and three years (Figure 3.3).

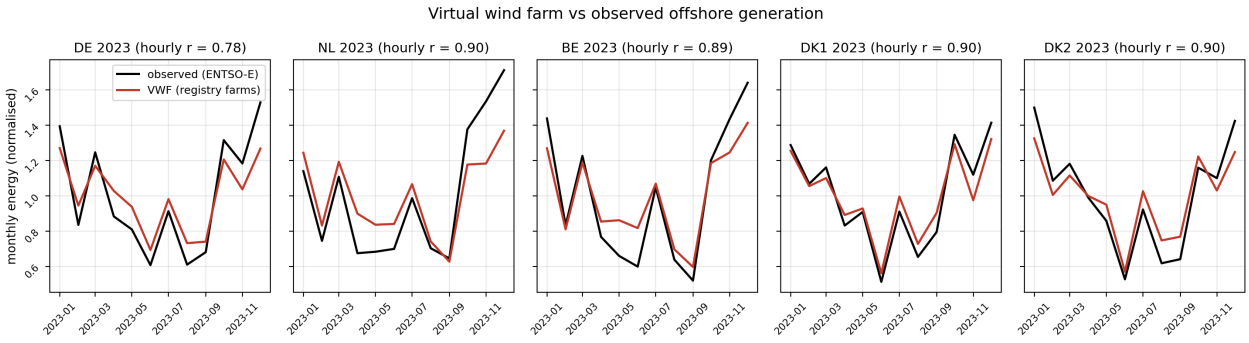


Figure 3.3: The modelled wind farms checked against observed offshore generation in 2023. Monthly energy from the modelled farms, normalised (red), against the metered fleet in the same zone (black). The median hourly correlation is 0.88 and the monthly correlation 0.95.

The timing of the output is what matters, because capture value weights prices by when a farm generates. The modelled output does sit above the metered figures in absolute terms, since it does not

subtract every loss a real farm incurs. This difference in level does not affect the results, because capture value is a ratio with generation on both sides, and the farm enters the model's inputs only through correlations, which are unchanged when output is scaled; a constant error in the level therefore cancels.

3.4.2 Feature set at bidding zone resolution

Each farm-month gets seven features, following Jacquet's design [107] with one upgrade, resolution: each is computed for the farm's *market area*, the bidding zone where zone-resolved data exist (DK1, DK2, SE4) and the country elsewhere, so the seven Danish farms see DK1 and DK2 conditions rather than a Danish average. Together they turn the cannibalisation mechanism of Section 2.2 into measurable quantities: how crowded the farm's selling hours are, how exposed its market is, and how much surplus the system can absorb.

Features 1 and 2 are the hourly correlations of the farm's output with its area's total variable renewable generation and with its load,

$$\rho^{\mathrm{VRE}}_{f,m} = \mathrm{corr}\big(g_{f,t},\, G^{\mathrm{VRE}}_{z,t}\big), \qquad \rho^{\mathrm{load}}_{f,m} = \mathrm{corr}\big(g_{f,t},\, L_{z,t}\big), \qquad t \in m. \tag{3.2}$$

The first measures crowding, a farm producing when every other renewable produces selling into depressed prices; the second measures alignment with demand, which lifts value. They enter separately so Chapter 4 can read their contributions apart; their signed difference forms the pattern penalty in the analytical equation of Section 3.4.4. Feature 3 is penetration, the monthly renewable market share $S_{z,m} = E^{\mathrm{VRE}}_{z,m}/E^{\mathrm{load}}_{z,m}$, which locates the market on the cannibalisation curve. Features 4 and 5 are monthly gas and carbon prices, which set the level of the thermal merit order and so how far prices fall when wind displaces it. Feature 6 is flexibility, storage capacity over renewable capacity, the system's ability to move surplus out of windy hours.

Feature 7 is interconnection, and it departs from both predecessors on Efthymiadou's evidence that realised interconnector *utilisation* rather than installed capacity dampens price volatility [111, 107]:

$$I_{\mathrm{int},z} = \frac{\sum_b \overline{|F_b|}}{\sum_b \max_t |F_{b,t}|}, \tag{3.3}$$

the mean absolute flow on each external border, divided by that border's capacity and summed across borders. The economic logic is that an interconnector relieves cannibalisation only when it carries the windy-hour surplus out of the zone, so its use and not its rated size is what lifts capture value. Exact definitions, the dropped neighbour-synchronicity discount and the index's limitations are recorded in Appendix B.

The two models represent cannibalisation differently: the regression model encodes it through these features, learned from data, while in the system model it emerges by construction as added wind shifts the merit order. Collinearity is clean (largest variance inflation factor 2.16), so the feature contributions in Chapter 4 read as near-independent mechanisms. Figure 3.4 shows how each of the seven has moved since 2017, averaged within each basin. This is the record the model learns from, and the starting point from which the projection of Section 3.4.5 moves them.

3.4.3 Algorithm selection and validation

In this work five algorithm families are tested: gradient boosting (XGBoost), random forests, support vector regression, a degree-two polynomial, and ordinary least squares. Their out-of-fold comparison is tabulated in Appendix I (Table I.1); gradient boosting is the strongest of the five and carries the analysis. The training unit is a bidding-zone market observation: a virtual farm at a representative site in a zone,

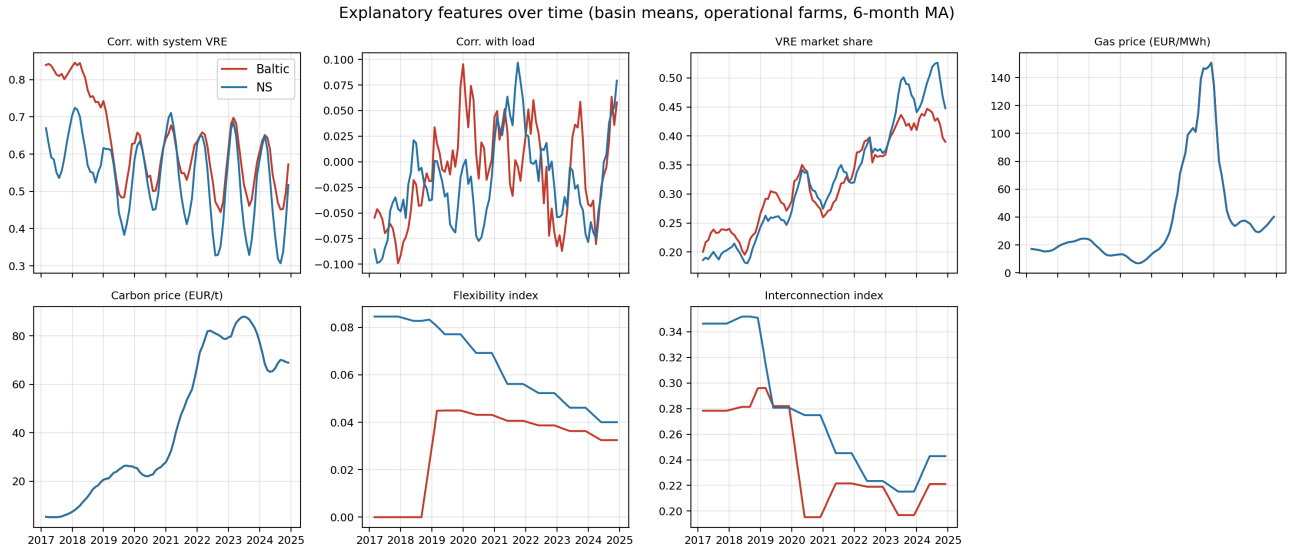


Figure 3.4: Basin-mean evolution of the seven features for operational farms, 2017 to 2024 (six-month moving averages). North Sea in blue, Baltic in red.

described by the seven features and labelled with its monthly capture value. By the principle of Section 3.2 a zone with observed prices is a valid observation whether or not a farm operates there, so the panel is all 31 sites over 2017 to 2024, 2,706 zone-months. This widens the enclosed Baltic from one operational price zone to seven. The operational farms remain the validation subset. No hyperparameters are tuned, so the comparison with Jacquet's untuned model is like for like.

Gradient boosting builds shallow decision trees in sequence, each fit to the error the ensemble still makes and added at a small learning rate, so nothing about the functional relationship is assumed in advance, and it captures nonlinear interactions a single equation would miss, for instance that interconnection lifts value only once penetration is high enough to need it.

Validation is layered, because one scheme can flatter a panel model, and all of them rest on one discipline, *out-of-fold prediction*: a prediction made by a model that never trained on that point. The schemes differ only in what they withhold.

- **Random five-fold cross-validation** withholds individual farm-months. It reproduces Jacquet's scheme, but for panel data it leaks: other months of the same farm stay in training, so the model is never asked about an unfamiliar farm.
- **Grouped five-fold cross-validation**, the headline scheme, withholds entire farms. Each farm-month is predicted by the one model copy blind to its farm, and the stitched series, every farm-month predicted once, is what all grouped metrics in Chapter 4 score.
- **A chronological split** trains through 2022 and tests on 2023 to 2024. It withholds the future and stress-tests extrapolation across the post-crisis regime change.
- **Leave-one-year-out** fits on every other year and predicts the held-out one. Where the single chronological split conflates two failures, inability to forecast any year and inability to cross the 2021 to 2022 regime break, this separates them year by year, matching the multi-year temporal suites of both predecessors.

- **Basin-transfer fits** train on one sea and test on the other, to probe whether the learned relationships are basin-specific.
- **A variant excluding 2022** mirrors Jacquet's removal of the gas-crisis months.

The stitched out-of-fold series is exported for the three-way comparison: when Section 3.6.1 sets the regression against observation in a Danish zone, the prediction comes from a model that never saw a Danish farm.

3.4.4 Interpretation of the fitted model

Two tools are used to read an explanation out of the fitted model. The first is SHAP, short for SHapley Additive exPlanations, which splits each prediction into a contribution from each of the seven quantities and is computed on the data the model was fitted to.

SHAP gives the size of each contribution but not the shape of the relationship, so Jacquet's analytical equation is refitted as the second tool [107]. Its starting point is that a farm selling in random hours would have a capture value of one, and cannibalisation is what subtracts from that one:

$$\mathrm{CV} \;=\; 1 \;-\; B\,\frac{M_{\mathrm{WF}}}{M_{\mathrm{BZN}}}\,S \;-\; C\,M_{\mathrm{WF}}\left(0.202\,p_{\mathrm{CO_2}} + p_{\mathrm{gas}}\right)^{\gamma}, \tag{3.4}$$

where S is the renewable market share and p_{gas} and $p_{\mathrm{CO_2}}$ are the monthly gas and carbon prices, folded together by the emission intensity of gas. The first subtracted term represents cannibalisation through the pattern penalty of Section 3.4.2, $M_{\mathrm{WF}} = \mathrm{sign}(\rho_{\mathrm{VRE}} - \rho_{\mathrm{load}})\,|\rho_{\mathrm{VRE}} - \rho_{\mathrm{load}}|^{\alpha}$, which grows when a farm's output follows the system's renewables more closely than it follows demand, is scaled by the market share S, and is divided by the buffering term $M_{\mathrm{BZN}} = (I_{\mathrm{int}} + I_{\mathrm{flex}})^{\delta}$, in which interconnection and flexibility absorb the surplus. The second subtracted term represents the distortion of capture value by fuel prices.

Each constant controls one part of the mechanism: α sets how sharply the penalty grows as a farm's output becomes more crowded, B sets the overall strength of the cannibalisation term, and δ sets how strongly interconnection and flexibility protect against it. The constants are fitted separately for each basin by least squares within fixed bounds, with confidence intervals obtained by resampling whole farms; the fitted values are in Appendix B.

The fitted constants cannot be compared between the basins one at a time, because different combinations of them produce almost the same curve: a stronger crowding penalty, through a larger α or B, can be offset by stronger buffering, through a larger δ, and the fit cannot fully separate the two. What the fit does pin down is the behaviour of the equation as a whole. The two basins are therefore compared on what their fitted equations predict, namely how much capture value falls when penetration rises, calculated for both basins at the same market conditions so that the comparison reflects the equations and never a difference in circumstances.

3.4.5 Projection with the regression model: approach and limitations

Projecting a farm that has not been built means giving each of the seven quantities a future value.

- The two pattern correlations are held at the site's recent values, since they follow from its weather and its position on the grid [107].
- Penetration is scaled by the growth in renewable generation relative to the growth in demand in the TYNDP scenario.

- Gas and carbon prices are taken from the scenario.
- Flexibility grows with the battery fleet in the scenario.
- Interconnection is held at its recent value. This is mildly optimistic, since transfer capacity is unlikely to keep pace with wind, and a sensitivity is reported alongside.

Twelve sets of inputs are built for each year, one per month, and the predicted capture values are averaged, so the projection keeps the site's own seasonal pattern.

The decisive test is whether the model is being asked about conditions it has seen. A projection is trusted only if all seven quantities lie inside the range covered by the training data, because outside that range the model has nothing to learn from.

A check on each quantity separately would be insufficient, however, because a projection can sit inside the range of every quantity taken one at a time and still be a combination that never occurred, as the seven do not vary independently. A stricter test, following Jacquet, asks whether the projection lies inside the region the training data occupy in all seven dimensions at once, using its convex hull. Where a projection fails, the prediction is repeated with the offending quantity capped at its training maximum, which shows whether the answer depends on the part of the range the model has never seen.

The equation of Section 3.4.4 is exempt from this limit, being a formula rather than a set of decision trees, and can therefore be evaluated at any input, including values outside the training range. It is projected on the same scenario inputs given to the system model, which gives a third independent estimate. It differs from Jacquet's use of it in one way. He holds the pattern correlations fixed, whereas in this work the Baltic projection lets the correlation with system wind rise towards the level the built-out Baltic zones already show, and lets interconnection utilisation fall as wind grows faster than transfer capacity (Appendix B).

3.5 The system model (PyPSA-Eur)

3.5.1 Price formation in PyPSA-Eur

PyPSA-Eur is the European configuration of PyPSA [81], an open workflow that assembles a model of the European power system from public data. The transmission grid comes from ENTSO-E maps, the power plants from open databases, the weather from the same ERA5 data used for the virtual wind farm, and demand from historical load. The thousands of connection points in the real grid are grouped into a smaller number of representative nodes, which is the setting that trades spatial detail against the size of the problem to be solved.

The model finds the cheapest way to meet demand in every hour of a year, subject to the capacity of every plant and every line. That problem is a linear program, so the solution found is the true optimum rather than a good approximation, even with tens of millions of variables. It is solved with the barrier algorithm in Gurobi.

Two of its outputs are used in this thesis. The first is how much every generator produces in every hour. The second is the shadow price at every node, which is what it would cost to serve one more unit of demand there in that hour. In a windy hour the last unit needed comes from wind, which costs almost nothing to run, so the price falls towards zero. In a still hour it comes from a gas plant, which sets a much higher price. A real market clearing on marginal cost prices in the same way, which is why the shadow price is used as the wholesale price. Practical details are in Appendix B.

3.5.2 Scope and configuration

The model covers the fourteen countries around the two basins. They are clustered into thirty nodes, with more nodes given to the markets that carry most of the offshore wind, and solved at full hourly resolution. Thirty nodes is finer than one node per country, so Denmark separates into its two bidding zones, and congestion within a zone as well as between zones is priced at each node.[1]

Transmission is held at today's capacity throughout. Congestion is priced as it exists now, which makes the projection conservative. Section 3.6.4 solves the opposite case, with transmission free to expand.

Generation fleets are set to the capacity installed in each modelled year, and topped up where the database falls short of national statistics. For 2023 and 2024, fuel and carbon prices are set to the observed annual averages.

3.5.3 Historical baselines and calibration

Three past years are solved. 2023 is the main validation year. 2019 is added so that the model is tested both before and after the energy crisis, in a cheap-gas world and an expensive one. 2024 is added because it provides validation points at the highest penetrations yet observed, which is what the bias correction of Section 3.6.2 is fitted on.

The calibration is judged on price levels and on capture value. Price levels are compared zone by zone against observed day-ahead prices, using a normalised absolute error, because percentage errors are undefined when prices are negative. Capture value, the quantity the thesis uses, is checked separately, and a third value is calculated alongside it to locate any error: the model's own generation valued at observed prices. If that hybrid value matches observation while the fully modelled value does not, the modelled prices carry the error and the modelled generation is sound.

Solver settings are the same for the past years and for the projections, so the bias correction is applied to prices formed in exactly the same way as the ones it was fitted on. The settings are listed in Appendix B.

3.5.4 Projection to 2030 and 2040

The projection solves six scenarios from the 2024 TYNDP trajectories on the validated 2023 network. Four follow National Trends, the pathway that tracks existing national policies: for each of 2030 and 2040, one solve installs the Baltic offshore capacity the plan schedules, and one an accelerated variant in which the Baltic builds further. The remaining two repeat the accelerated 2040 case under the plan's other demand pathways, Distributed Energy and Global Ambition, which electrify more of Europe's energy demand. All six are listed with their results in Table 4.4.

Two assumptions govern demand. The base year is 2019, the last year before the crisis depressed consumption, and demand in each scenario year is that base scaled by the country's TYNDP growth factor, with documented substitutes for the two countries the scenario tables omit. Capacity targets are adopted in full: where a scenario prescribes more capacity than the clustered nodes can physically host, the whole target is still installed and the excess is recorded, so no capacity is dropped without record.

Six further corrections close the remaining gaps between renewable-only scenario tables and an electricity-only model, and Table 3.2 states each with its implementation and rationale. All act in the same direction, preventing the model from generating surplus zero-cost electricity that would make capture value look higher than it should be.

[1] In a zone represented by more than one node, the denominator of capture value is the price averaged across that zone's nodes, which corresponds to the single zonal price of Equation 2.1, while the numerator weights each generator by the price at its own node. The two are identical wherever offshore wind is present at all of a zone's nodes, which is the case in the single-node Baltic zones that carry the headline result. A difference arises only in zones that also contain inland nodes with no offshore wind.

Table 3.2: The six corrections applied to the scenario inputs before the 2030 and 2040 solves. Each closes a gap between renewable-only scenario tables and an electricity-only model.

Correction	What is changed, and why
A. Coal phase-out	German and Polish coal and lignite capacity is reduced year by year in line with the legislated closure schedules, which the scenario tables leave out.
A2. Polish nuclear	Poland's announced 3.9 GW nuclear programme is added from 2040, since the scenario tables add no thermal capacity at all.
B. Offshore caps	Danish, Finnish and Swedish offshore capacity is capped at the nationally planned levels, removing capacity that the scenario tables wrongly assign to these zones through planned energy islands.
C. DK1/DK2 split	The Danish offshore total is split 60/40 between the DK1 and DK2 bidding zones by 2040, because the default allocation by wind resource alone gives DK2 too little.
D. Hydrogen demand	Electrolysers with a fixed hydrogen demand are added in each country, alongside a priced import alternative that sets the level at which producing hydrogen is worthwhile. This gives surplus electricity a use; without one, an electricity-only model produces implausibly many hours priced at zero.
E. Demand base year	Demand is projected from a 2019 base using the TYNDP growth factors, with documented substitutes where factors are missing, because the factors are quoted relative to 2019 and two countries are absent from the scenario tables.
F. Shortage pricing	Demand response is added at 5 % of peak load, offered at 400 €/MWh, with involuntary curtailment at the 4,000 €/MWh price cap, so that hours of shortage are priced the way a real auction prices them.

3.6 Validation against observed prices

3.6.1 Comparison with observed ENTSO-E prices

For every zone and year in which offshore generation was observed, which is five zones across 2019, 2023 and 2024, the same capture value is estimated three times: from observation, by the system model, and by the regression model. The regression estimate comes from a copy that never saw that zone's farms.

These three estimates are mutually independent, since the regression estimate uses nothing from the farm it is predicting, the system estimate uses no market prices at all, and the observed value comes from outside both models. Agreement between them is therefore informative, and where they disagree, the pattern of the disagreement shows which model is at fault. This is a different exercise from the hybrid value of Section 3.5.3, which mixes modelled generation with observed prices in order to locate an error inside the system model.

3.6.2 Bias correction procedure

The comparison above shows the system model reading high in small congested zones, and by more where more offshore wind has been built (Chapter 4). That relationship is what the correction uses. A straight line is fitted by least squares through the measured overstatements against the zone's offshore

penetration, meaning its offshore generation as a share of consumption, and the fitted line is subtracted from the model's output:

$$\mathrm{CV}_{\mathrm{corr}} = \mathrm{CV}_{\mathrm{raw}} - \left(\hat{a} + \hat{b}\,\pi\right), \tag{3.5}$$

where π is the zone's modelled offshore penetration and $\hat{a}$ and $\hat{b}$ are the fitted intercept and slope. The model's own penetration is used, not the observed one, because the correction is applied to the model's own output. This quantity is narrower than the all-renewables share used as a feature by the regression model.

3.6.3 Decomposition and scoring of the difference between the models

Section 3.6.1 establishes that the two models differ and four further tests establish why: where they agree, where they part, how each compared to the observed counterparts, and how their projections compare.

Channel decomposition. Both models compute capture value with the same formula (Equation 2.1) from different inputs. The regression model uses the virtual farm's generation valued at observed day-ahead prices, and the system model uses its own generation valued at its own prices. Substituting one input at a time, and averaging over both possible orders, divides the difference between their answers exactly into a generation part and a price part, with nothing left over and no dependence on which substitution is made first. The price part divides again by re-pricing against observed prices with negative values set to zero, which isolates the effect of the linear program being unable to produce them. A fourth part carries the way the network is represented, and a small remainder absorbs the out-of-fold fit and the averaging of prices across nodes.

Scoring both models on targets neither has seen. Comparing the two directly would favour the regression model, since it is fitted to the record it would be scored on. It is therefore refitted with the target withheld, using the grouped schemes of Section 3.4.3, once leaving out each bidding zone and once leaving out each year. Whole blocks are withheld rather than individual observations, because capture value is correlated in space and in time, and withholding single points would leave near-duplicates in the training set [99]. Both models are then scored on the same zone-months against three measures.

Mean absolute error and root mean squared error are reported on the natural scale of capture value, which is dimensionless and bounded, so the usual objections to percentage errors do not apply [96]. Signed bias records the direction of each model's error. A skill score compares both against the benchmark of assuming that a farm simply earns the average price. Because zone-months within a zone are not independent, the difference between the two models is tested by a paired comparison with a bootstrap that resamples whole zones [98].

Accuracy alone would not settle the question, since two models can be equally accurate and have different errors. Each model's error is also separated, following Theil [97], into three parts: a consistent offset, a failure to match how much the observations vary, and mistiming, which is the error that remains once level and spread are both right. The arithmetic is in Appendix G.

Backtesting the projection. The test above scores the models but not the act of projecting, which also requires future inputs to be built. The regression model is therefore refitted on data up to 2021 and asked for 2023 and 2024 twice: once from the inputs that occurred, and once from inputs constructed the way a projection must construct them, with weather terms set to each farm's pre-2022 average and system terms carrying the pre-2022 monthly shape. The system model is fitted to nothing, so it is already a projection, and it is scored on the same targets. The difference between the two regression runs is the cost of building the inputs, and the difference between either of them and the system model is the like-for-like comparison. The constructed inputs use the annual levels that occurred, and not those a forecaster in 2021 would have assumed, so the test grants perfect foresight of the scenario.

Comparing the two projections. Both models are run on the same TYNDP scenarios, so the only difference between their answers is the model. They are compared at 2030, where both are still inside the observed range, and at 2040, where the regression model is not. Because 2040 cannot be checked against observation, both projections are also placed beside published estimates of capture value at high penetration, recording the renewable share at which each published figure was measured. Recording that share matters, because a benchmark measured at a share on which the regression model was trained cannot separate the two models, since a model reproducing conditions inside its training range will match it regardless of its behaviour beyond that range, and only a benchmark from an already saturated market is decisive.

The exclusion of DK2. DK2 is left out of the scored set, on a criterion fixed before the results were seen. Its prices are set partly by plants that must keep running and by combined heat and power, neither of which a linear program represents, and this shows in its 264 observed hours of negative prices in 2023 against the system model's zero. It remains in the regression model's training data, and every headline figure is also reported with DK2 included.

3.6.4 Robustness, model runs and reproducibility

The projected values rest on assumptions about fuel prices, the transmission grid, the pricing of shortage, and the six scenarios chosen, and the sensitivity of the results to each of these is tested in turn.

- **Gas prices.** The assumed gas price is varied by re-solving three buildouts across five gas prices, which separates the effect of the buildout from the effect of fuel costs.
- **Transmission.** The grid assumption is inverted by re-solving all six scenarios with transmission free to expand, which gives the bound opposite to the grid held at today's capacity (Section 3.5.2).
- **Shortage pricing.** The treatment of shortage is tested by recalculating every capture value with the shortage-priced hours removed, meaning those in which supply is tight and the price rises to the levels set by demand response or the price cap, which establishes that the result does not rest on a few extreme prices.
- **Scenario choice.** The dependence on the six chosen scenarios is tested with a fast approximation of the system model, trained on 76 solved networks, which prices a scenario in milliseconds and so allows far more scenarios to be examined than could be solved directly.

The results of all four sensitivity analyses, together with the validation of the fast approximation, are collected in Appendix D.

Every model run behind Chapter 4 is listed in Table K.1 of Appendix K, so each number can be traced to the run that produced it, and the whole procedure is scripted from the raw downloads to the final tables. A single command re-solves everything in about three hours, every reported number can be regenerated from public data, and the solver settings and assumptions are fixed in Appendix B.

3.7 Value-adjusted cost of energy

Capture value on its own does not tell an investor whether to build, so this section turns it into a cost. The measure used is the value-adjusted levelised cost introduced in Section 2.5.4: the levelised cost divided by capture value, which gives the average wholesale price a site must earn to break even. It is used because capture value enters it directly, so a projected fall in value becomes a rise in the price a site needs.

3.7.1 Mapping levelised cost

Costs are mapped at the resolution of the wind resource rather than that of the system model. Three geospatial datasets are used: capacity factor from the Global Wind Atlas at roughly two kilometres [102], water depth from GEBCO [101], and distance to shore, calculated on the same grid.

Capital cost rises with both depth and distance, since deeper water requires heavier foundations, with a step change beyond 60 metres where fixed-bottom foundations give way to floating ones, and sites further from shore require longer export cables. Capital cost is modelled in segments, one for each of these regimes, anchored to the PyPSA-Eur cost database and IRENA's 2024 review [81, 103] and set out in Appendix B. It is annualised at a 7 % cost of capital over a life of 27 to 30 years.

The levelised cost is the annualised cost divided by the energy a megawatt of capacity delivers in a year,

$$\mathrm{LCOE}(d,\ell,y) \;=\; \frac{\mathrm{CRF}\,\lambda(y)\,\mathrm{CAPEX}(d,\ell) + f_{\mathrm{OM}}(\ell)}{\mathrm{CF}\cdot 8760}, \tag{3.6}$$

and it is computed for every cell at all three years. The full set of maps is in Appendix L.

3.7.2 Combining cost with capture value

The value-adjusted cost (Equation 2.3) is calculated for each cell by dividing its levelised cost by the capture value of the zone it sits in, taken from the solved networks.

The same inputs give a second quantity, the net value of a site, which is the revenue it captures less its annualised cost,

$$\mathrm{NV}(d,\ell,y) \;=\; \underbrace{\mathrm{CF}\cdot 8760\cdot \mathrm{CV}\cdot \bar{p}}_{\text{captured revenue}} \;-\; \big(\mathrm{CRF}\,\lambda(y)\,\mathrm{CAPEX} + f_{\mathrm{OM}}\big). \tag{3.7}$$

The value-adjusted cost says what price a site needs. The net value says how much money it makes or loses at a given price. Both are reported for a merchant project, meaning one that sells at market prices with no support contract.

3.7.3 Pricing interconnection and robustness checks

Capture value enters the denominator of the value-adjusted cost, so any change in capture value changes the average wholesale price at which a site's revenues cover its costs. The effect of transmission on that break-even price is isolated by computing Equation 2.3 for 2040 twice, once with the per-zone capture values of the grid held at today's capacity and once with those of the grid free to expand, while the levelised cost is held identical in both. Transmission is a separate investment that leaves the cost of building and operating a farm unchanged, so only the revenue side of the calculation moves, through capture value, and the difference between the two computations is reported for each basin as the reduction in the break-even price, in €/MWh.

Whether the value-adjusted cost rises or falls to 2040 is then tested across 27 combinations of three cost assumptions: the weighted average cost of capital at 5, 7 and 9 %, a multiplier on capital cost of 0.8, 1.0 and 1.2, and a 2040 learning rate of 5, 12 and 25 %. The absolute cost level cancels when the 2040 value-adjusted cost is divided by the 2025 one, so the direction depends only on whether cost falls faster than capture value. That same ratio also gives directly the rate of cost decline that would be needed to hold the value-adjusted cost flat.

3.8 Methodological justification

Each choice above follows from one of the research questions.

- **Two models rather than one**, to compare them and decide which is better suited to representing the present and forecasting the short term, and which to forecasting the long term.
- **PyPSA-Eur for the system model**, because it is the only option combining open code and open data, the full European network, resolution at the level of bidding zones, and hourly dispatch.
- **Every scenario solved twice**, with the grid at today's capacity and with it free to expand, because the difference between the two isolates the part of the erosion caused by congestion, which is what the second research question asks.

The limits of each of these choices are recorded in Appendix B.

4 Results

This chapter reports the results in the order of the research questions. Section 4.1 establishes the measured record. Sections 4.2 and 4.3 validate the two models against observation, examine why they differ, and settle which is used for what, and Section 4.4 then projects the committed buildout to 2030 and 2040. Section 4.5 explains what drives the erosion and how much of it system design removes, and Section 4.6 carries the results into the decisions they inform, the break-even price a site must earn and where to build.

4.1 The empirical record of capture value

This section establishes where the two basins stand today, on eight years of observed prices.

4.1.1 Measured capture value, 2017 to 2024

Revenue per megawatt of installed capacity has three parts that change over time: how much a farm generates, the average price in its market, and capture value. Figure 4.1 separates them for 2017 to 2024. Generation is flat across the period, and the average price rises and falls with the gas crisis, moving in the same direction in every market. Only capture value declines steadily, and it is the term this chapter measures.

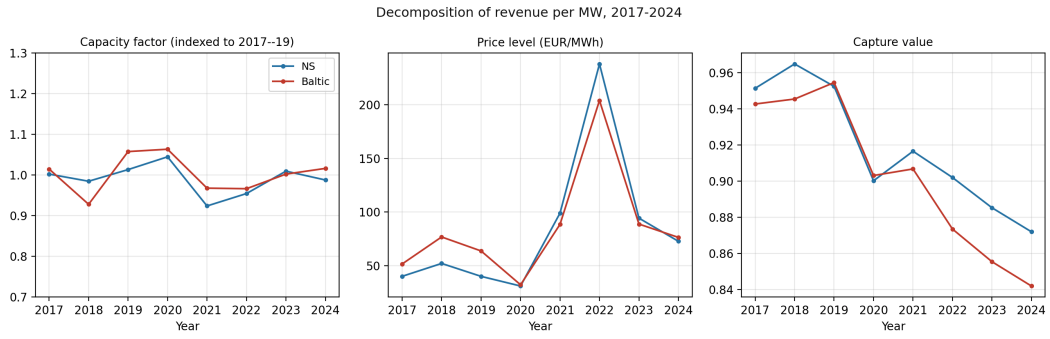


Figure 4.1: The three parts of revenue per megawatt, 2017 to 2024. Generation (left) is flat and the average price (centre) follows the gas crisis, while capture value (right) declines steadily.

Figure 4.2 shows capture value for the four groups of farms defined in Section 3.3.2. Over 2023 and 2024 the Baltic zones that have built little offshore wind capture 0.883 of the average price and the North Sea 0.879, which puts the two level. The lower Baltic average comes from the other two groups: DK2 and SE4, which have already built out, at 0.766, and Finland at 0.568, where nuclear generation holds prices down.

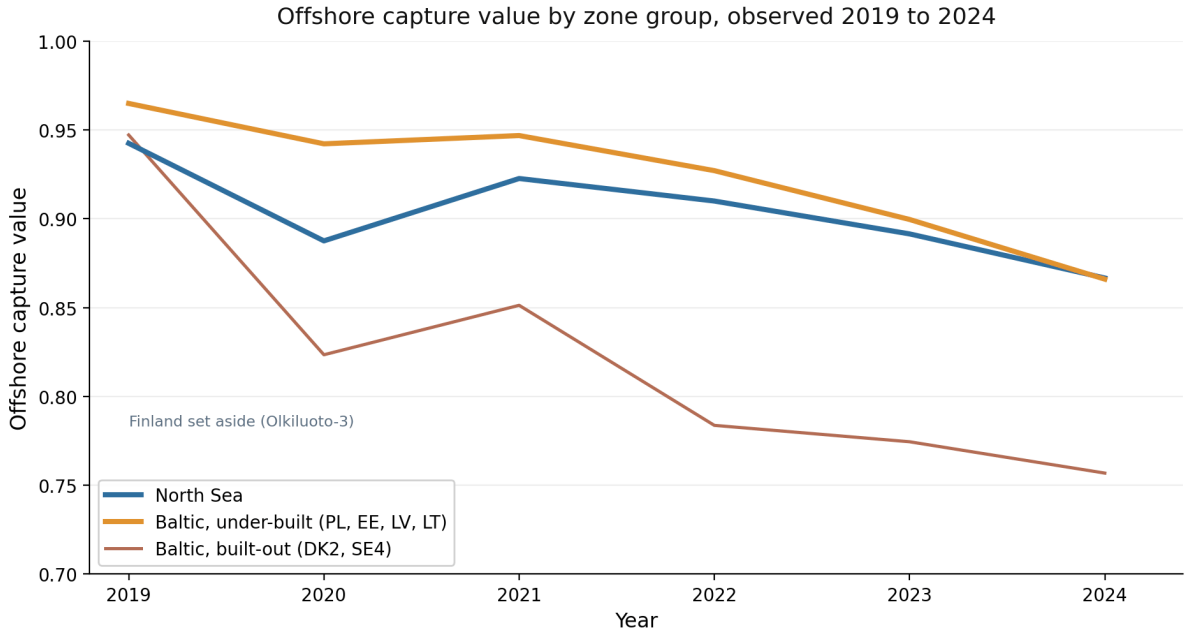


Figure 4.2: Observed capture value by group of zones, 2019 to 2024. The Baltic zones with little offshore wind capture the same share as the North Sea, while DK2 and SE4, which have built out, capture less than both.

Table 4.1 adds the measured values to the expectations set out in Chapter 2, and most of those expectations are incorrect. Germany was expected to be the most eroded market, yet its offshore farms capture 0.904. Its offshore fleet generates at different times from its much larger onshore fleet, and it is the onshore fleet that pulls the national figure down to 0.76 [12]. Denmark's national 0.822 combines two very different price zones, which are separated below.

The Baltic was expected to capture close to 1.0, above the North Sea, because little capacity has been installed there. The measured values are level with the North Sea. Poland captures 0.922, slightly above it. Estonia, Latvia and Lithuania capture 0.86 to 0.88, slightly below. The group averages 0.88 against the North Sea's 0.879, with almost none of the planned capacity built. Sweden at 0.729 and Finland at 0.568 sit lower, but each rests on a single farm in a market dominated by nuclear and hydro generation, where a surplus of constant output holds prices down as much as wind does. They are read as signals about their price zone more than about the farm, and the comparison holds without them. The hydro premium predicted by Hirth [2] does not appear in the Baltic after 2022.

Ordering the same zones by how much offshore wind they have built turns the comparison into a sequence. DK2 and SE4, the most built out, capture least. The zones with little capacity earn what the North Sea earns. The two basins are following the same relationship between buildout and capture value, and the Baltic entered it at the same level, with DK2 and SE4 showing the stage it has already reached.

Two comparisons show that what sets the value is the price zone a farm sells into. Four German farms stand in Baltic water but sell into the German market, and they capture 0.91, the same as German farms

Table 4.1: Expectations from Chapter 2 compared with measurement: farm-level capture values for 2023 and 2024, from modelled farm output and observed zonal prices. The Swedish and Finnish rows rest on one farm each.

Market	Basin	Expected in Chapter 2	Measured capture value	Farm months	Outcome
Germany	NS	Eroded, declining	0.904	144	Milder than expected
Netherlands	NS	Eroded	0.890	96	As expected
Denmark	NS/BS	Eroded, with some buffering	0.822	168	Averages two different zones
Sweden	BS	Favourable (hydro)	0.729	24	Lower than expected
Finland	BS	Favourable	0.568	24	Lower than expected, one farm
Poland	BS	Very favourable	0.922	48	Close to expectation
Baltic states	BS	Close to 1.0	0.86–0.88	72	Level with the North Sea

in the North Sea. The weather is Baltic, the market is open, and there is no discount. Denmark gives the reverse case: DK2 captures 0.784 and DK1 0.854, in the same country and under the same support scheme, separated only by the price zone their output must be sold in.

4.2 Validation of the two models

The supporting tables for the comparisons in this section are in Appendices I and J.

4.2.1 Performance of the regression model

The regression model was tested on price zones that were excluded from the inputs used to fit it. On those zones it accounts for 69 % of the variation in capture value across farms and months. Its estimates are within 0.04 of the observed value in five of the seven enclosed Baltic zones and within 0.07 in Sweden. The one clear miss is Finland, at 0.10, where nuclear generation holds prices down in a way the seven market conditions used as inputs do not describe.

Measuring those conditions separately for each bidding zone, instead of once per country, raises the score by nine points. Prices are set within zones, and treating DK1 and DK2 as a single Danish market would hide the difference between them, which is central to this thesis.

The model's limit emerges once it is fitted on data up to 2022 and asked to estimate 2023 and 2024, where it performs worse than simply assuming that every farm has the average capture value. The energy crisis pushed gas and carbon prices beyond anything in its training data. Leaving out one year at a time confirms this: the model works well inside the range of conditions covered by its training inputs, and fails across the break between 2021 and 2022.

4.2.2 Performance of the system model

The system model is solved with the transmission grid held at its present capacity, so congestion is priced as it exists today. This makes the absolute price level less accurate, which matters little in this

case. Capture value is a ratio between the price a farm earns and the average price in its market, so an error shared by both largely cancels.

Where the grid is strong the ratio matches observation closely. In Germany the model gives 0.908 against an observed 0.909, and the Netherlands is similar. Where the grid is congested the model reads too high, and the gap widens as congestion increases: 0.03 too high in Belgium and 0.06 in DK1. These are differences in capture value itself, so 0.06 means the model credits Danish offshore farms with six per cent more of the average price than they earned. Great Britain and Norway do not pass this check at the resolution used and are excluded from every aggregate.

Because the model's error runs in one direction only, always crediting farms with more of the average price than they received, it is a bias in the statistical sense, and a bias can be traced to its source. The model produces two outputs that could carry it, the generation of each farm and the price in each hour, and the two are separated by valuing the model's own generation at the prices that were actually observed. Most of the error disappears under that substitution, which shows that the modelled generation is close to correct and that the bias sits in the modelled prices. The same bias appears again in 2019 and in 2024, so it is stable from year to year, and an error that is stable and always in one direction can be measured and subtracted, which is what the correction of Section 3.6.2 does.

The one zone that cannot be corrected is DK2, where the model reads 0.14 too high. That zone had 264 hours of negative prices in 2023, which occur because some plants keep running even when prices are below zero: combined heat and power stations that must supply heat, and generators whose subsidy makes it worth paying to stay online. The system model is a linear optimisation and does not represent decisions of that kind, so it produces no negative prices at all. This limitation belongs to the class of model, so DK2 is set aside (Appendix D).

4.2.3 Allocation of models to analytical tasks

When the two models are compared with the same observed record, the regression model proves the more accurate of the two: across fifteen combinations of price zone and year, its estimates deviate from the observed capture value by 0.011 on average, against 0.054 for the system model. This is expected, as the regression model learns from observed prices, so it captures what markets actually do, including negative prices. The system model builds prices from costs, and misses what costs cannot explain, such as plants that must keep running. For the present it is therefore the regression model that is more reliable.

Each model is used where the evidence supports it. The regression model measures the present, explains why one farm earns less than another, and supports short-term projection. The system model carries the projection to 2030 and 2040, because it derives its prices from costs, and because it alone can be solved again under a different grid, gas price or level of flexibility (Table 4.2).

Table 4.2: The two models across the six tasks the thesis requires of them. Errors are mean deviations from observed capture value over the fifteen zone-year validation points; scenario ranges span the four Baltic 2040 scenarios.

Question	Task	Result	Conclusion
Accuracy on the observed record	Reproduce the capture value observed between 2017 and 2024	The regression model deviates from observation by 0.011 on average and the system model by 0.054, with the system model's error growing with congestion	Regression model
The most cannibalised zone	Reproduce DK2 in 2023, where 0.784 was observed	The regression model gives 0.798, an error of 0.014, while the system model gives 0.932, an error of 0.148	Regression model
Negative prices	Reproduce hours in which the price falls below zero	The regression model learned these hours from observed prices, while the system model builds prices from generation costs, which cannot fall below zero	Regression model
Congestion	Represent congestion and flows between countries	The regression model carries them only as input variables, while the system model computes a price at every grid node, so congestion appears directly	System model
Conditions beyond the record	Estimate capture value at renewable shares beyond the historical record	Across the four 2040 scenarios the regression model's answers span only 0.008, while the system model's span 0.428 and respond to the buildout	System model
Changed assumptions	Estimate the effect of a different grid, gas price or level of flexibility	The regression model contains no representation of the power system to change, while the system model is solved again with the assumption changed	System model

4.3 Sources of divergence between the models

Section 4.2 establishes where each model can be trusted. This section explains why they differ, since passing the projection from one model to the other is only defensible if the difference is understood. The full arithmetic is in Appendices J and G.

4.3.1 Agreement within the observed record

Over the four zones the 2023 record can check, the two models give almost the same answer: 0.890 against 0.885 in capture value, a difference of 0.005. Two models that share no code, one learning from eight years of observed prices and one rebuilding prices from generation costs, land within half a point of each other. Zone by zone the agreement is looser, with differences running from 0.001 at Belgium to 0.056 at DK1.

The two models differ in only two inputs, the generation series each uses and the prices each applies to it, so the difference between their answers can be split between those two causes. Both parts are small, and one adds while the other subtracts. The generation part adds 0.004, because the system model's output is smoother than a farm-specific weather series, and a smoother profile earns slightly more of the average price. The price part subtracts 0.014, because the system model's prices stay flatter across windy

hours and so understate how far prices fall when wind blows. The way the network is represented adds 0.004 on average, and 0.050 at DK1 alone. Forbidding negative prices is the largest structural difference between the two models, and it changes the answer least, at 0.003, because the 221 to 315 negative hours observed are too few to move a yearly average.

4.3.2 Out-of-sample skill and the choice of model

Both models were applied to the same Baltic zones with only the projection year changed, and their estimates differ by 0.035 in 2030 and by 0.348 in 2040, ten times as much. The widening has an identifiable source in the scope of the regression model's training inputs. In 2030 the assumed market conditions still lie within the range covered by those inputs, so the 0.035 is a genuine disagreement between the models about how prices form, whereas by 2040 the conditions have moved outside that range and the regression model returns almost the same estimate regardless of the scenario assumptions. This loss of response accounts for 0.313 of the 0.348, so ninety per cent of the 2040 difference arises because one model has stopped responding to its inputs.

Which assumption pushes the regression model outside its range can be identified, because the scenarios vary four inputs. Of the four, only the share of electricity generated by renewables moves far beyond the training data, rising to roughly fifteen times today's level in the Baltic, and on this input alone fourteen of the thirty-six projected cases fall outside the range the model was fitted on, while a check across all seven market conditions together places all thirty-six outside it. Repeating the projection with the carbon price held at its highest observed value leaves the estimates unchanged, which rules carbon out and identifies renewable share as the binding assumption.

Judged month by month over the years already observed, the two models are equally accurate. Both were scored on 144 combinations of price zone and month that neither had been built from. The regression model is out by 0.041 in capture value on average and the system model by 0.046, a gap too small to be meaningful given how few independent price zones there are. This sits oddly beside the annual comparison of Section 4.2.3, where the regression model was five times closer, and the reason is the averaging. The regression model's error is in which months are high and which are low, and those errors cancel once a year is averaged.

What separates them is the kind of mistake each model makes (Table 4.3). The system model reads too high almost every time, by 0.041 on average, and an error that always points the same way can be measured and subtracted, which is what the correction in Section 3.6.2 does. The regression model reads slightly low, and its error is the mistiming described above, for which there is no single number to subtract.

Fitted on nothing after 2021 and asked for 2023 and 2024, the two models are again equally accurate, at 0.049 each, because those two years still lay largely within the conditions the regression model had learned from (Appendix G).

Table 4.3: Error composition of the two models on 144 zone-months withheld from both, in units of capture value. A positive bias means the model reads high.

Model	Average error	Bias	Where the error comes from
Regression model, tested on zones it never saw	0.041	−0.026	Which months are high and which are low
Regression model, tested on years it never saw	0.046	−0.018	Which months are high and which are low
System model	0.046	+0.041	Reading too high in almost every case

The projection of Section 4.4 therefore uses the system model, with the correction applied to it as a refinement.

4.4 Projected capture value under the committed buildout, using the system model

Sections 4.2 and 4.3 established that the system model is the one that can be used beyond the observed record. This section uses it to project capture value forward, as the offshore wind already planned around the two seas is built. The capacities are those of the 2024 Ten-Year Network Development Plan, so this is the buildout that governments and system operators have already committed to, and the projection runs to 2030 and 2040. The values reported are those the system model produces directly. The correction of Section 3.6.2 is applied afterwards as a refinement.

The projected values depend on the whole power system around each farm, so Figure 4.3 shows the installed capacity assumed at each year.

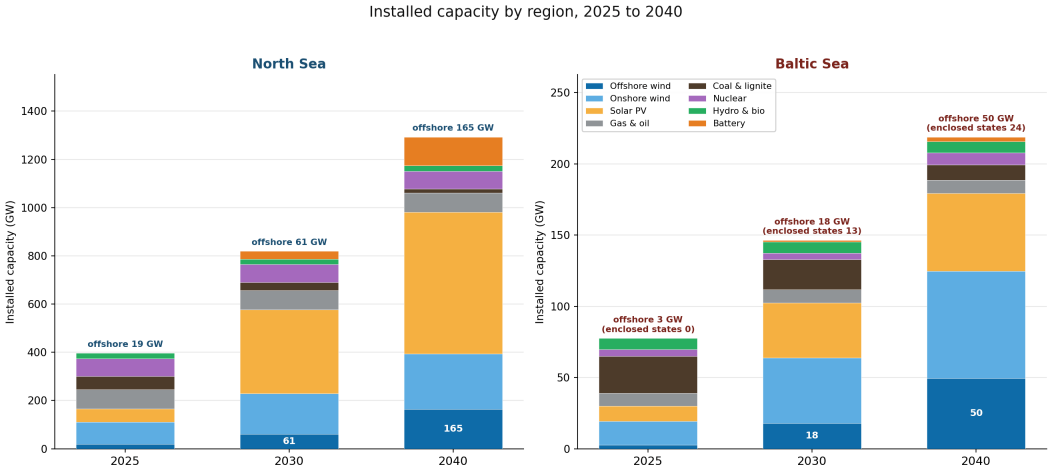


Figure 4.3: Installed capacity to 2040 under the TYNDP trajectories. Offshore wind is assigned to the sea each farm stands in, and all other capacity to the fleet of the bordering region.

Around the North Sea, offshore wind grows from 18.5 to 164.8 GW, nine times its present size. Around the Baltic it grows from 2.6 to 49.5 GW, nineteen times its present size. The Baltic starts much smaller and grows twice as fast, and that difference drives the results below.

Gas, nuclear and hydro capacity remain at similar levels in all three years, and only coal declines, so conventional plant still sets the price in hours with little wind.

4.4.1 Projected capture value to 2030 and 2040

The system model was solved for six scenarios from the 2024 Ten-Year Network Development Plan, listed in Table 4.4. They cover 2030 and 2040, a scheduled and an accelerated pace of Baltic buildout, and the plan's three demand pathways. National Trends, the pathway that tracks existing national policies, is the one quoted in the text, and the table gives the rest. Finland is set aside, as in the measured record.

Baltic capture value falls steeply under National Trends, from 0.88 today to 0.47 by 2040, while the North Sea holds 0.57, and most of the fall arrives after 2030. The faster the Baltic builds, the further it falls: in the accelerated case it reaches 0.36. These values come from the model directly, with no correction applied, and in every 2040 scenario the Baltic ends below the North Sea.

Applying the correction for the overstatement measured in Section 4.2.2 (the model reads too high in congested zones) deepens each 2040 value by roughly a further 0.1. The correction is a refinement, and the conclusion does not depend on it, since its size rests almost entirely on DK2, and removing that zone leaves a nearly constant adjustment (Appendix D). Where a single number is needed, this thesis uses

0.47, the uncorrected National Trends case on today's grid, at which the Baltic has lost close to half of its present value.

Two further checks leave this conclusion unchanged, as expanding the transmission grid softens the fall without undoing it (Section 4.5.3) and the assumed level of hydrogen demand barely moves it. The basin average also conceals differences between zones, with Estonia falling furthest and Poland least (Appendix C), but every zone that builds out loses value.

Table 4.4: Projected capture value by scenario: generation-weighted basin averages from the system model on today's grid, no correction applied. Finland is set aside.

Year	Demand pathway	Baltic buildout	Baltic offshore share of consumption	North Sea capture value	Baltic capture value
2030	National Trends	scheduled	0.24	0.84	0.79
2030	National Trends	accelerated	0.29	0.84	0.75
2040	National Trends	scheduled	0.33	0.57	0.47
2040	National Trends	accelerated	0.51	0.57	0.36
2040	Distributed Energy	accelerated	0.38	0.63	0.60
2040	Global Ambition	accelerated	0.40	0.70	0.57

4.4.2 Spatial distribution of projected capture value

Figure 4.4 shows the capture value a farm would earn at each buildable location in 2040. The 2025 and 2030 maps are in Appendix I.

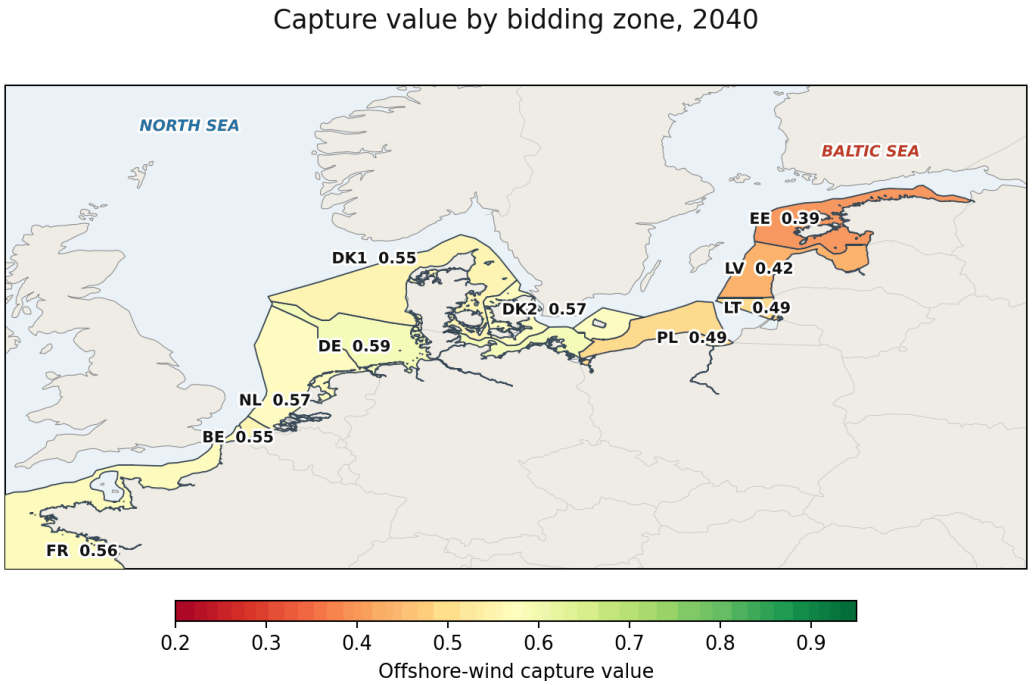


Figure 4.4: Capture value at each buildable location in 2040, with the transmission grid held at today's capacity. The North Sea holds close to its present level, while the enclosed eastern Baltic falls furthest.

In 2025 capture value is high and much the same across the whole shelf. By 2030 it has fallen everywhere, with no clear difference between the two seas. By 2040 the two have separated: the eastern Baltic is lowest, while the North Sea holds its level despite taking far more new capacity. What decides the fall is how easily a basin can move its surplus away.

Appendix L repeats these maps under cost-optimal transmission, alongside levelised cost, value-adjusted cost, net value and revenue density at all three years. Comparing the frozen and expanded versions shows the split of Section 4.5.4 geographically.

4.5 Drivers of value erosion and its recoverable component

This section examines what drives the fall in capture value and how much of it depends on the configuration of the surrounding system. The first part identifies what sets the severity of the fall: the differences between individual farms, the shape of the relationship between buildout and value, and the season in which the loss concentrates. The second part quantifies that dependence, by re-solving the scenarios with transmission free to expand and by decomposing the fall into a congestion component and a structural component.

Three further checks are reported in Appendix D. The same geography appears from hourly generation alone, with no prices used at all. Removing the hours priced by scarcity leaves the fall largely intact. And allowing demand to shift in time recovers as much capture value as expanding the grid does.

4.5.1 Farm-level drivers

This part asks why one farm earns less than another in the same month. Carbon and gas prices cannot explain it: they are Europe-wide monthly prices, identical in every zone, so they move all farms together. They still appear important in the fitted model, because the 2021 to 2022 fuel crisis moved fuel prices and capture value at the same time.

Comparing each farm with the average of all farms in the same month removes that common movement (Figure 4.5). Carbon's share then falls from 29 to 4 % and gas from 14 to 3 %. Crowding takes their place: how closely a farm's output follows the wind of the system around it explains 32 % on its own, and the zone's existing wind capacity a further 13 %. A farm generating when everything else is generating sells into a market already full of wind, and that is the largest single cause of a low capture value.

The rest is the system's ability to absorb that surplus. How well demand matches generation, how much flexibility exists, and how much can be exported together explain about 48 %, each by moving surplus out of the windiest hours, either across a border or into storage. The Baltic, with less export capacity, is the more exposed basin. Fuel prices still set the overall price level, but the differences between farms come from crowding and from how much surplus the system can absorb.

A much simpler equation, fitted separately to each basin, checks this result (Equation 3.4). One term in it measures how heavily a farm is penalised for generating at the same time as everything else. That penalty is 0.74 in the Baltic against 0.37 in the North Sea, and the Baltic is the steeper of the two in 99 of 100 resamplings of the farms. Capture value falls about 1.6 times as fast per unit of buildout in the Baltic as in the North Sea today, and about 1.9 times as fast once wind dominates the system. Fifteen farms is a small basis, so this is indicative, but it agrees with the system model, which shows a moderate fall by 2030 and a severe one by 2040. The fit is in Appendix H.

4.5.2 Seasonality of the erosion

Wind in the two basins peaks together in winter and separate in summer (Figure 4.6). Their output moves together at a correlation near 0.4 in winter, falling to 0.13 in July. The Baltic can therefore export

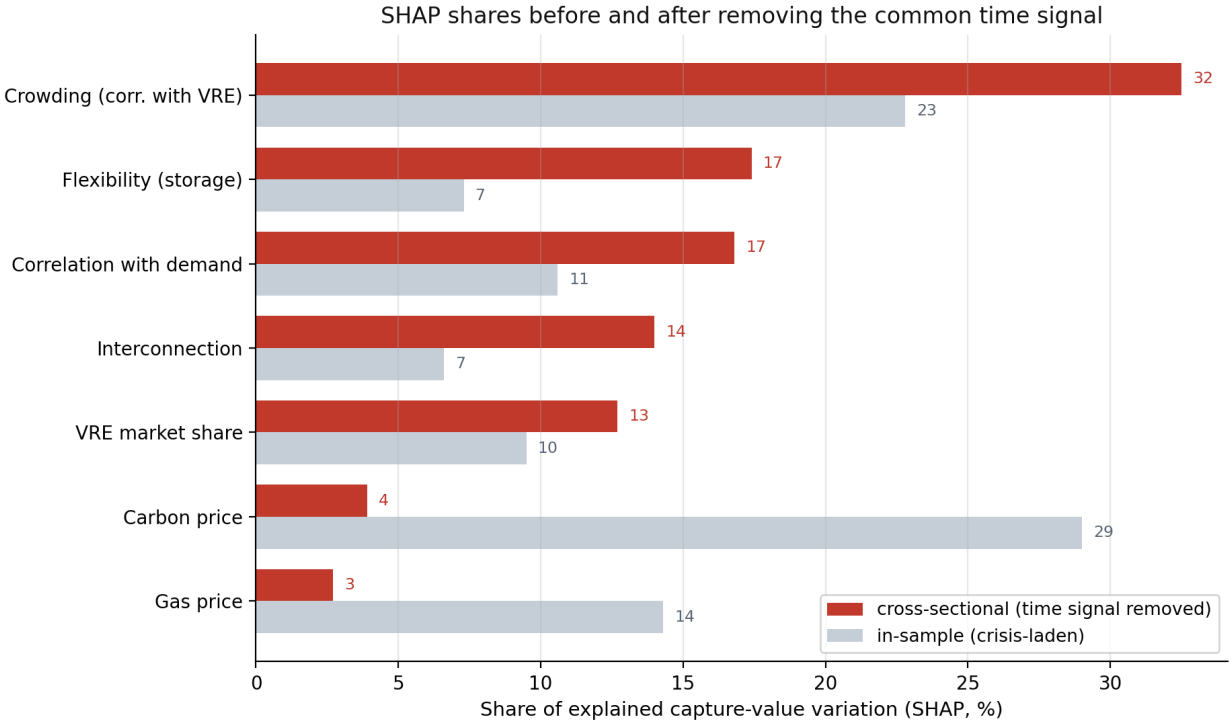


Figure 4.5: Importance of each driver of capture value, measured by SHAP. Grey shows the values as fitted; red shows them after each farm is compared with the average of all farms in the same month, which removes movements common to every zone.

its surplus in summer, but not in winter, when the same Atlantic weather reaches both seas. Export capacity is least useful in the season when output is highest.

That matters because winter is when the farms generate, at about 70 % of capacity against 40 % in summer, yet today they record the same capture value in both seasons. By 2040 the loss falls almost entirely in winter, to 0.56 in the North Sea and 0.50 in the Baltic, while summer stays at 0.91 and 0.83 (Figure M.3, Appendix M). Most of the year's output arrives in winter, so the winter fall is what produces the annual one, and any remedy must work in that season: interconnection, seasonal storage, or demand that can shift into windy winter hours.

4.5.3 The effect of transmission expansion

The projection so far holds the grid at today's capacity, which is a pessimistic assumption, since transmission will be built over the fifteen years to 2040. To find the opposite bound, all six scenarios were solved again with transmission free to expand wherever doing so lowers total system cost (Figure 4.7).

Expansion changes nothing at 2030, because the congestion it would relieve has not yet appeared. By 2040 it matters a great deal. Under National Trends the Baltic recovers from 0.47 to 0.66 while the North Sea moves from 0.57 to 0.64, so the basin that trails on today's grid ends slightly ahead once the grid is expanded. Every Baltic zone shares in the gain, Poland most of all.

Two conclusions follow from this comparison: the Baltic's disadvantage relative to the North Sea is largely attributable to congestion and disappears once transmission is expanded, whereas its loss relative to its own present value persists under either grid assumption, since the fall from 0.88 today is severe in

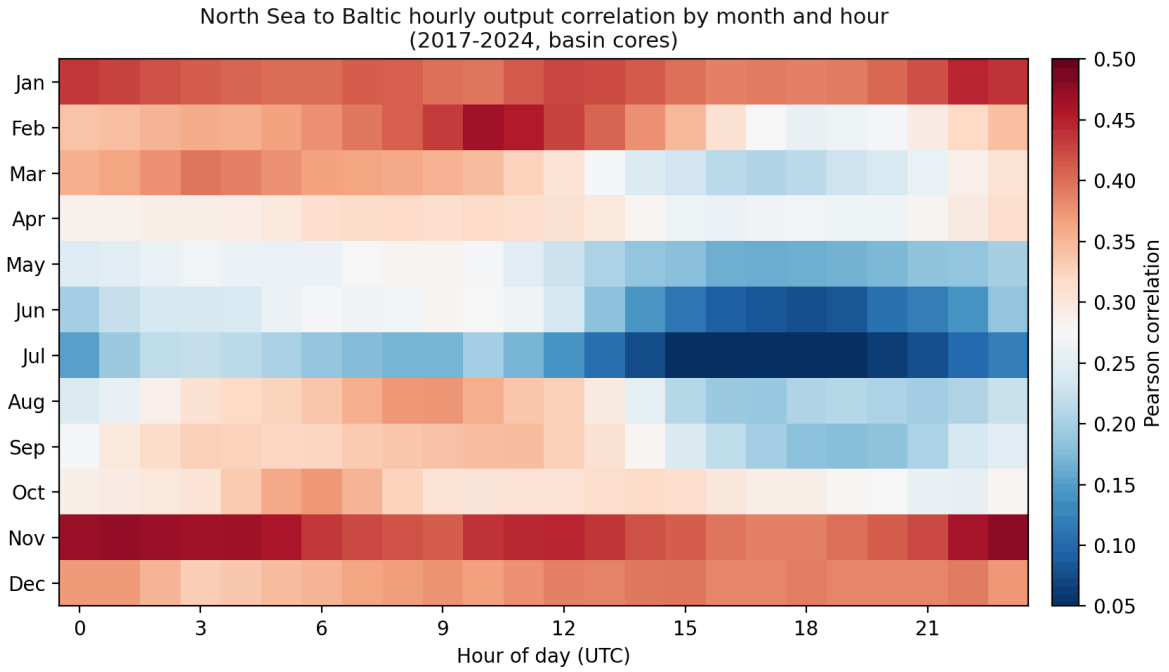


Figure 4.6: Correlation of hourly wind output between the two basins, by month and hour of day, 2017 to 2024.

both.

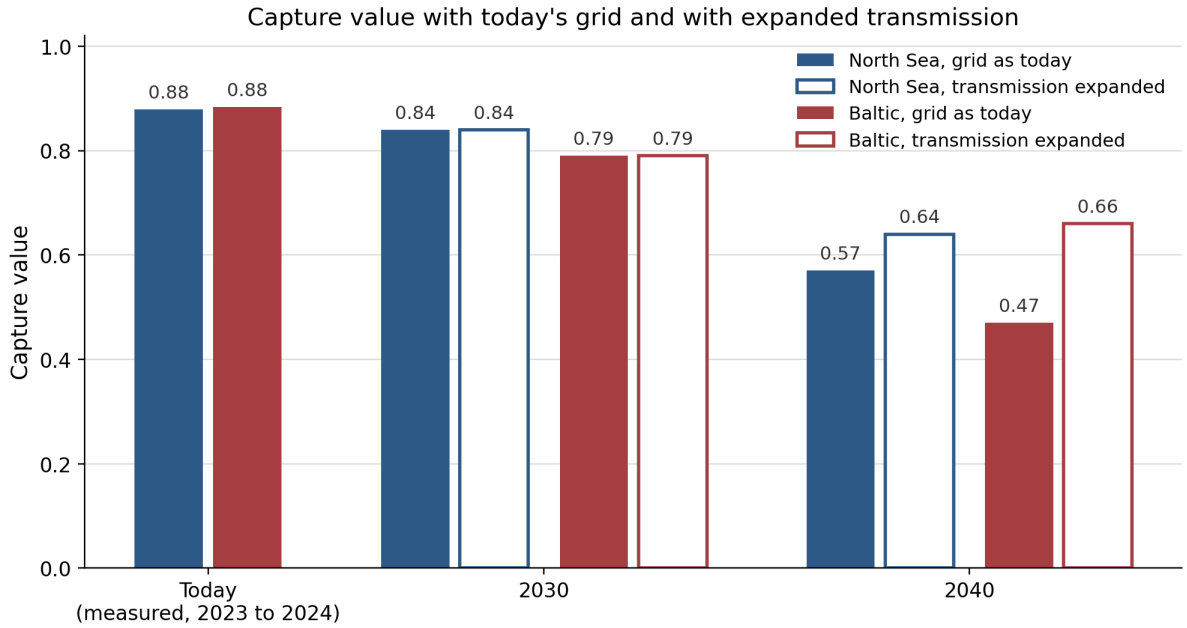


Figure 4.7: Capture value in both basins with the grid held at today's capacity and with transmission free to expand, under National Trends with the scheduled buildout. The bars for today are the measured 2023 to 2024 values, where only one grid exists.

4.5.4 Decomposition into congestion and structural components

The previous section establishes that transmission expansion matters, but it does not establish how much of the total fall congestion accounts for. That share is the number the policy conclusion rests on, because the congestion part of the fall can be removed by building lines, while the part caused by the buildout itself remains whatever is built.

The two model runs already provide this separation, since today's observed capture value is the starting point, the run with transmission free to expand gives the value a basin would hold if the grid were never a constraint, and the run with the grid frozen gives the value it holds now. The fall between today and 2040 therefore divides in two:

$$\underbrace{\mathrm{CV}_{\mathrm{today}} - \mathrm{CV}_{\mathrm{frozen}}}_{\text{total fall}} = \underbrace{\left(\mathrm{CV}_{\mathrm{today}} - \mathrm{CV}_{\mathrm{optimal}}\right)}_{\text{structural, which no grid removes}} + \underbrace{\left(\mathrm{CV}_{\mathrm{optimal}} - \mathrm{CV}_{\mathrm{frozen}}\right)}_{\text{congestion, which an expanded grid removes}} \tag{4.1}$$

Both basins start level, at 0.887 in the North Sea and 0.889 in the Baltic, within half a point of the measured 0.879 and 0.883. By 2040 on the frozen grid the North Sea has fallen by 0.314 and the Baltic by 0.418. Of the North Sea's fall, 0.250 is structural and 0.064 is congestion, a split of 80 to 20. Of the Baltic's, 0.225 is structural and 0.193 is congestion, a split of 54 to 46 (Figure 4.8).

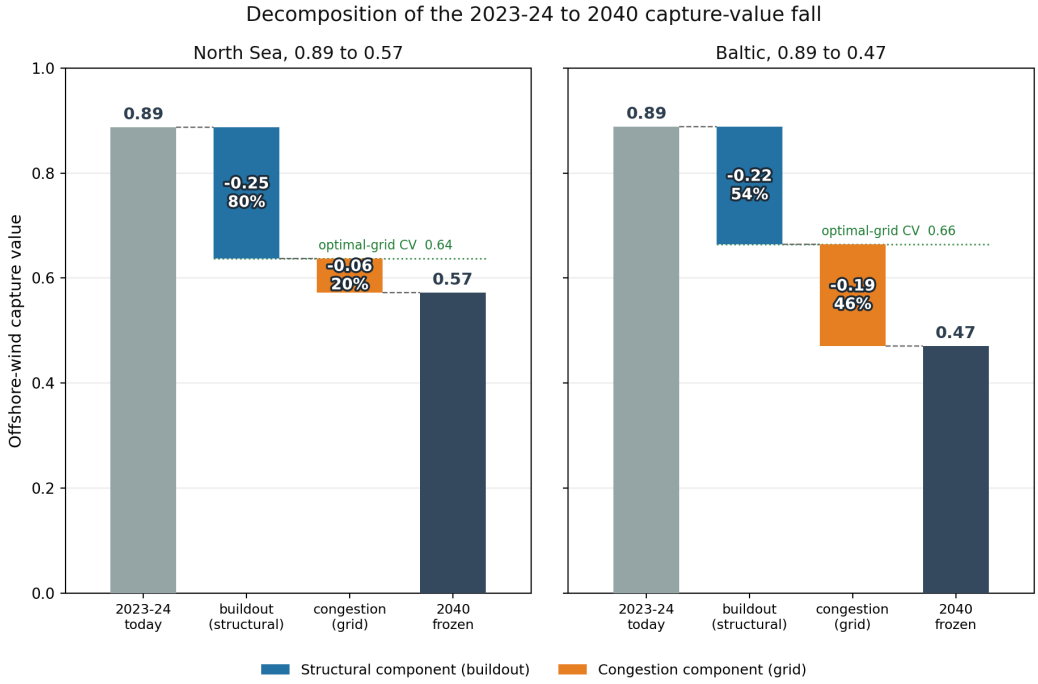


Figure 4.8: Each basin's fall in capture value between today and 2040, divided into the part caused by the buildout itself, which no amount of transmission removes, and the part caused by congestion, which an expanded grid removes. Both basins start at about 0.89.

The Baltic's congestion penalty is three times the North Sea's, which is what an enclosed basin should show. The more surprising result is the structural fall: at 0.225 against the North Sea's 0.250, the part no transmission can recover is the smaller of the two. The share of the Baltic's disadvantage that appears to come from enclosure is congestion, and congestion is decided by what is built. How the two components differ in who pays for them, while they remain, is taken up in Section F.3.

4.6 Value-adjusted cost of energy and site selection

The chapter has measured capture value and explained why it falls. This section turns it into the number an investor uses. Combining capture value with the levelised cost of energy (Equation 2.3) gives the average wholesale price a site must earn over its life to break even, and that price is mapped across both basins at the same resolution as the wind and seabed data.

4.6.1 Levelised cost and the break-even price

Building offshore wind is getting cheaper. Across the seabed where farms could be built, the median levelised cost falls from 74 to 66 €/MWh between 2025 and 2040. The price a site must earn to break even moves the other way, rising from 87 to 121 €/MWh (Figure 4.9), because capture value falls faster than cost does. The share of that seabed clearing a market price near 70 €/MWh falls from about 35 per cent to under 1 per cent.

Cheaper turbines do not save the economics, because offshore wind keeps improving when measured per megawatt-hour produced, yet gets worse when measured per megawatt-hour of value delivered, and it is the second that a merchant project, one selling at market prices without a support contract, is paid for.

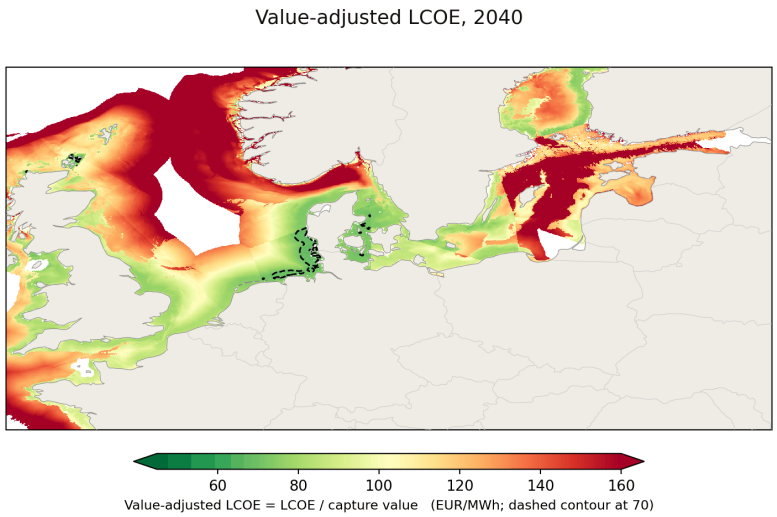


Figure 4.9: The average wholesale price each site must earn to break even in 2040, calculated as levelised cost divided by capture value. The dashed contour marks a market price of about 70 €/MWh; sites inside it are viable at that price.

The direction does not depend on the cost assumptions. Across all 27 combinations of discount rate, capital cost and learning rate, the break-even price rises to 2040 in every case (Figure M.10). The reason is that the cost level cancels out: the break-even price is cost divided by capture value, so what decides the direction is only whether cost falls faster than value. For the break-even price to stay level, offshore capital costs would have to fall by about 36 per cent by 2040, three times what the cost database projects.

Comparing the revenue a farm captures against its annual cost gives the same picture in absolute terms. About 48 per cent of the buildable seabed is profitable at market prices in 2025, and roughly 12 per cent by 2040. Falling value drives that shrinkage, since costs are falling over the same period (Appendix L).

4.6.2 Monetised value of interconnection

The break-even price is levelised cost divided by capture value, so the grid result of Section 4.5.3 reappears in terms of money. The 2040 break-even price was recalculated with the frozen and the expanded capture values, holding offshore cost identical in both. Expansion lowers it everywhere, and four times more in the enclosed Baltic than in the open North Sea: by about 8 €/MWh in the North Sea, by 33 in the enclosed Baltic, and by 48 in the Danish corridor that carries the redirected flows (Figure 4.10).

In revenue terms the Baltic gain alone is worth roughly 1.2 billion euros a year, close to half the annual cost of the whole transmission programme, and that is before the congestion rent of Section F.3.1 is counted. Interconnection therefore cuts the break-even price where the cut is most needed.

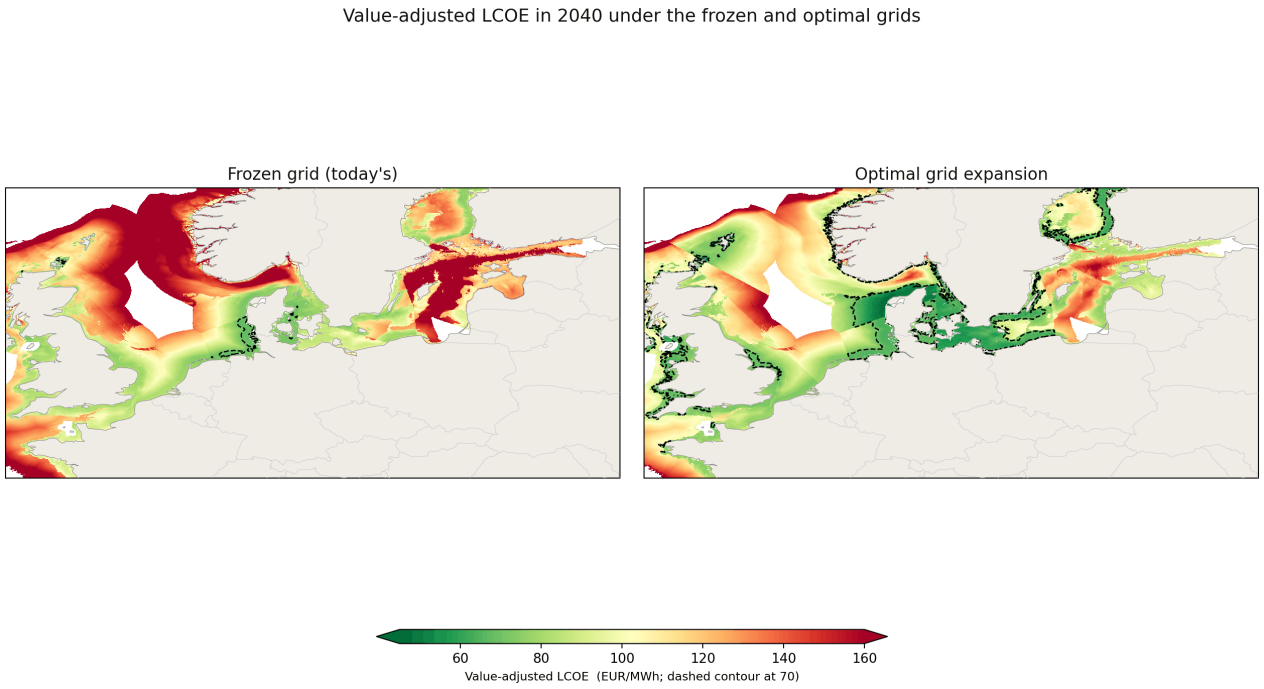


Figure 4.10: The 2040 break-even price with the transmission grid held at today's capacity (left) and free to expand (right), with offshore cost identical in both. The price falls most where the Baltic is enclosed.

4.6.3 Effect of capture value on site ranking

Ten candidate development areas, one in each offshore bidding zone, were ranked twice in each year: once on levelised cost alone, and once on the break-even price that includes capture value. Comparing the two orderings shows how much capture value changes where a developer would build (Figure 4.11).

The ranking by levelised cost alone and the ranking by break-even price agree almost perfectly today and drift apart over time, with a rank correlation of 0.99 in 2025, 0.95 in 2030 and 0.90 by 2040 (Appendix C). Ranking on cost alone is therefore adequate for a project being built now, and increasingly misleading for one being built into the 2040s, which is the period over which offshore wind is financed.

The individual movements identify the misranked areas, and the clearest case is the cheap eastern Baltic: Estonia has one of the lowest levelised costs on the shelf at 51 €/MWh, but at a capture value of 0.39 its break-even price is 130 €/MWh. The Danish areas rise, because their capture value holds up.

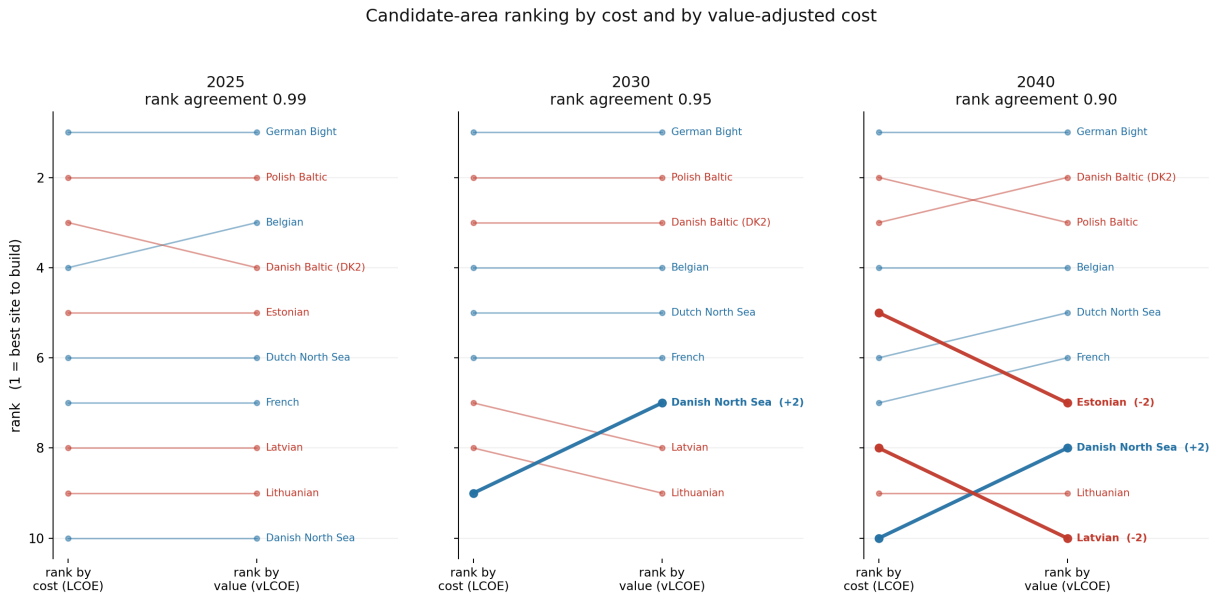


Figure 4.11: Candidate development areas ranked on cost alone against the break-even price, 2025 to 2040. Areas that hold their position lie on the diagonal, and bold lines mark moves of two places or more, in blue for the North Sea and red for the Baltic.

The German Bight ranks first on both measures in every year, having the strongest wind in shallow water close to shore.

Widening the comparison to 169 candidate locations across both seas gives the same answer. Whether areas are judged on value for money or on absolute profit, the German Bight and the shallow western Baltic come first, and the cheap eastern Baltic scores well on cost and poorly on value (Appendix L).

Cost alone remains a defensible basis for the current round of projects, and a progressively worse one as the buildout advances, with the error concentrated in the cheap eastern Baltic sites it would promote.

Who bears the resulting loss is a separate question, worked through in Appendix F. Splitting each basin's loss into the part a support contract absorbs and the part it does not shows that the two basins differ beneath similar headline numbers: 18.5 % of Baltic energy on today's grid earns nothing at all, while the whole of the North Sea's loss falls outside what a contract compensates. Building the grid both reduces the loss and changes who carries it.

5 Discussion

This chapter answers the three research questions against the evidence of Chapter 4, sets those answers against the literature and gives the limits of the work and what follows from them.

5.1 Answers to the research questions

The first question asks how capture value in the enclosed Baltic compares with the open North Sea today, and how it changes under the committed buildout to 2030 and 2040. The two basins are level today: offshore farms in the under-built Baltic zones capture 0.883 of the average wholesale price against 0.879 in the North Sea, which is to say that a farm in either basin received close to 88 per cent of the price a plant generating continuously would have received over the same period.

The fall is modest to 2030 and steep after it. By 2040, under National Trends, the pathway of the official plan that tracks existing national policies, the Baltic falls to 0.47 while the North Sea holds 0.57, so a basin that begins level has lost close to half of the value it captures today. That outcome depends on the network built alongside the farms, since with transmission free to expand the Baltic recovers to 0.66 against the North Sea's 0.64. How much grid is built therefore decides how far the Baltic falls behind, although even under the most favourable grid tested it still captures about 0.2 points less than it does today. Figure 5.1 shows the measured record and the projection in one view.

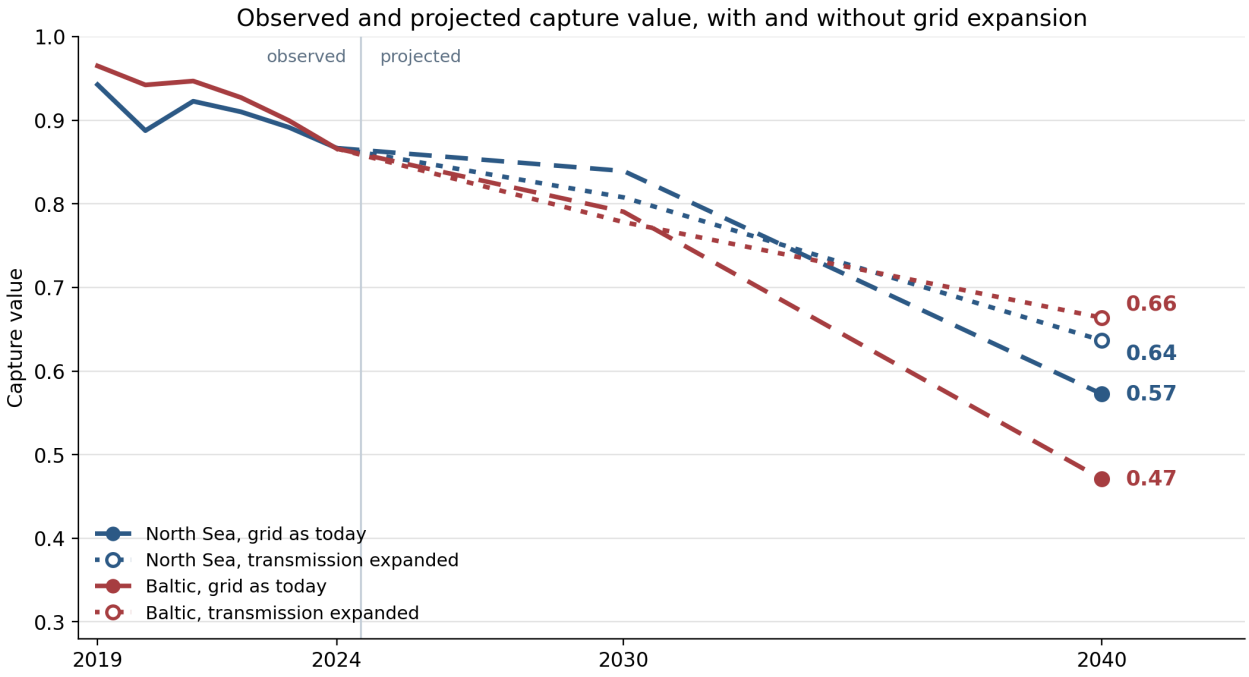


Figure 5.1: Capture value for the North Sea and the Baltic, measured to 2024 (solid) and projected to 2040 under National Trends with the scheduled buildout, on today's grid (dashed) and with transmission free to expand (dotted). The Baltic lines cover the zones that build out, Poland, Estonia, Latvia and Lithuania.

The second question asks which mechanisms set the rate at which capture value is eroded, whether they differ between an enclosed and an open basin, and how much of that erosion system design can remove. Between individual farms the largest single cause is crowding, meaning how closely a farm's output coincides with that of the rest of the system, since a farm generating when everything else generates is selling into hours in which the price has already been driven down. The same mechanisms operate in both basins and are simply stronger in the Baltic, where the penalty for generating alongside everything else is twice as large as in the North Sea.

A substantial part of the resulting loss can be removed by building differently. Of the 0.42 the Baltic loses under National Trends, 0.19 is congestion that an expanded grid removes while 0.23 comes from the buildout itself, so nearly half of the loss follows from the network around the basin. Allowing demand to shift in time recovers a comparable amount to expanding the grid, 0.28 against 0.19, which means the two levers available to a system planner are of similar size (Table 5.1).

The third question asks whether a model trained on historical observation can project these futures, whether a system model is required, and how the two should be combined and validated against each other. A model trained on observation cannot project them, and the

Table 5.1: The Baltic fall attributed, in the 2040 National Trends scenario unless stated otherwise. The last two rows are the two design levers.

Component	Capture value	Section
Total fall, today to 2040 on today's grid	−0.42	§4.4.1
the buildout itself, which no grid removes	−0.23	§4.5.4
congestion, which an expanded grid removes	−0.19	§4.5.4
Scarcity pricing inside the stressed scenarios	up to −0.15	§D.2
Winter concentration (winter 0.50, summer 0.83)	seasonal	§4.5.2
Recovered by expanding transmission	+0.19	§4.5.3
Recovered by fully flexible demand	+0.28	§D.3

reason lies in how the two models are built.

The regression model learns a relationship between seven market conditions and capture value from prices that have already occurred, which is why it reproduces the recent record more closely than the system model does. That accuracy is inherited from the record, and it holds only while conditions resemble the ones the model was fitted on. Once the system changes fundamentally, the association the model learned no longer describes the system it is being asked about. The system model is built the other way round: it re-derives each hour's price from the fleet, the network and demand, so it can respond to a scenario that has never occurred.

The measurements support this account of the two models, since over the observed record the two are equally accurate, each missing observed capture value by about 0.04 on average across 144 combinations of zone and month that neither had been fitted on (Table 4.3). Beyond that record 90 per cent of their disagreement at 2040 comes from the regression model no longer responding to the scenario, while the two continue to agree on how prices are formed. To locate that boundary, the regression model was refitted on data ending in 2021 and asked for 2023 and 2024. The two models projected those years equally well, since most of the targets still lay inside the conditions the regression model had seen, so the limit is the edge of the observed record rather than the number of years ahead.

Their construction assigns their roles (Table 3.1). The regression model, fitted to 2,706 monthly observations from 31 farms, measures the present and forecasts over horizons short enough that market conditions remain recognisable. The system model, solved hourly across thirty nodes under official capacity trajectories, is the only instrument that can price the 2040 fleet, and it carries the projection with its measured overstatement subtracted.

5.2 Findings in relation to the literature

The flexibility offered by hydro was expected to protect the Baltic from a sharp decline in capture value, and it does not. Hirth finds that systems with a large hydro fleet hold wind value better than systems built on gas and coal, by about eighteen percentage points of capture value once wind reaches 30 % of supply, because reservoirs absorb the surplus produced on windy days [2]. The Baltic has that hydro, so on Hirth's reasoning its zones should capture more of the average price than the North Sea's, whereas the measured record puts the two level at 0.883 against 0.879 with almost none of the Baltic's offshore pipeline yet built (Table 4.1). The expectation was built into the design of this thesis, since the archetype table of Chapter 2 marked the hydro-buffered markets as favourable before the measurement was made. The explanation is one of scale, since the storage available around the Baltic is small relative to the surplus a full buildout would produce, and once that storage is full the remaining power has nowhere to go, because the basin has little capacity to export it. Hirth's effect operates within a market, and the

difficulty in the Baltic is moving power out of one.

The projection reported falls further than any published for a European offshore basin. The Baltic's central 0.47 in 2040 sits below the worst North Sea case Jacquet reports [107], and it does so at the resolution at which prices are set, where earlier work reports national averages. The same work identifies winter as the season in which cannibalisation concentrates, and the results sharpen that finding, because the two seas are windy at the same time in winter, so export capacity is least useful in the months when output is highest, and by 2040 almost the entire annual loss falls in that season, at 0.50 in winter against 0.83 in summer.

Efthymiadou's finding for the United Kingdom, that it is how much interconnection is actually used, and not the presence of the capacity alone, that steadies prices [111], is extended here, since relieving the transmission constraint lifts the most stressed 2040 scenarios by about 0.2 in capture value while allowing demand to move in time is worth a comparable amount. Those two changes to the physical system are the largest levers identified anywhere in this work. Stiewe et al. represent the influence of neighbouring markets as a single fixed coefficient for each zone [12], which holds that influence constant over time, whereas in this case the same penalty grows as more capacity is built and can be separated into the part congestion causes and the part the buildout causes.

Every published benchmark sits above the values projected, whether Hirth's range of 0.5 to 0.8 at a 30 % share [2], Germany's measured 0.76 at 35 % [12, 24], or the roughly 0.6 of Jacquet's worst North Sea case [107]. Each of those was measured in a more open market and at a lower level of wind than the Baltic reaches in these scenarios, so the difference between them reflects where each sits on the same curve. The panel literature reinforces that reading, having established that the decline steepens as penetration rises [23, 25], which extends the published trajectories into the range projected.

The two models behaved as the literature predicts in one respect and not in the other. A model fitted to data is conventionally expected to understate the spread of the observations, while a model built from physical and cost relationships is expected to carry a consistent offset [97]. The system model behaves as expected, reading consistently too high by 0.041 on average, but the regression model does not, since it reproduces the spread of the observations accurately and its error lies almost entirely in which months it places high and which low. What it has learned, in other words, is how much cannibalisation there is without learning when it occurs, which is a more forgiving failure when the task is to measure today's level than when the task is to rank one month against another.

5.3 Limitations and future directions

The complete register of assumptions is given in Appendix B. The limitations capable of altering a reported value are set out below, each with the work that would resolve it.

The bias correction is fitted on a single high-penetration zone. DK2 is the only validation point at an offshore share comparable to the projected Baltic, and removing it leaves a correction that no longer increases with buildout, so the correction is reported as a refinement and the values quoted throughout come from the uncorrected system model (Appendix D). Resolving this requires further validation points at high penetration, which will become available as the Baltic fleet is commissioned.

The system model solves a linear dispatch and does not represent generating units that must remain online or that bid below their marginal cost. This is the source of the overstatement in DK2, and replacing the assumed interconnector ratings with the capacities offered to the market corrects DK1 without correcting DK2. Representing unit commitment and must-run heat production explicitly would address it directly, at a substantial increase in computational cost.

The accounting of who bears the loss covers generated energy only and does not include curtailed volume, which amounts to a third of North Sea wind output on the 2040 network held at today's capacity.

Including curtailed energy, and solving at intermediate levels of network expansion so that the planned network is priced alongside the present and the fully expanded cases, would complete that accounting.

Great Britain and Norway do not pass validation at the spatial resolution used and are excluded from every reported aggregate, although the three British farms remain in the regression model's training data up to 2020, when British prices left the ENTSO-E platform. Great Britain is a suitable subject for a separate application of the same design, since its market record is public and nothing after 2020 has been used.

Only DK2 and one Swedish farm have operating offshore capacity in the enclosed Baltic, so the remaining zones are represented by the validated virtual wind farm, since local measurement does not exist there. The regression model nevertheless predicts six of the seven enclosed zones to within 0.07 of the observed value when those zones are withheld from fitting, and metered output from the farms now under construction would allow this to be tested directly.

The projection holds the correlation between wind patterns and the utilisation of interconnection fixed at present values, and the fast approximation used for the wider scenario sweep becomes optimistic at the highest levels of buildout. For both reasons the values reported come from the direct zone-by-zone solves, with the present network and the fully expanded network bounding the result.

Two extensions follow from this work. The first is geographic. The design uses open data, an open resource model and an open system model throughout, so it can be applied wherever hourly prices and a network representation exist, and extending it beyond the two basins studied in this work would allow offshore potential to be assessed on the value of the electricity produced rather than on the wind resource alone, across a considerably wider set of markets. The second concerns sector coupling. The sector-coupled solve reported is a single sensitivity, and it shows that flexible demand recovers as much capture value as network expansion does, which makes the assumption about future flexibility one of the largest open questions in the result. Running the complete scenario set in PyPSA-Eur-Sec, rather than as a bracket applied to one case, would establish how much of the projected erosion survives once heat, transport and hydrogen demand are optimised alongside electricity.

6 Conclusion

In 2024 the German day-ahead price fell below zero for 459 hours, and this thesis set out to establish what happens to the value of offshore wind once hours like these become a sea basin's ordinary operating state. Europe is committed to the buildout that will produce them: Baltic offshore capacity is planned to grow from 2.6 to 49.5 GW by 2040, a nineteenfold increase against a ninefold increase around the North Sea, while support moves away from fixed payments towards contracts that leave generators exposed to the wholesale price. What a farm earns therefore depends on the price prevailing in the hours it generates, and those are the hours its neighbours generate too.

This thesis measured the share of the average wholesale price that offshore wind actually receives, its capture value, for every bidding zone around the two seas, and projected it to 2030 and 2040. Over 2023 and 2024 the Baltic zones that have built little so far captured 0.883 of the average price, level with the North Sea's 0.879. By 2040, on today's grid, they fall to 0.47 under National Trends, the pathway of the official plan that tracks existing national policies, while the North Sea holds 0.57, so close to half of present value is lost. The ordering of the two basins reverses within fifteen years.

The largest single cause at the level of an individual farm is crowding, meaning how closely a farm generates at the same time as everything else, and the penalty for it is twice as steep in the Baltic as in the North Sea. At the level of the system the fall divides in two. Of the 0.42 the Baltic loses, 0.23 comes from the buildout itself and 0.19 from congestion, and the congestion part disappears once the grid is

expanded: with transmission free to grow the Baltic recovers to 0.66 against the North Sea's 0.64. Nearly half the loss, in other words, is a consequence of what gets built around the farms.

Which of the two models could make that projection had to be established rather than assumed. Over the years that can be checked against observation the statistical model and the system model are equally accurate, and their estimates of capture value agree to within 0.005. Beyond those years ninety per cent of their disagreement at 2040 is the statistical model ceasing to respond to the scenario, because it inherits the present instead of reconstructing it. The statistical model therefore measures the present and the near term, and the system model, which rebuilds prices from plants, costs and the network, carries the projection.

The first of this work's four contributions is a systematic comparison of offshore capture value between the North Sea and the Baltic at the resolution of the bidding zones in which prices are set, carried from today's measurement through to the 2040 buildout. The second sets a statistical model and a system model against observation and against each other on the same measured quantity, and establishes by measurement the point at which a projection must pass from one to the other. The third divides the loss of capture value twice, into the part no network removes and the congestion part an expanded network does, and into the shares of the loss borne by the farm and by the state under the contracts that now dominate support. The fourth computes a value-adjusted cost of energy for each location rather than for each technology, so that a measured fall in value can be read as the price a specific site must earn.

What the Baltic buildout earns will be decided by the investments delivered alongside it. Expanding transmission removes most of the congestion loss, and the income the congested grid already generates, which European law sets aside for exactly this use, would cover the planned transmission programme several times over. Allowing demand to move into the windy hours is worth as much again. The value of Baltic offshore wind is therefore set by decisions that are still open, and most of that capacity has not yet been built.

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A Farm Registry and Data Inventory

The study set is the 31 offshore wind farms of Table A.1, sixteen in the North Sea and fifteen in the Baltic, spanning operational, under-construction, and planned sites. Each farm's coordinates fix the ERA5 cell its virtual wind farm samples (Section 3.4.1); capacity and commissioning year are carried for reference and never enter the capture-value ratio, which is normalised so that rated capacity cancels. Operational farms, where metered generation exists, are the validation subset; under-construction and planned sites enter only as price-anchored panel observations (Section 3.4.3). For recently commissioned farms the same applies to their pre-commissioning years: Hollandse Kust Zuid came online in 2023, so its earlier panel rows rest on the validated virtual wind farm rather than metered output, the method's standard treatment of not-yet-built capacity. The four German-licensed farms that stand in Baltic water (zone DE_LU) are listed under the Baltic by geography but sell into the open German price, so they capture North-Sea-like value.

B Assumptions Register

Every modelling choice that shapes a number in this thesis is listed here with its value, its source, and where its influence is tested or bounded. Items marked (L) reappear in the consolidated limitations of Chapter 5.

B.1 Shared data and definitions

1. **One capture-value definition.** Generation-weighted price over time-mean zone price, on common hours only; zone price is the unweighted mean over the zone's network nodes; Denmark splits at 11°E into DK1/DK2. Used identically by every component (Section 3.2).

2. **Basin membership is geographic.** A farm belongs to the sea its foundations stand in; DK2 and SE4 are Baltic-facing although their markets sit in the Nordic synchronous area (Section 3.3.2). Corroborated by weather-only clustering (Section D.1).

3. **ENTSO-E Transparency is the single market-data source** (2017 to 2024): prices for 22 zones, load, generation by type, flows, capacities. Dutch renewable generation is rescaled to physical capacity factors (documented under-reporting); British day-ahead prices end with Brexit, so GB farms contribute no farm-months after 2020. (L)

4. **ERA5 is the weather truth** (regression model: nearest 0.25° cell via Open-Meteo; system model: 0.3° yearly cutouts via atlite). Its offshore wind-speed bias is not corrected explicitly; the Hayes power-curve shift absorbs it, and the end result is validated against metered generation (median hourly correlation 0.88, monthly 0.95, fifteen zone-years).

Table A.1: The 31-farm study registry: market area, location, capacity, commissioning year, and status. Coordinates are the farm centroid used to select the ERA5 sampling cell. Sources are developer disclosures, the Crown Estate, and national energy agencies.

Farm	Country	Zone	Lat	Lon	MW	Year	Status
North Sea (16 farms)							
Horns Rev 3	DK	DK1	55.69	7.73	407	2019	Operational
Anholt	DK	DK1	56.60	11.21	400	2013	Operational
Borssele 1-2	NL	NL	51.70	3.02	752	2020	Operational
Borssele 3-4	NL	NL	51.70	2.95	731	2020	Operational
Hollandse Kust Zuid	NL	NL	52.30	4.20	1,525	2023	Operational
Gemini	NL	NL	54.04	5.97	600	2017	Operational
Borkum Riffgrund 2	DE	DE_LU	53.97	6.55	450	2019	Operational
Veja Mate	DE	DE_LU	54.31	5.86	402	2017	Operational
Hornsea 1	GB	GB	53.88	1.79	1,218	2019	Operational
Hornsea 2	GB	GB	53.95	1.90	1,320	2022	Operational
C-Power	BE	BE	51.55	2.93	325	2013	Operational
Norther	BE	BE	51.50	2.95	370	2019	Operational
Saint-Brieuc	FR	FR	48.82	−2.55	496	2024	Operational
Dogger Bank A	GB	GB	54.75	1.85	1,200	2025	Under constr.
Princess Elisabeth Zone	BE	BE	51.50	2.60	3,500	2030	Planned
North Sea Energy Island	DK	DK1	56.20	7.50	3,000	2033	Planned
Baltic (15 farms)							
Rodsand 1 (Nysted)	DK	DK2	54.55	11.71	165	2003	Operational
Rodsand 2	DK	DK2	54.55	11.80	207	2010	Operational
Kriegers Flak DK	DK	DK2	54.97	12.90	605	2021	Operational
Wikinger	DE	DE_LU	54.83	14.07	350	2017	Operational
Arkona	DE	DE_LU	54.78	14.12	385	2019	Operational
Baltic 2	DE	DE_LU	54.99	13.16	288	2015	Operational
Baltic Eagle	DE	DE_LU	54.79	13.92	476	2024	Operational
Lillgrund	SE	SE_4	55.51	12.78	110	2007	Operational
Bornholm Energy Island	DK	DK2	55.10	14.60	3,000	2032	Planned
Saare-Liiv	EE	EE	58.30	22.50	3,000	2035	Planned[a]
Liivi Bay	LV	LV	57.80	23.50	1,000	2030	Planned
Klaipeda offshore	LT	LT	55.80	20.60	1,400	2030	Planned
Baltic Power	PL	PL	55.00	17.40	1,140	2026	Under constr.
Baltyk II / III	PL	PL	55.05	17.20	1,440	2027	Planned
FEW Baltic II / Olhava	FI	FI	64.80	24.50	1,280	2030	Planned

[a] Saare-Liiv falls on a coastal ERA5 cell and is flagged accordingly.
Data coverage: Dutch renewable generation is rescaled to physical capacity factors (documented ENTSO-E under-reporting); British day-ahead prices end with Brexit, so the three British farms contribute panel data only through 2020.

5. **TYNDP 2024 is the scenario authority** for capacities, demand, and fuel and carbon prices. The parse covers renewables only; the corrections of Table 3.2 repair the documented consequences. (L)

B.2 The system model

6. **Scope.** Fourteen countries, 30 clusters with focus weights on offshore-heavy markets; Sweden and Norway remain coarse (three clusters), so SE is reported with an all-Sweden caveat and NO is excluded from headline aggregates alongside GB. (L)

7. **Resolution.** Full hourly year (8,760 snapshots; the framework drops the 2024 leap day by convention).

8. **Frozen transmission** (`transmission_limit: v1.0`): line and link capacities fixed at today's volume, the conservative bound; the optimistic bound is computed explicitly (optimal expansion moves clean 2040 scenarios +0.04 to +0.06, stressed ones +0.18 to +0.19).

9. **Year-correct fleets.** Conventional plants filtered by commissioning and retirement dates; renewable capacities topped up to the run year's national statistics (IRENASTAT via powerplantmatching).

10. **Fuel-price overrides** (annual means): 2023 gas 41.0 €/MWh$_{th}$, coal 16.0, lignite 5.5, oil 75.0; 2024 gas 34.7, coal 14.0, lignite 5.5, oil 74.0; 2019 uses the 2020 technology-cost defaults. Sources: TTF and EUA from the project's monthly series, coal from ARA annual means, oil from Brent.

11. **Carbon prices**: 25 / 84 / 67 €/tCO_2 for 2019 / 2023 / 2024 (EUA annual means), applied through emission pricing.

12. **Perfect competition, perfect foresight, linear dispatch.** No unit commitment, start-up costs, minimum loads, must-run district-heating CHP, or strategic bidding. Consequence: shadow prices are too flat in windy hours in small congested zones. This is not left as an assumption: it is measured (DK2 bias +0.143 in 2023, +0.141 in 2024), explained by the hybrid decomposition, and corrected with propagated uncertainty (Section 3.6.2). (L)

13. **Interconnectors are fully available every hour** (no TSO de-rating or reservations). Same direction and same treatment as the previous item: absorbed into the measured, corrected bias. (L)

14. **Scarcity ladder** (correction F): demand response sized at 5 % of each node's peak load offered at 400 €/MWh; unserved energy at the 4,000 €/MWh day-ahead cap; residual unserved energy below 0.05 % of demand, reported per scenario (Section D.4).

15. **Scenario corrections A to F** as enumerated in Table 3.2, including the Danish offshore split (65/35 in 2030 to 60/40 by 2040), offshore caps (DK 50, FI 25, SE 35 GW), and Poland's announced 3.9 GW of nuclear by 2040.

16. **Demand trajectories**: scenario demand is rebased from crisis-2023 to 2019 before TYNDP growth; the two countries missing from the scenario tables receive documented growth fallbacks. All scenario horizons are solved on 2023 weather and hourly load shapes, so the only thing that changes across scenarios is the buildout. (L)

17. **Hydrogen floor** (correction D): a parametric stand-in for sector coupling, with per-country electrolysers, fixed hydrogen demand, and a priced import backstop near 35 to 42 €/MWh; halving and raising its demand by 50 % moves the 2040 Baltic capture value by at most 0.013, and by under 0.01 in the clean DE and GA scenarios (Appendix D). (L)

18. **Solver parity.** Gurobi barrier without crossover, identical tolerances, for every baseline and every scenario, because the bias correction is fitted on interior-point duals and must be applied to prices formed the same way.

19. **Bias correction.** Linear in the model's own offshore penetration; fourteen of fifteen validation points (Netherlands 2019 excluded as a low-confidence observation); clamped at penetration 0.50;

corrected values floored at zero. The penetration slope is anchored on DK2, the only validation zone above 31 % penetration: leave-one-zone-out collapses it from 0.233 to 0.045 with an interval crossing zero (Appendix D), so the correction is reported as a refinement of the uncorrected system model, which remains the source of the result. (L)

20. **Gas-price grid**: 15 to 120 €/MWh$_{th}$ across three buildouts; exposure slopes computed over every offshore zone, with basin aggregates excluding GB and NO.

21. **Sector-coupled variant**: PyPSA-Eur's sector configuration at twelve-hour resolution, thirty clusters, 2019 weather, overnight 2040, frozen grid, with heating, hydrogen, storage and electric transport co-optimised; country totals pinned to the TYNDP 2040 fleet (Belgium capped at its 4.9 GW sea-space potential); the electricity-only control is identical in every setting. The weather cutout spans 47°N northward, truncating southern France identically in both arms. (L)

22. **Surrogate emulator**: eight inputs; fifty Latin-hypercube training solves at three-hour resolution (full-hourly equivalence checked: mean absolute capture-value difference 0.024, up to 0.147 only at the extreme 139-GW corner); validation by held-out whole scenarios; the Baltic correction parsed from the canonical fit. (L)

B.3 The regression model

22. **Virtual wind farm**: logarithmic extrapolation from the 10 m and 100 m ERA5 levels to the 150 m hub height of the IEA 15 MW reference turbine [109], whose power curve enters as a hand-keyed table of the published values interpolated on a 0.25 $\mathrm{m\,s^{-1}}$ grid (cut-in 3, rated 10.59, cut-out 25 $\mathrm{m\,s^{-1}}$); Hayes adjustments $\sigma = 1.17\ \mathrm{m\,s^{-1}}$ and $w_0 = 0.71\ \mathrm{m\,s^{-1}}$ (calibrated on 57 metered European farms), applied as a Gaussian convolution of the curve followed by a rightward shift; output hard-zeroed outside the 3 to 25 $\mathrm{m\,s^{-1}}$ operating envelope after smoothing, since the convolution otherwise leaks output into storm and calm hours. The archived eight-year panel predates this re-zero; hours outside the envelope are rare enough to bound its effect below half a percent of capture value; the released code applies the re-zero. Farms closer than one reanalysis cell share identical output, so near-neighbour pairs are excluded from synchronicity comparisons.

23. **Monthly capture value** requires at least 500 overlapping farm-price hours; capacity factors are normalised, so rated capacity cancels in the ratio.

24. **Training set**: the bidding-zone market panel, every site with observed prices over 2017 to 2024 (all 2,706 zone-months across 31 sites). The operational farms are the validation subset, the only sites where generation can be checked against metered output. (L: metered offshore exists only in DK2 and SE4, so the enclosed-Baltic price-only zones are anchored on observed prices and a validated wind profile rather than local metered generation.)

25. **Market-area resolution**: features computed at bidding-zone level where zone-resolved series exist (DK1, DK2, SE4), country level elsewhere, recorded per row. (L)

26. **Flexibility index numerator**: openly reported pumped hydro plus battery capacity where ENTSO-E reports it, a documented substitution for the predecessor's proprietary battery and demand-response data. (L)

27. **Interconnection index**: realised utilisation, the mean absolute cross-border flow over each external border's capacity (proxied by its yearly maximum flow), summed across borders. The predecessors'

neighbour-synchronicity discount is dropped on evidence (Section 3.4.2); static within each year, blind to hour-by-hour congestion. (L)

28. **Fuel features**: monthly TTF gas and EU ETS carbon averages.

29. **Collinearity**: variance inflation factors on the 2,706-row panel peak at 2.16 (carbon), far below the conventional alarm level of five; feature 7 sits at 1.21.

30. **No hyperparameter search**; fixed settings for all five algorithm families: gradient boosting with 400 trees of depth 4 at learning rate 0.05, random forest with 300 trees of depth 8, and support-vector, polynomial and linear regression at library defaults (library versions in Appendix K); the untuned predecessor result is the comparator.

31. **2022 retained** in training; an ex-2022 variant is reported alongside, mirroring the predecessor's crisis treatment.

32. **Projection feature choices**: pattern correlations held at the site's recent monthly shape; penetration scaled by TYNDP renewable growth over demand growth, anchored on the observed present; interconnection held at its recent value (no future border capacities exist in the scenario tables, a mildly optimistic simplification, with a conservative-bound sensitivity that lets utilisation decline as renewables grow); flexibility grows with the scenario battery fleet. All guarded by the strict all-features range check, the joint convex-hull support test, and the carbon-clamp sensitivity. (L)

B.4 The comparison with observation

33. **Independence of the three-way comparison's columns**: the regression column is out-of-fold (the model never trained on the zone's farms); the system column uses no market price data; observation arbitrates.

34. **Hybrid decomposition**: model generation priced at observed prices isolates whether an error lives in the wind or in the price structure.

35. **Aggregation difference**: the regression model's zone capture value is a simple mean over farm-months while the system model's is energy-weighted; a small, documented difference.

36. **The DK2-analogue premise**: the system model's bias transfers along the penetration axis from validated zones to projected ones. This is tested directly: the bias replicates across 2023 and 2024, the fit's slope interval excludes zero, and no single zone-year moves the leave-one-out range.

B.5 The analytical equation in full

Because an additive split cannot express the shape of the mechanism, Jacquet's analytical equation is refit as well [107]. It is not a black box, nor an arbitrary curve: it is a parametric form assembled, term by term, from the cannibalisation mechanism of Section 2.2. A farm that sold at random hours would earn the average price, a capture value of one; cannibalisation is what subtracts from that one. The equation writes that subtraction as two physical channels, a cannibalisation term and a fuel-price distortion term, so that every constant in it is a named lever on a specific channel rather than a free coefficient fit for its own sake:

$$\mathrm{CV} \;=\; 1 \;-\; B\,\frac{M_{\mathrm{WF}}}{M_{\mathrm{BZN}}}\,S \;-\; C\,M_{\mathrm{WF}}\left(0.202\,p_{\mathrm{CO_2}} + p_{\mathrm{gas}}\right)^{\gamma}. \tag{B.1}$$

where S is the renewable market share and p_{gas}, p_{CO_2} the monthly fuel and carbon prices, folded together by the 0.202 tCO_2/MWh emission intensity of gas. The first term is cannibalisation: the wind-pattern penalty $M_{\text{WF}} = \text{sign}(\rho_{\text{VRE}} - \rho_{\text{load}})\,|\rho_{\text{VRE}} - \rho_{\text{load}}|^{\alpha}$ grows when a farm tracks the system's renewables more than its demand. It is scaled by penetration S and divided by the buffering term $M_{\text{BZN}} = (I_{\text{int}} + I_{\text{flex}})^{\delta}$, in which interconnection and flexibility absorb it. The second term is fuel-price distortion.

The constants carry the mechanism. The pattern-penalty exponent α sets how sharply the penalty escalates as a farm's wind aligns with the system's. The **cannibalisation scale** B is its overall strength: it multiplies the whole crowding-times-penetration term, so a larger B means a given amount of crowding and penetration erodes more value. The **buffering exponent** δ governs how strongly interconnection and flexibility reduce that erosion: the penalty is divided by $(I_{\text{int}} + I_{\text{flex}})^{\delta}$, so a larger δ makes interconnection and flexibility more protective per unit of capacity. C scales the fuel-price distortion term and γ sets its curvature. They are fit by bounded least squares, separately for the pooled data, the North Sea, and the Baltic, with a 200-resample farm-block bootstrap for intervals. Because the constants trade off inside shared terms, no single one is read in isolation; the basins are compared by the quantity they jointly produce, the implied slope of capture value against penetration, at common reference conditions.

B.6 The spatial cost model in full

Offshore capital cost rises with both water depth and distance to shore: deeper water demands heavier foundations, and farther sites demand longer export cables. Writing c_0 for the turbine-and-array base, $\phi(d)$ for the foundation, and a linear export term beyond a near-shore allowance ℓ_0,

$$\text{CAPEX}(d,\ell) \;=\; c_0 \;+\; \phi(d) \;+\; \gamma\,\max(\ell - \ell_0,\,0), \qquad \phi(d) = \begin{cases} a_f\,\max(d - d_0, 0), & d \le d_1, \\ a_f(d_1 - d_0) + c_{\text{fl}} + a_{\text{fl}}(d - d_1), & d > d_1, \end{cases} \tag{B.2}$$

with $c_0 = 1.85$ M€/MW, $d_0 = 25$ m, $d_1 = 60$ m, fixed-bottom slope $a_f = 12$ k€/MW/m, a floating premium $c_{\text{fl}} = 1.0$ M€/MW appearing past d_1, floating slope $a_{\text{fl}} = 6$ k€/MW/m, export term $\gamma = 10$ k€/MW/km, and $\ell_0 = 25$ km. The piecewise jump at d_1 is the transition from fixed-bottom to floating foundations. Levels are anchored to the offshore cost database that ships with PyPSA-Eur and to IRENA's 2024 cost review [81, 103]. Capital is annualised at a capital recovery factor $\text{CRF} = w(1+w)^n/[(1+w)^n - 1]$ with a weighted-average cost of capital $w = 7\%$ over an economic life n of 27 to 30 years. Technology learning enters through a factor $\lambda(y)$ normalised to unity in 2025 and falling on the database's offshore trajectory (capital cost about 2505 to 1916 €/kW over 2020 to 2050). Fixed operations cost is $f_{\text{OM}}(\ell) = 75 + 0.25\,\ell$ k€/MW/yr.

C Extended Results and Per-Zone Projections

The headline projection of Section 4.4 reports the Baltic at basin-mean resolution. Because the thesis rests on within-basin spread, Figure C.1 resolves it. The figure shows each Baltic zone that builds out (Poland, Estonia, Latvia, Lithuania), with its modelled capture-value history, its bias-corrected 2030 and 2040 projections, and vertical bars for the spread across scenarios. Finland is shown but set aside as before: its decline comes from the price collapse that followed the Olkiluoto-3 nuclear startup, not from wind cannibalisation. The history is modelled, since metered revenue does not exist for these zones: they have little or no operating offshore capacity, and most rest on a single planned-farm proxy (Poland on two). Each zone's history weights its observed prices by the virtual-farm output. The regression model returns a constant at these horizons, so the projections are system-model projections, corrected per zone on the fit of Section 3.6.2. The zones diverge sharply by 2040, from Lithuania near 0.40 to Estonia near

0.20, a range the basin mean conceals and the reason a developer cannot read a single Baltic number off the headline.

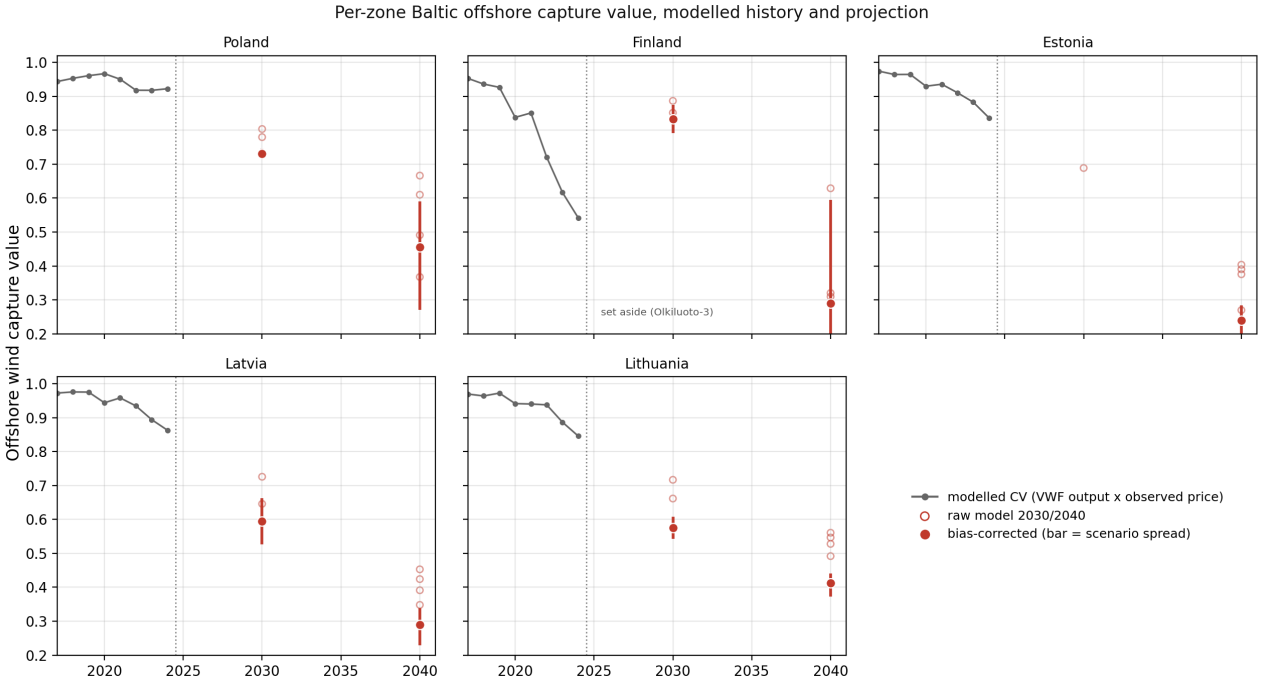


Figure C.1: Per-zone Baltic offshore capture value, modelled history (2017 to 2024, grey; observed zone prices weighted by virtual-farm output, as no metered revenue exists) to bias-corrected projection (red), for the zones that build out (Finland shown but set aside, nuclear-driven). Open markers are the raw model; filled markers the bias-corrected central value; vertical bars the spread across scenarios at each horizon. The regression model stops responding at these horizons; these are system-model projections.

C.1 Full algorithm and validation-scheme comparison (regression model)

Table C.1 extends the headline metrics of Table I.2 to all five algorithm families under every validation scheme. Gradient boosting leads on every genuine generalisation test. The polynomial's large negative out-of-fold scores show how violently a closed form extrapolates: that instability is why the tree is the operational model and the analytical equation only an interpretable backup.

C.2 Baseline capture-value calibration, three years

Table C.2 gives the per-zone calibration behind Section 4.2.2 and the bias correction. The bias (PyPSA minus observed) is near zero in the large North Sea zones and grows with congestion, peaking in DK2 in both 2023 and 2024; these fifteen points are exactly what the correction of Appendix D is fitted on. Because the baselines use observed fleets, they are unaffected by the projection re-solve.

C.3 The sector-coupled arms, per zone

Table C.3 details the matched twelve-hour pair behind Table D.3: the electricity-only and sector-coupled solves of the pinned 2040 TYNDP fleet, zone by zone. The cost-optimal arms behind them build

Table C.1: Out-of-fold R^2 of all five algorithm families under every validation scheme, at bidding-zone feature resolution. A dash marks schemes run only for the headline model. These metrics are independent of the system-model re-solve.

Validation scheme	XGBoost	Random forest	SVR	Poly(2)	Linear
Random K-fold (Jacquet scheme)	0.752	0.682	0.583	0.437	0.330
Leave-farm-out (headline)	**0.686**	0.636	0.497	−19.9	0.270
Leave-farm-out, excluding 2022	0.692	—	—	—	—
North Sea → Baltic	0.379	0.331	0.196	−12.2	0.230
Baltic → North Sea	0.139	0.204	−0.56	−1.06	0.020
Time split (train ≤2022, test 23–24)	−0.03	—	—	—	—

Table C.2: System-model capture value against observation, per zone and year, on common hours. Observed pairs metered offshore generation with observed prices; PyPSA is fully modelled; the hybrid prices modelled generation at observed prices, localising any error to the wind or the price. Bias is PyPSA minus observed.

Zone	Observed	PyPSA	Hybrid	Bias
2019 (pre-crisis)				
DE	0.923	0.984	0.929	+0.061
NL	0.977	0.986	0.978	+0.010
BE	0.920	0.983	0.956	+0.064
DK1	0.924	0.986	0.944	+0.062
DK2	0.933	0.987	0.954	+0.054
2023 (headline validation)				
DE	0.909	0.908	0.896	−0.001
NL	0.881	0.883	0.897	+0.002
BE	0.869	0.896	0.910	+0.027
DK1	0.854	0.917	0.869	+0.064
DK2	0.784	0.927	0.815	+0.143
2024 (most recent)				
DE	0.910	0.941	0.875	+0.031
NL	0.900	0.919	0.884	+0.019
BE	0.893	0.928	0.913	+0.035
DK1	0.843	0.947	0.837	+0.104
DK2	0.808	0.949	0.821	+0.141

52 and 127 GW of offshore respectively and hold capture value near 0.8; the pinned pair carries the comparison because its fleet matches the main projection. At 2030 the sector-coupled solve gives a North Sea of 0.81 and a Baltic of 0.86, within a few points of the electricity-only projection, so the flexibility effect, like the grid's, appears only once the buildout binds. DK2 is excluded: the pinning fixes country totals, so its small allocated fleet is not comparable across arms (Section D.5).

C.4 Who bears the loss, by zone

Table C.4 resolves the three regimes of Section F.3 zone by zone: scarcity-priced Poland, whose low capture value coexists with a small unpaid share; the locally oversupplied eastern Baltic, whose high unpaid shares the grid barely relieves; and the open North Sea, whose shares the grid does relieve.

Table C.3: The matched 2040 twelve-hour solves, per zone: capture value and the share of wind energy produced into zero-priced hours, electricity-only against sector-coupled.

	Capture value		Zero-price share (%)	
Zone	**Electricity**	**Sector**	**Electricity**	**Sector**
DE	0.40	0.72	57	25
NL	0.48	0.70	42	19
BE	0.71	0.82	3	10
DK1	0.38	0.74	77	20
FR	0.80	0.84	6	1
PL	0.65	0.85	18	28
EE	0.39	0.77	45	20
LV	0.41	0.82	41	20
LT	0.37	0.81	43	21

Curtailment is the share of available energy spilled, the volume channel that capture value cannot see and Article 13 compensates.

Table C.4: The 2040 zero-price exposure by zone: capture value, the share of wind energy in zero-priced hours, and the curtailed share of available energy, on the frozen and the cost-optimal grid.

	Frozen grid			Optimal grid	
Zone	**CV**	**Zero-price %**	**Curtailed %**	**CV**	**Zero-price %**
DE	0.59	47	44	0.63	34
NL	0.57	56	36	0.58	44
BE	0.55	5	0	0.62	23
DK1	0.55	69	18	0.84	36
FR	0.56	32	20	0.68	28
PL	0.49	6	1	0.74	32
EE	0.39	42	2	0.53	39
LV	0.42	37	5	0.59	33
LT	0.49	30	12	0.59	31

C.5 The candidate areas, ranked

Table C.5 lists the ten development areas behind Figure 4.11 at the 2040 horizon: levelised cost, capture value, the resulting break-even price, and the two rankings whose divergence the body reports (a rank correlation of 0.99 in 2025 falling to 0.90 by 2040).

D Robustness of the Projection

The headline projection of Section 4.4 rests on the uncorrected frozen-grid system model, with the penetration bias correction and the parametric hydrogen-demand assumption reported as refinements. This appendix tests both, prices the gas exposure, and sweeps the wider scenario space with the fast approximation of the system model; none of it moves the conclusion.

Table C.5: The ten candidate development areas in 2040, ordered by value-adjusted cost. The last column gives each area's value rank in 2025, so a change between the two final columns is the reordering the horizon produces.

Area	**LCOE** €/MWh	**CV**	**vLCOE** €/MWh	**Cost rank**	**Value rank**	**Value rank 2025**
German Bight	41	0.59	69	1	1	1
Danish Baltic (DK2)	46	0.57	80	3	2	4
Polish Baltic	43	0.49	88	2	3	2
Belgian	49	0.55	89	4	4	3
Dutch North Sea	61	0.57	107	6	5	6
French	68	0.56	120	7	6	7
Estonian	51	0.39	130	5	7	5
Danish North Sea	78	0.55	142	10	8	10
Lithuanian	71	0.49	144	9	9	9
Latvian	69	0.42	164	8	10	8

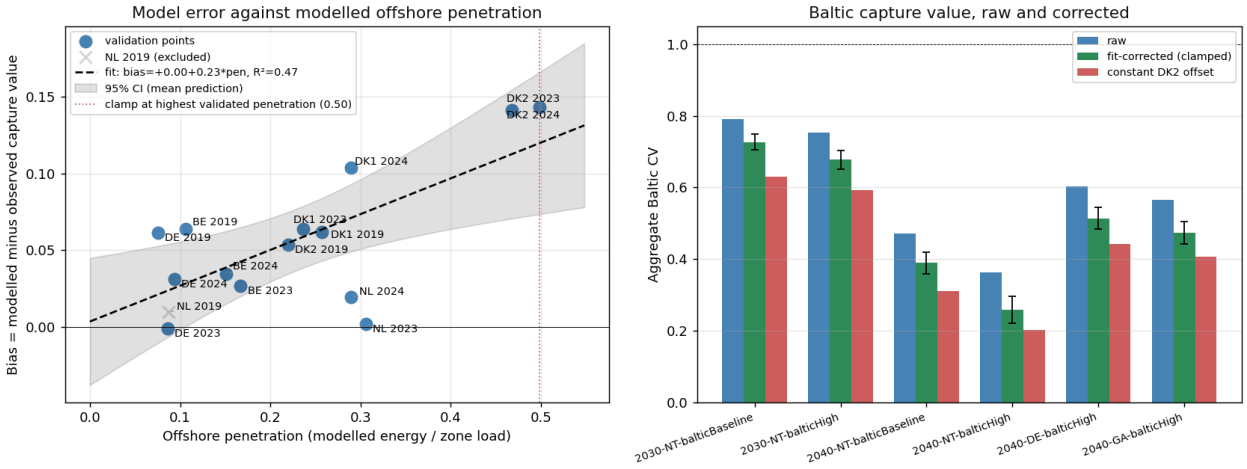


Figure D.1: The bias correction and its effect: the validation bias against modelled penetration, anchored by DK2 (left), and raw against corrected Baltic capture value per scenario (right).

The bias correction is anchored on DK2

The correction of Section 3.6.2 fits the system model's capture-value overstatement against its own offshore penetration on fourteen validation points (five zones in 2019, 2023 and 2024, the Netherlands 2019 excluded). Only the two DK2 entries lie above 31 % penetration, so the slope is a line between a structureless low-penetration cloud and a single high-penetration zone (Figure D.2). Leave-one-zone-out makes this explicit (Table D.1): dropping any zone except DK2 leaves the slope between 0.22 and 0.28 with its 95 % interval clear of zero, while dropping DK2 collapses it from 0.233 to 0.045, sends the interval across zero, and drops the penetration term's R^2 from 0.47 to 0.02. The penetration dependence is, statistically, the statement that DK2 has a large bias at a high penetration.

The conclusion nonetheless survives, because the correction is not what produces it. The uncorrected model already lands at 0.36–0.60 in 2040; the DK2-removed correction is a near-constant offset of about 0.05 that takes it to 0.31–0.56; the full correction reaches 0.26–0.51. All three agree on a loss of at least a third of present capture value, and near half in the central scenario. DK2 is kept as the anchor

precisely because it is the only existing high-penetration, Baltic-facing zone. That uniqueness makes it the right analogue, and it is also why the correction cannot be claimed as a calibrated law. The analysis is reproduced by `1.19_bias_correction_robustness.py`.

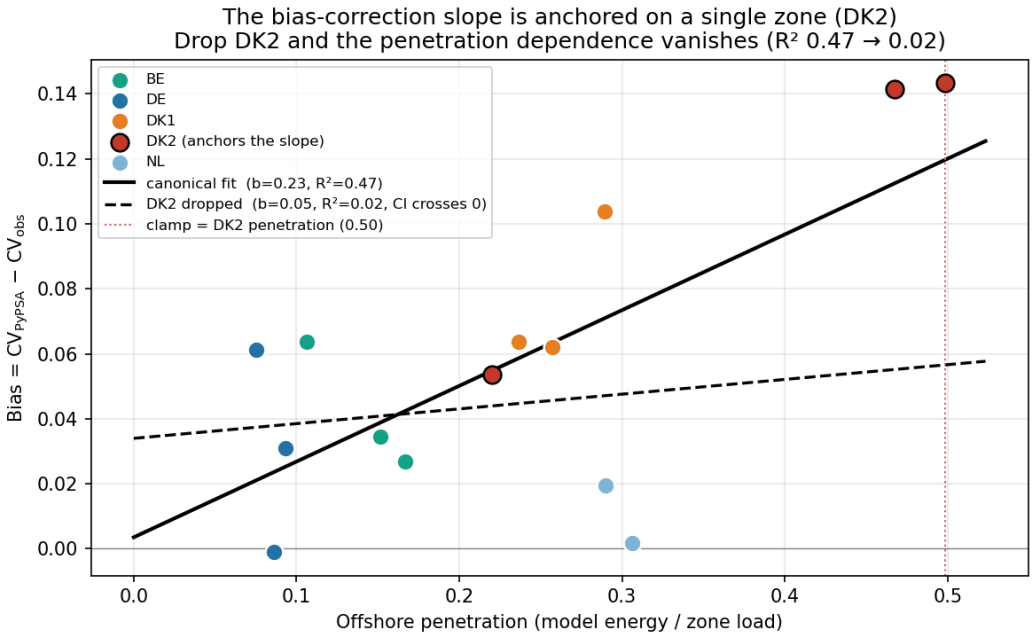


Figure D.2: The fourteen validation points of the bias correction (coloured by zone), the canonical fit, and the fit with DK2 removed. Without DK2 the penetration dependence vanishes (R^2 0.47 to 0.02) and the slope's interval crosses zero.

Table D.1: Leave-one-zone-out and (last three rows) leave-one-year-out refits of the bias slope. The slope and its 95 % interval are stable to dropping any zone except DK2, whose removal collapses both.

Refit	n	**Slope**	**95 % CI**	R^2
Canonical (all 14)	14	0.233	[0.08, 0.39]	0.47
Drop DE	11	0.262	[0.05, 0.48]	0.46
Drop NL	12	0.278	[0.18, 0.38]	0.79
Drop BE	11	0.251	[0.06, 0.44]	0.50
Drop DK1	11	0.224	[0.04, 0.40]	0.46
Drop DK2	11	**0.045**	**[−0.21, 0.30]**	**0.02**
Drop 2019	10	0.303	[0.11, 0.49]	0.63
Drop 2023	9	0.200	[−0.00, 0.40]	0.43
Drop 2024	9	0.195	[−0.04, 0.43]	0.36

The DK2 price shape is outside the model class

The bias correction treats the DK2 overstatement as a measured error. The alternative, repairing the dispatch until DK2 comes out right, was tried exhaustively and failed, which is what justifies the treatment (Table D.2). The blocking mechanism is structural: DK2 retains about 2,900 MW of export capacity into

a Nordic system holding 62 TWh of reservoir hydro, so a local surplus always finds a positive-priced sink and the zonal price never collapses. The observed shape, 264 negative-price hours in 2023, is produced by unit commitment, must-run combined heat and power, and subsidised bidding, none of which a linear program contains. Figure D.3 shows the diagnosis: the model's generation is right and its prices are too flat in exactly the windy hours. The model that does land DK2 is the regression model, out of fold at 0.798 against the observed 0.784.

Table D.2: The DK2 re-dispatch experiments. No dispatch-side change reproduces the observed 0.784. Values are DK2 capture value unless stated.

Experiment	**Change**	**Outcome**
Faithful re-solve	none	0.927 to 0.968: the price is not even unique
Negative wind bids	−60 €/MWh region-wide	no change: hydro absorbs the surplus
Derate the Öresund alone	3,396 to 1,700 MW	backfires: mean price 38 to 98 €/MWh, re-pinned to German gas
Myopic hydro	reservoirs capped at 24 h	explodes: 0.58 at a 1,207 €/MWh mean price
Market NTCs, both corridors	DK2–SE 1,700, DE–DK1 2,500 MW	DK1 error +0.06 to +0.01; DK2 unmoved at +0.15
Rolling-horizon hydro	twelve monthly solves, 14-day lookahead	backwards: 0.932 to 0.957
Must-run heat × negative bids (nine-point grid)	minimum load 0.20 to 0.50, bids 0 to −10 €/MWh	0.947 to 0.958 across all nine; zero negative-price hours

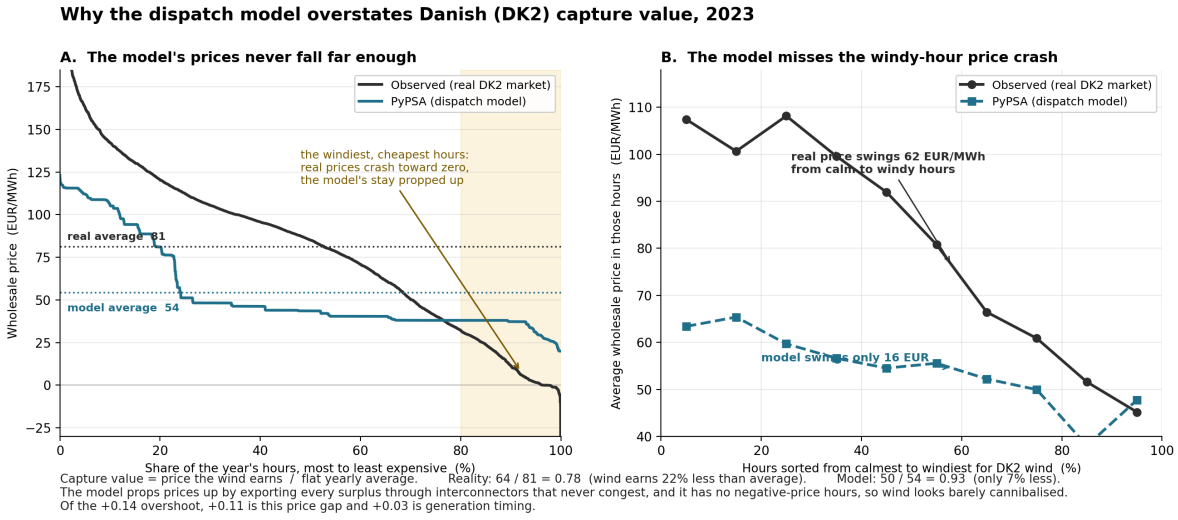


Figure D.3: The DK2 diagnosis. Pricing the model's own generation at observed prices recovers the observed capture value, so the wind is right and the error lives entirely in the price shape, which stays flat in the windy hours where real DK2 prices collapse.

The hydrogen-demand assumption does not drive the result

Correction D adds a parametric hydrogen layer that holds prices off zero at high renewable shares; its demand is assumed, so the 2040 result is tested against it by re-solving every 2040 scenario with the electrolyser demand and capacity scaled to half and one-and-a-half times the canonical level (Figure D.4). Across that ±50 % range the energy-weighted Baltic capture value moves by at most 0.013 in any scenario and by under 0.01 in the two cannibalisation-dominated DE and GA scenarios; the 2040 band shifts only from 0.36–0.60 at the central floor to 0.37–0.60 at either extreme. Load-shed volumes are unchanged across the variants, so the small sensitivity that does exist runs through the price floor and not through adequacy: more hydrogen demand lifts the windy-hour prices where wind earns, so capture value rises slightly. The floor sets the depth of the erosion at the margin; it does not create it. The sweep is reproduced by `1.20_h2_floor_sensitivity.py`.

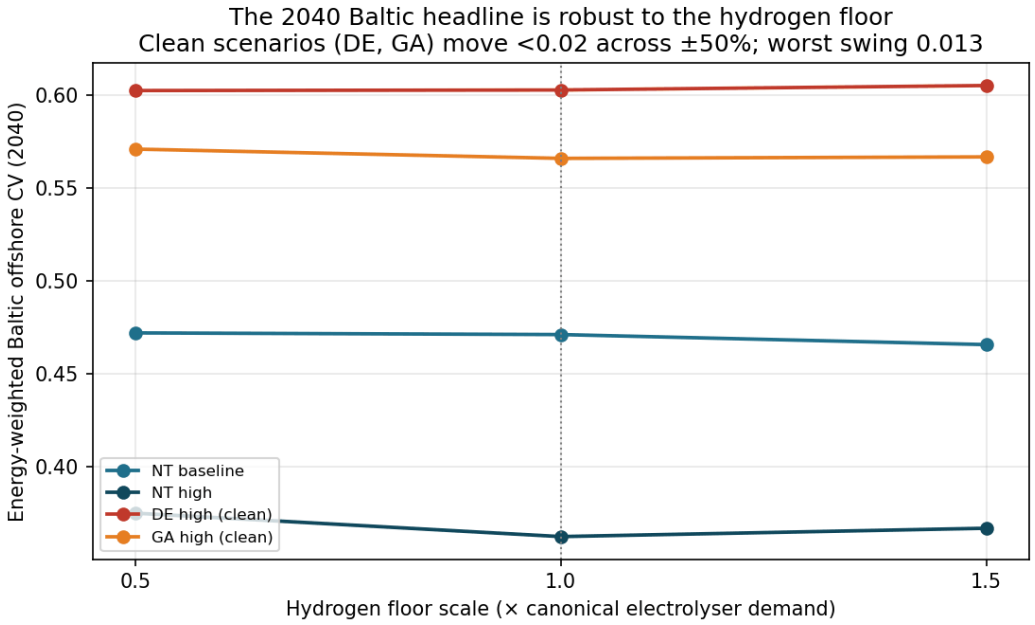


Figure D.4: The 2040 Baltic capture value against the scale of the assumed hydrogen demand (half to one-and-a-half times the central electrolyser demand). The clean DE and GA scenarios are nearly flat; the NT scenarios carry a small, floor-mediated dependence.

The gas-price grid

One input remains untested: the gas price that set every European record in 2022. Re-solving three buildout scenarios across five gas prices, 15 to 120 €/MWh, prices the exposure directly (Figure D.5). The price grid is coarse, fifteen networks, so the slopes indicate direction only. In 2030 the North Sea is the more exposed basin and the Baltic is already near-decoupled: gas still sets the price often in the thermal North Sea, rarely in the windy Baltic hours. Even end to end, the full crisis sweep moves the 2030 North Sea capture value by only 0.07, less than one step of the buildout. By 2040 both basins decouple almost entirely, their prices set by renewables, the demand-response tier, and hydrogen production. Gas risk is real, asymmetric, and second-order. The buildout drives Baltic value, and the fuel price only modulates it.

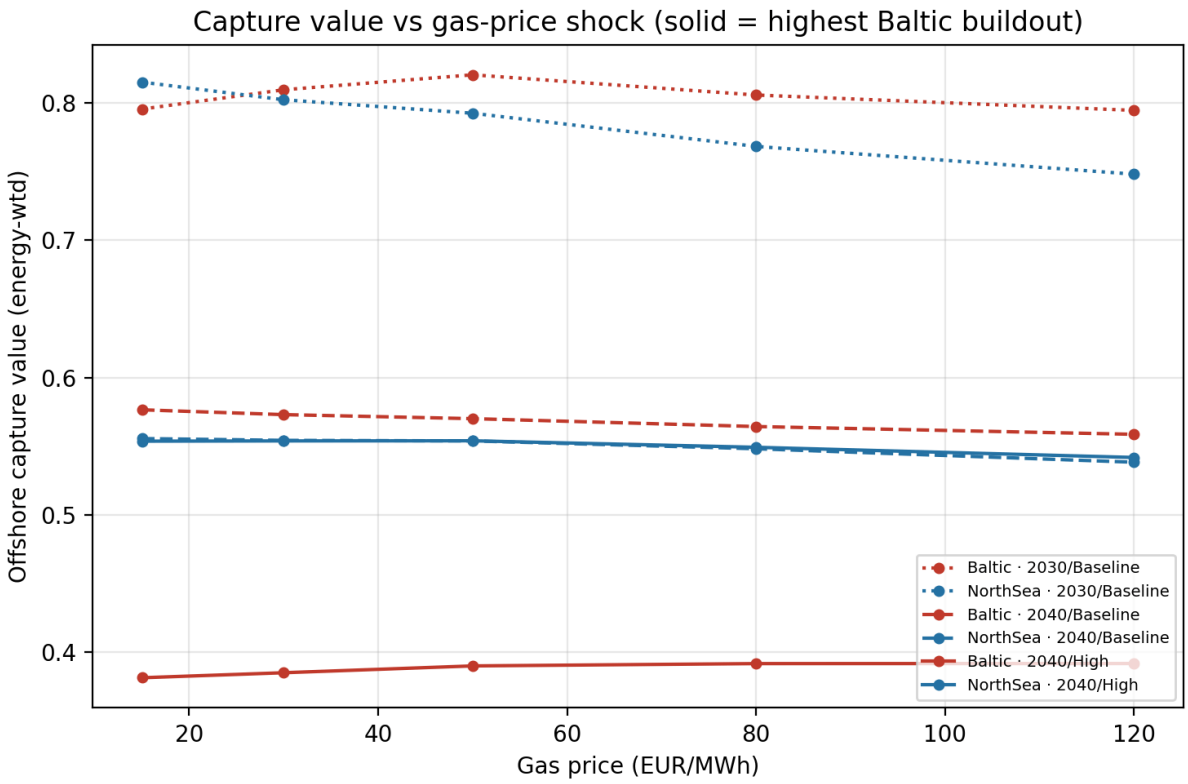


Figure D.5: Capture value across the gas-price grid, by basin and scenario. The North Sea's exposure is visible in 2030; the Baltic's curves are nearly flat, and by 2040 both basins decouple from gas.

The wider scenario sweep

The fast approximation of the system model is a gradient-boosted model trained on 76 solved networks (the 55 dedicated training solves plus the 21 reused full-resolution baseline and scenario solves of Table K.1), its inputs spread across the scenario space by Latin hypercube sampling. Its training solves run three-hourly; re-solving design points at full resolution confirms the approximation is benign in the policy-relevant range and only optimistic at the highest levels of buildout.

Its held-out skill is adequate for exploration (R^2 0.86 for capture value): it reproduces the 2040 scenarios to within 0.05 and leans optimistic beyond about 100 GW of Baltic offshore. The headline is left to the direct solves throughout.

Across that space, buildout drives the erosion and gas barely moves it. The Monte-Carlo draw (Figure M.8), reaching beyond the plan to 150 GW, puts the median Baltic capture value at 0.28 and the North-Sea-to-Baltic asymmetry near 2.3 times. The envelope samples buildouts well beyond the official plans, so its centre of mass is more aggressive than any TYNDP scenario; that the direct solves sit in the upper half of the distribution means the headline range is, if anything, conservative. Nowhere in the explored space does heavy Baltic buildout coexist with a healthy Baltic capture value.

D.1 Weather-only corroboration

Features and capture values share market data, so the mechanism could be circular. The weather is not. Cluster the farms by hourly output alone, no prices and no market labels, and the geography sorts

them on its own (Figure D.6): North Sea zones correlate at 0.44, the under-built Baltic states at 0.61, the built-out DK2 and SE4 at 0.87, and the two basins are largely independent at 0.28. The grouping reflects meteorology, and it points forward: the under-built zones, the ones the buildout will fill, are a tightly correlated bloc. The same signal repeats inside countries: a compact fleet blows nearly as one (Belgium 0.96, Poland 0.94) while a spread one does not (Germany 0.64, Denmark 0.67). Germany's farms even split into a North Sea and a Baltic bloc, so the basin is the right boundary for the analysis (Appendix M). The physical picture behind the enclosure penalty is now complete: Baltic wind arrives together, into a smaller market with weaker export capacity.

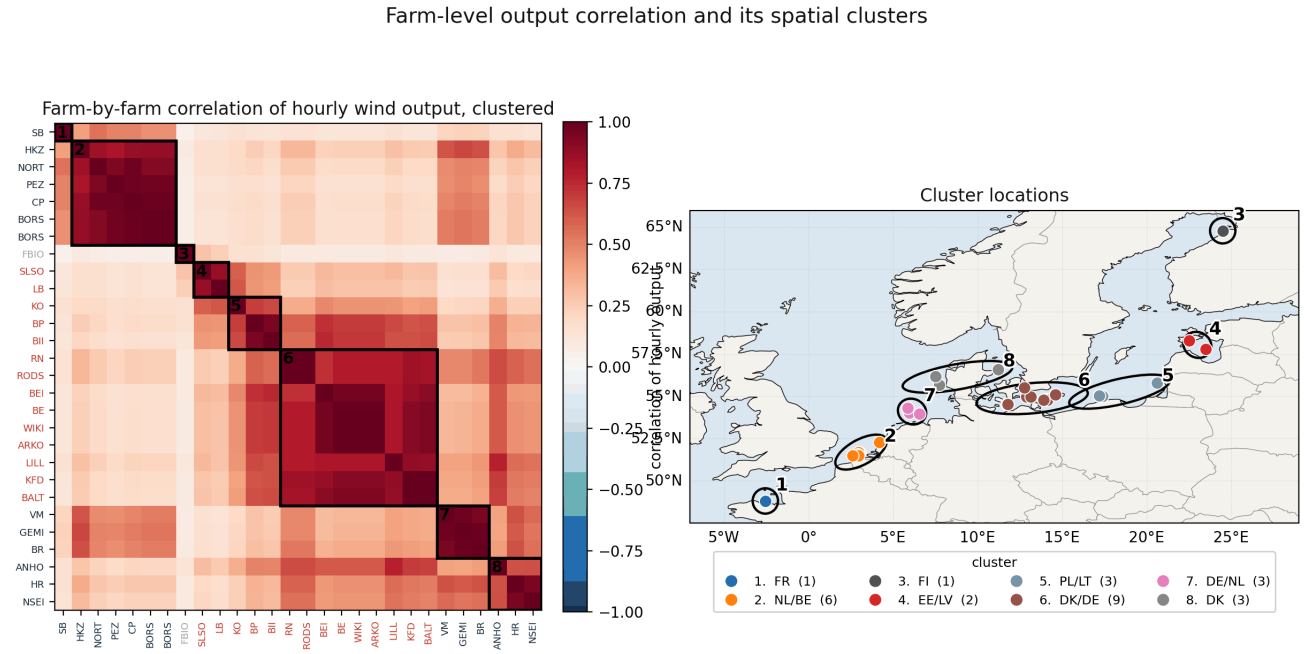


Figure D.6: Weather-only corroboration: hourly-output correlation clustered (left) and the clusters mapped (right). Within-basin correlation is high and cross-basin low: North Sea 0.44, under-built Baltic 0.61, built-out 0.87, cross-basin 0.28.

That proximity reflects the mechanism itself. Plotting every farm pair's output correlation against its separation collapses both basins onto a single decay curve, correlation falling away with a fitted length of 422 km (Figure M.11). The Baltic pairs obey no different physics; they sit on the near, steep end of the same curve because the basin's farms are closer together (median 285 km against 347, a few long pairs notwithstanding). Distance sets synchronicity, and the Baltic's geography fixes the distance.

D.2 Scarcity and gas sensitivities

The first rival explanation is that the 2040 erosion is adequacy stress rather than cannibalisation. Recomputing every capture value with scarcity-priced hours removed rules it out (Figure M.9): the 2030 and clean 2040 scenarios barely move, and the two NT scenarios, whose +0.15 adequacy component sits almost entirely in Poland, still reach only 0.51 to 0.62, far below today's 0.88. The floor of the range comes from cannibalisation alone.

Gas is the other suspect input, and it is second-order too. Re-solving three buildouts across five gas prices, 15 to 120 €/MWh, moves the 2030 North Sea capture value by only 0.07 end to end, less than one step of the buildout. By 2040 both basins have decoupled from gas almost entirely, their prices set by

renewables, demand response and hydrogen production (Appendix D). Baltic value is set by the buildout, with fuel prices a second-order influence.

D.3 Sector coupling and flexible demand

The grid is the spatial escape for surplus wind; the temporal escape (heat, hydrogen and storage absorbing windy hours locally) is excluded from the electricity-only model by construction. A sector-coupled variant therefore tests it against an electricity-only twin identical in every setting (Section D.5), so the difference between the two solves is the value of flexibility alone.

Table D.3: The value of flexible demand: the matched twelve-hour 2040 solves, TYNDP fleet pinned, frozen grid. Zero-price columns give the share of wind energy produced into zero-priced hours.

	Capture value		**Zero-price share**	
2040 solve (12 h)	**North Sea**	**Baltic**	**NS**	**Baltic**
Electricity only	0.51	0.54	45 %	29 %
Sector-coupled	0.74	0.82	19 %	25 %
Flexibility effect	+0.23	+0.28		

Flexibility recovers most of what the frozen grid destroys (Table D.3): the isolated effect, +0.23 and +0.28, is the same order as the grid lever, and it is again the Baltic that gains most, ending above the open basin as local electrolysis absorbs its surplus. In 2030 the sector world sits within a few points of the electricity-only one, so flexibility, like transmission, pays only once the erosion binds. Per-zone values are in Appendix C.

Two facts keep the headline in place: the zero-price problem shrinks but does not vanish, 19 to 25 % of wind energy still unpaid in the sector-coupled solves, so the contract question survives full flexibility; and the twelve-hour resolution smooths the price shape both arms see, so these numbers bracket the hourly headline rather than replace it. Read beside the two grid solves, the levers agree: the frozen electricity-only world is the floor, and either escape lifts the Baltic off it further than the North Sea.

D.4 Pricing scarcity

A stressed scenario must be able to not serve demand, and the price of that choice shapes every average. Correction F sets the ladder: demand response at five per cent of peak load offered at 400 €/MWh, and unserved energy at the 4,000 €/MWh cap, so scarcity hours price as the real market would. Section 3.6.4 measures how much of each scenario's erosion is scarcity rather than cannibalisation.

D.5 The sector-coupled variant

The electricity-only model can move surplus wind only through space, over the grid; real systems also move it through time and into other demands. To measure how much of the projected erosion flexible demand removes, the system model is rebuilt in PyPSA-Eur's sector-coupled configuration, with heating, hydrogen, storage and electric transport co-optimised alongside electricity for an overnight 2040 system. The full-sector problem is tractable only at twelve-hour resolution, so its control is an electricity-only twin solved at the same twelve hours with every other setting identical, both arms pinned to the same TYNDP fleet. The difference between the two solves is then the value of flexibility itself. Two limits qualify the reading: twelve-hour steps smooth the price shape both arms see, and country-level pinning makes small zones such as DK2 incomparable across arms (Section D.3).

E The Two-Estimator Test

Section 4.3.1 states why the two models diverge beyond the observed record. This appendix carries the underlying test, which asks whether the divergence follows from how each model is built rather than from any disagreement about how prices form. The error types named here are structural descriptions of each model's construction; the measured composition of each model's error, on held-out targets, is the Theil partition reported in Section 4.3.2.

The division of labour has a mathematical basis, and it was tested (Figure E.1). The regression model is a bounded interpolator: four hundred trees give a piecewise-constant surface whose output cannot leave the training range and whose gradient is exactly zero outside it. The system model is a linear program whose price is the shadow price of the nodal energy balance, re-derived from the merit order every hour with no training range at all, but floored at zero. Table E.1 sets the two error structures side by side.

Table E.1: The two error structures compared. Each row is measured in this appendix or Appendix D; the mean signed errors are against observed 2023–24 capture values.

	Regression model	**System model**
Error type	Bounded interpolation	Simplified market
Mean signed error	−0.005	+0.057
Sign and shape	No systematic sign	Always overstates; grows with congestion
Largest where	Outside the training support (all 36 projection rows)	Small, hydro-coupled zones (DK2 +0.14)
Detected by	Range and convex-hull checks	Three-way comparison
Consequence	Returns a constant from 2040	Corrected by the measured fit

Three consequences follow, and the runs confirm each (Table E.2). Outside the range of its training data the regression model returns a constant: across the four Baltic 2040 scenarios its answers span 0.008 of capture value where the system model spans 0.43, a fifty-four-fold difference. The two agree where both can be checked, to 0.06 in sample and 0.04 at 2030, and diverge to 0.35 at 2040, where the projection leaves the training support. And the system model's error has a predictable sign and order: it overstates every validated zone, most where zones are small and hydro-coupled, DK2 above DK1 above Belgium above Germany and the Netherlands, which is exactly the shape the bias correction of Section 3.6.2 subtracts.

One expectation fails, and the failure is informative. The erosion was expected to accelerate with penetration; the fitted response is the opposite, a saturating curve, because the zero price floor caps how much a marginal windy hour can still lose once the price already sits at zero. The saturation means the system model, if anything, understates cannibalisation at extreme penetration, so the 2040 numbers are upper bounds on capture value, conservative in the same direction as the measured bias. The two models' errors are of different kinds, and the division of roles places each model where its error is smallest. Section 4.3.2 measures those kinds directly on held-out targets.

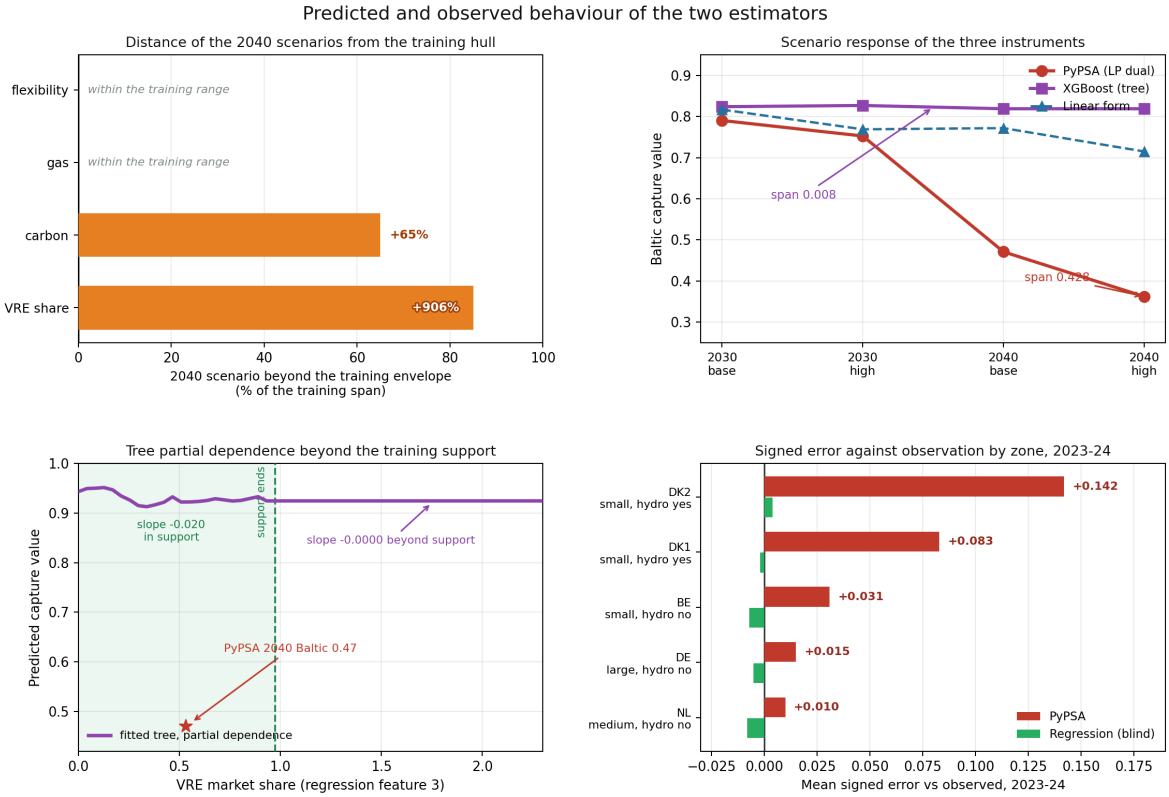


Figure E.1: Four checks of the two-estimator argument against the runs. Top left: every 2040 scenario leaves the training envelope, renewable share furthest. Top right: across the four Baltic scenarios the regression model moves 0.008 where the system model moves 0.43. Bottom left: the regression model's fitted response is exactly flat beyond its support, so the divergence comes from leaving the range of the training data and does not reflect curvature. Bottom right: the system model's in-sample error is largest in small, hydro-coupled zones; the regression model's is near zero, which fixes the sign and order of the bias correction.

Table E.2: The two-estimator test in numbers. Response spans are across the four Baltic 2040 scenarios; divergence is the mean absolute gap between the models' Baltic estimates; curvature is the fast-approximation fit across 76 solved networks with a cluster-bootstrap interval; errors are against observed 2023–24 capture values.

Response span: tree / linear / analytical / system model	0.008 / 0.102 / 0.179 / 0.428
Divergence: in sample / 2030 / 2040	0.062 / 0.035 / 0.348
Penetration-squared coefficient (95 % CI)	−0.193 [−0.215, −0.169]: concave
Mean signed error: system / regression	+0.057 / −0.005
System error by zone	DK2 +0.14 ¿ DK1 +0.08 ¿ BE +0.03 ¿ DE +0.02 ¿ NL +0.01

F Portfolio and Distributional Analyses

These analyses serve Objective 5 and the question of who bears the loss. They are reported here in full; their conclusions are carried in Sections 4.6.3 and 5.1.

F.1 Spreading across both basins

Objective 5 asks whether allocating capacity across both basins beats concentrating it in the North Sea; the two channels of Section 2.3 are tested separately because they give opposite answers (Figure F.1).

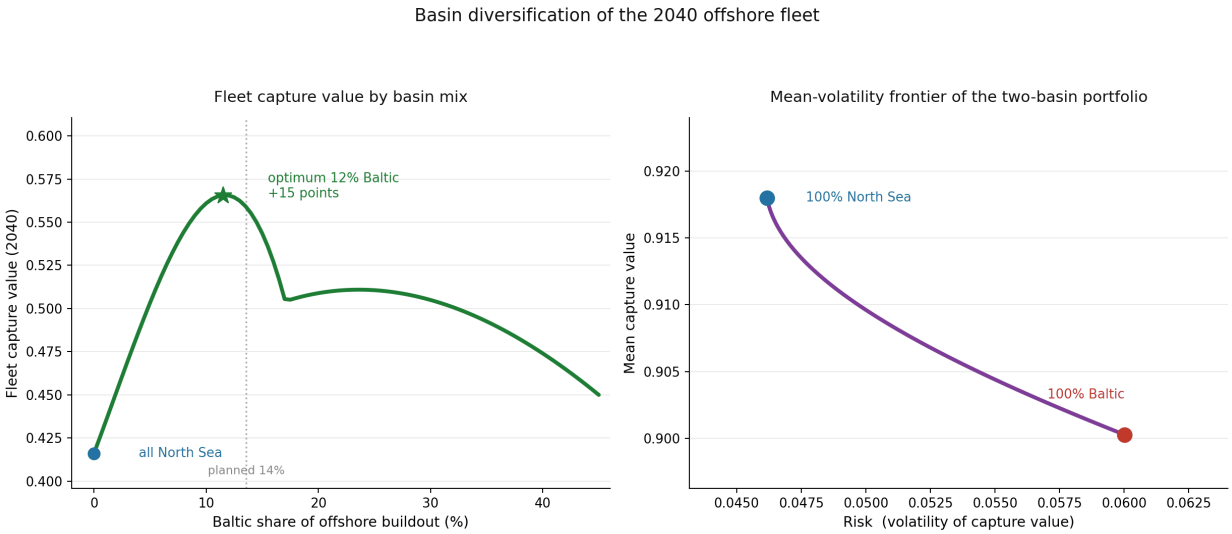


Figure F.1: The two portfolio channels. Left: aggregate 2040 capture value of a fixed 623 TWh of offshore generation as a function of the share allocated to the Baltic, holding the total constant. Right: the mean-variance frontier built from monthly capture values, 2017 to 2024.

The first channel is a cannibalisation hedge, and it works. Holding total offshore generation fixed and varying only its split, an all-North-Sea allocation cannibalises itself down to an aggregate capture value of 0.416, while shifting 12% to the Baltic, which starts earlier on its curve, lifts the aggregate to 0.566, a gain of 36%; the planned 14% split is within half a point of that optimum.

The second channel is a risk hedge, and it fails. Monthly capture values over 2017 to 2024 make the North Sea simultaneously the higher-mean and lower-variance basin (0.918 at 0.046 against 0.900 at 0.060), and the two series co-move at a correlation of 0.80, so the minimum-variance portfolio is 100% North Sea and spreading delivers no volatility reduction at all.

The 0.28 wind correlation of Section 4.5 explains the split: wind is decorrelated, so moving capacity eases cannibalisation, but market coupling propagates price levels across borders. The benefit of spreading sits entirely in the mean: the portfolio is less cannibalised, while its volatility barely changes.

F.2 Method: decomposing capture value by price state and pricing congestion rent

Who bears the loss: decomposing capture value by price level

A capture value states how much revenue is lost. It does not state who loses it, and under the support instruments reviewed in Section 2.3 the answer differs by mechanism. A two-sided contract for difference tops any positive price up to the strike, and volumes constrained off by the network are compensated under Article 13 of Regulation (EU) 2019/943 [44], so both losses fall on the state. In a zero or negative price hour the contract is suspended, following GB practice from allocation round four and the 2024 market design reform [11], so that loss falls on the farm and no compliant scheme may cover it. What decides the bearer is therefore the price *level*, not the physical cause.

Capture value factorises to match. Writing z for the share of generation occurring in hours priced below a threshold ϵ, and CV^{+} for the capture value computed over the remaining hours,

$$\mathrm{CV} = (1-z)\,\mathrm{CV}^{+}, \qquad \text{so} \qquad \underbrace{1-\mathrm{CV}}_{\text{total loss}} = \underbrace{z}_{\text{farm-borne}} + \underbrace{(1-z)\big(1-\mathrm{CV}^{+}\big)}_{\text{state-borne}} \tag{F.1}$$

and the split is exact, with nothing left over. The farm-borne term is energy earning nothing with the contract suspended; the state-borne term is the depression of positive prices that the contract absorbs. Both are computed hour by hour on the solved networks for 2030 and 2040 under the frozen and cost-optimal grids, with $\epsilon = 1$ €/MWh. Because the linear program floors prices at zero (Section 3.5.1), z is a conservative lower bound on the real farm-borne share, which in a market with a negative-price tail would be larger. Curtailed volume is reported separately, since it never enters capture value at all yet is compensated under the same Article 13. The observed anchor is the wind-weighted share of 2023 and 2024 generation falling in hours with a day-ahead price at or below zero. A required strike price, $\mathrm{LCOE}/(1-z)$, follows directly and states what a contract must pay if the farm is to recover its costs given that a share z of its output earns nothing.

Congestion rent and who funds the remedy

If interconnection is the lever, its affordability is a fair question, and the income that congestion itself generates is the natural place to look. For every branch the rent is the flow multiplied by the price difference across it, summed over the year:

$$R_{\ell} = \sum_{t} w_t \, f_{\ell,t}\big(\eta_{\ell}\,\lambda_{\mathrm{to}(\ell),t} - \lambda_{\mathrm{from}(\ell),t}\big) \tag{F.2}$$

with $f_{\ell,t}$ the flow, λ the nodal prices, w_t the snapshot weighting, and η_{ℓ} the efficiency, which is unity for lossless lines and the link efficiency for controllable interconnectors. Rents are aggregated by border and classified by whether each branch is internal to a basin or crosses its boundary. The same quantity is computed on the frozen and the cost-optimal network, so that the difference measures the rent that building the grid destroys, which is the merchant-transmission problem Joskow and Tirole [43] identify. Against that income sits the annualised capital cost of the transmission the optimal solve builds, taken as the added capacity multiplied by its annuitised cost. Because frozen-grid Poland prices several hundred scarcity hours, every rent figure is reported twice, once over all hours and once excluding hours above 390 €/MWh, the same threshold used for the scarcity sensitivity in Section D.4. The rent must not be added to the captured-revenue gain of Section 3.7.3: both arise from the same price convergence, and building the lines removes the rent, so the two must not be summed.

F.3 Who bears the loss, and what funds the cure

Every number so far has said how much value is lost; who loses it is a separate question with a different answer. Applying the factorisation of Equation F.1 splits each basin's erosion into the part a contract absorbs and the part it cannot; Figure F.2 draws that split for both basins and both grids.

The two basins look similar on the capture-value chart and are opposites underneath it. The frozen-grid Baltic factorises as $(1-z)\,\mathrm{CV}^{+} = 0.815 \times 0.578$: only 18.5% of its energy earns nothing, so almost two-thirds of its loss is depressed-but-paid, the kind a contract absorbs. The North Sea factorises as 0.533×1.075: its paid hours earn slightly above average, so there is no depression to top up and the entire loss is uncompensable; with CV^{+} above one the state term is a small premium, so the farm's share slightly exceeds the loss itself. The basin with the worse headline number has the better protected investor.

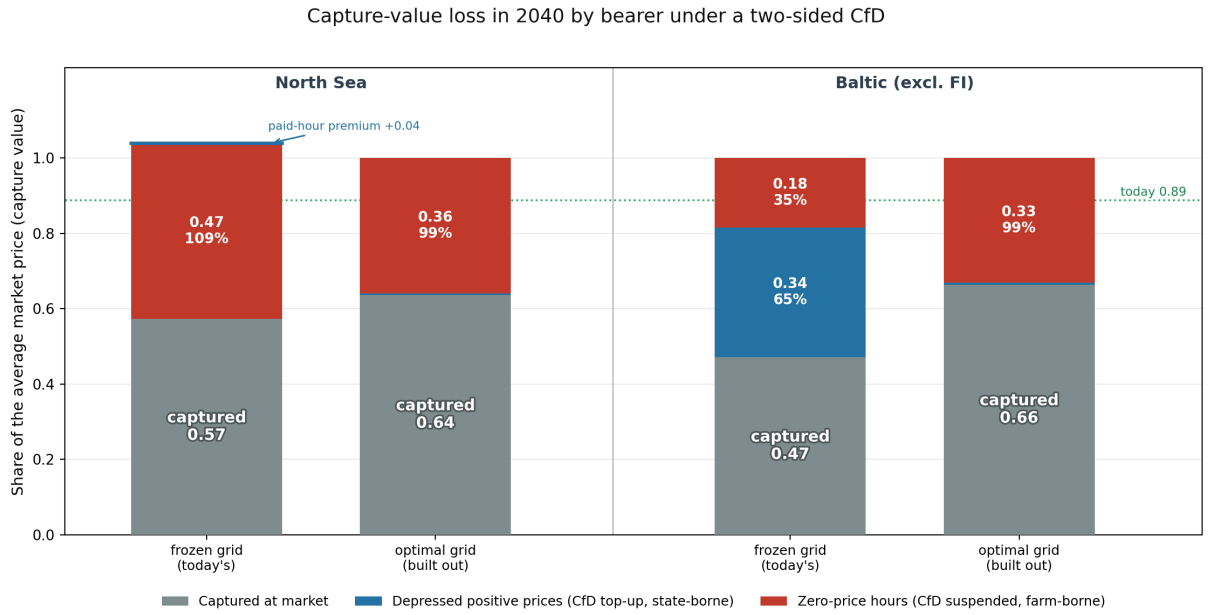


Figure F.2: The 2040 capture-value loss split by bearer. The grey block is revenue captured at market. The blue block is value lost to depressed but positive prices, which a two-sided contract tops up to the strike, so the state carries it. The red block is energy earning nothing, where the contract is suspended and the farm carries it. In the frozen North Sea the paid hours earn a small premium (+0.04), which offsets part of the zero-price loss, so the farm-borne share amounts to 109% of the net loss.

Volume shows the same split from the other side: the frozen-grid North Sea spills a third of its available 2040 energy against 3.6% in the Baltic, curtailment being a further state-borne bill invisible to capture value.

Building the grid does not simply reduce the loss; it changes its bearer. Under optimal expansion the depression factor snaps to about 0.99 in both basins, so what survives is purely the zero-price channel. The share of Baltic energy earning nothing *rises* from 18.5% to 33.1% even as capture value improves, because interconnection couples the basin to the wider system's surplus hours, and roughly 99% of the remaining loss becomes farm-borne. Expansion converts a fiscal problem into an investor one, a consequence absent from the transmission-benefit literature.

The zonal detail separates three regimes the basin averages hide (Appendix C): scarcity-priced Poland, the locally oversupplied eastern Baltic, and the open North Sea, most exposed but most relieved. The strike a contract needs follows the same split, $\mathrm{LCOE}/(1-z)$, roughly 71 €/MWh for Poland against 125 to 210 across the open zones, converging to 92 to 103 under expansion: an ordering that inverts the capture-value ranking for a contract-backed investor.

All farm-borne shares here are conservative: the linear program floors prices at zero, and the observed 2023 to 2024 anchors already show 5.9% of North Sea and 3.0% of Baltic generation at or below zero, so the real exposure is larger than reported.

F.3.1 Congestion rent and the cost of the cure

The recommendation that follows is to build transmission, and its affordability need not be a public spending question: congestion is itself a revenue stream, and Article 19 of Regulation (EU) 2019/943

already earmarks that revenue for this purpose [44]. Equation F.2 prices it. The rent accrues to the transmission operators, and none of it reaches generators: congestion is only ever a cost to a wind farm, and the question here is only whether the system's own congestion income covers the cure.

Table F.1: Annual congestion rent in 2040, in billion euro per year, computed on the frozen and cost-optimal networks. Figures are given over all hours and excluding scarcity hours above 390 €/MWh, since frozen-grid Poland prices several hundred such hours. The grid cost is the annualised capital cost of all transmission the cost-optimal solve builds across the fourteen countries.

	All hours (bn€ per year)	**Excluding scarcity** (bn€ per year)
System congestion rent, frozen grid	38.9	22.6
of which Baltic-related	11.8	4.3
System congestion rent, optimal grid	15.2	15.1
of which Baltic-related	2.1	2.1
Rent destroyed by building the grid	23.7	7.4
Annualised capital cost of the new lines	2.7	2.7
Baltic rent as a multiple of total grid cost	**4.4**×	**1.6**×

The conservative column is the one to quote: even excluding every scarcity hour, Baltic-related borders generate 4.3 bn€ per year against an annualised 2.7 bn€ for the entire fourteen-country expansion. The enclosed basin already produces, and by statute already earmarks, more than enough income to pay for the interconnection that would relieve it, and the concentration is specific: the German-Polish border alone carries 7.2 bn€ on the frozen grid. Poland's borders are where the money and the constraint coincide.

Two qualifications matter. First, the rent and the captured-revenue gain must not be added: building the lines destroys the rent that funds them, the merchant-transmission problem in its original form [43], so the rent is the pre-existing, ring-fenced funding pool and the welfare gain the justification. Second, expansion is not congestion-free: Baltic *border* rent collapses from 2.9 to 0.7 bn€ while *internal* rent barely moves, so interconnection resolves the enclosure and leaves the domestic bottlenecks standing.

G Forecast Checks on the Two Models

Each model's mean squared error is separated following Theil [97]:

$$\mathrm{MSE} = \underbrace{(\bar{\hat{y}} - \bar{y})^2}_{\text{bias}} + \underbrace{(s_{\hat{y}} - s_y)^2}_{\text{variance}} + \underbrace{2(1-r)\, s_{\hat{y}} s_y}_{\text{covariance}}, \tag{G.1}$$

whose three shares sum to one.

Section 4.3.2 states the conclusions of two checks on the models as forecasters. This appendix carries them in full.

G.1 Backtesting the act of projecting

Skill on held-out targets tests the models but not the projection, which additionally has to build its future inputs. The one horizon where that whole chain can be scored is the recent past: fitted on data to 2021 and asked for 2023 and 2024, across the same four zones, ninety-six zone-months (Table G.1).

Three results follow. The first is that building the feature vectors costs nothing. Predicting from constructed inputs, with the weather terms at each farm's pre-2022 monthly climatology and the system

terms carrying the pre-2022 monthly shape, is not worse than predicting from the realised features but slightly better, 0.049 against 0.053. The seasonal construction smooths month-to-month movements the model could not have anticipated in any case, so it acts as a mild regulariser. Whatever is wrong with the projection at 2040, it is not the construction of the inputs.

The second is that over this horizon the two approaches project equally well. The constructed-input regression model and the system model both score 0.049. Both over-predict, by 0.019 and 0.041, which is cannibalisation deepening faster than either anticipated. The system model is again the more bias-dominated of the two, at 0.33 of squared error against 0.09.

The third explains the first two and connects them to the projection proper. This backtest is only a mild excursion beyond the training data. Of the 2023–24 observations, 58 % lie inside the pre-2022 range on all seven features at once. The single feature breached at any scale is the carbon price, at 42 % of rows, which Section 4.3.1 showed to be inert. The regression model still largely holds support here, and where it holds support it projects as well as the system model does. That is the finding: the divergence at 2040 is not a general difference in forecasting ability between the two model families, since over a two-year horizon inside the data there is no such difference. It is specific to the loss of support, where 2040 leaves every one of the thirty-six projection rows outside the joint training range rather than 42 % outside on one inert feature.

Table G.1: Backtest of the projection: fitted on data to 2021, scored on 96 zone-months in 2023 and 2024. Shares are the Theil partition and sum to one. The constructed inputs use realised annual levels, so the test grants perfect scenario foresight.

				Share of squared error		
Prediction	**MAE**	**RMSE**	**Bias**	**Bias**	**Var.**	**Cov.**
Regression, realised features	0.053	0.066	+0.027	0.17	0.12	0.72
Regression, projected features	0.049	0.063	+0.019	0.09	0.33	0.58
System model	0.049	0.071	+0.041	0.33	0.16	0.51

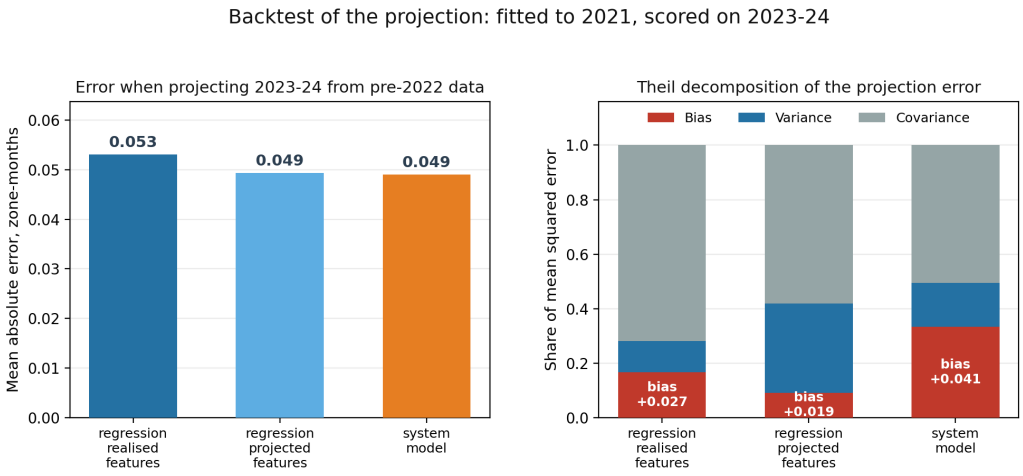


Figure G.1: Backtesting the projection. Left: error when each approach is fitted to 2021 and asked for 2023 and 2024; constructing the input vectors costs nothing, and the constructed-input regression model matches the system model. Right: the Theil partition, where the system model remains the more bias-dominated of the two.

G.2 The two projections against external evidence

Neither model can be scored at 2040, because 2040 has no observations. But the projections are not therefore unfalsifiable, because the published literature carries expectations against which both can be placed (Table G.2). The regression model, run forward exactly as its predecessor does (twelve monthly feature vectors per horizon, built from the scenario levels and the farm's own seasonal shape), returns a Baltic 2040 band of 0.72 to 0.85 with no farm inside the training range. The system model returns 0.36 to 0.60, centrally 0.47.

Read naively, this favours the regression model. Its band sits inside the published range and brackets Germany's measured 0.76 almost exactly, while the system model falls below every benchmark. The reading inverts once the penetrations at which those benchmarks were measured are attached to them. Hirth's band is estimated at a 30 % share and Germany's 0.76 at 35 %, both comfortably inside the regression model's training range. A model that cannot leave the regime it was fitted on will land among benchmarks drawn from that same regime whether or not it is right about 2040, so the agreement is not corroboration. It is the constant response of Section 4.3.1 producing a plausible number.

The benchmark that discriminates is the one measured in a market that is already saturated. Aurora's French offshore holds near 0.94 to 2060, which is an under-built market and the analogue of the Baltic today rather than the Baltic of 2040; its French solar, heavily built out, falls to about 0.59 and stays there [112]. That is the closest available analogue to what the Baltic becomes, and it lies inside the system model's raw range and below the regression model's band. The comparison is developed further in Section 5.2.

The decisive quantity, however, is neither estimate. Between a baseline Baltic buildout and a high one, the regression model's answer moves by 0.008 while the system model's moves by 0.428. A projection that barely changes between two different futures is not a forecast of the buildout, however reasonable its value looks beside the literature. This is what the constant response means in practice, and it is the reason the external evidence confirms the allocation of roles rather than establishing it.

Table G.2: The two projections for the Baltic in 2040, against published benchmarks. The penetration column shows where each benchmark was measured: every benchmark below the rule was measured inside the regression model's training range, and only the last in a saturated market.

Source	Capture value	Measured at
Regression model, 2040 Baltic	0.72–0.85	no farm inside training range
System model, 2040 Baltic	0.36–0.60 (central 0.47)	projected buildout
Hirth [2]	0.5–0.8	30 % share
Germany [12, 24]	0.76	35 % share
Jacquet, worst North Sea case [107]	≈ 0.6	North Sea buildout
Aurora, French offshore [112]	≈ 0.94	under-built market
Aurora, French solar [112]	≈ 0.59	**saturated market**

H The Analytical Equation

Section 4.5.1 states the conclusion of the per-basin analytical fit. This appendix carries it in full.

H.1 One equation, two basins

SHAP names the influential features; the analytical equation describes the mechanism's shape in closed form. Refit per basin, the same equation fits both seas and answers the second research question in its sharpest form: the pattern-penalty exponent is 0.74 in the Baltic against 0.37 in the North Sea, steeper in 99 of 100 farm-block bootstraps (Figure M.12). The absolute fit is low by design, fitted for interpretable shape rather than prediction, so the load-bearing result is the bootstrap-stable gap. Since the exponent trades off against the penalty's scale, the fair summary is the implied slope of capture value against penetration at common conditions: about 1.6 times the North Sea's today and 1.9 times at the dominance a buildout creates. On a fifteen-farm panel that is directional rather than definitive, but it is consistent with the system model's moderate-2030, severe-2040 trajectory (Section 4.4).

I Regression Validation and Earlier-Horizon Maps

This appendix supports Sections 4.2 and 4.4.2; the numbers are quoted in the body.

Table I.1: The five algorithm families, out-of-sample R^2 on the bidding-zone panel, untuned. Random K-fold withholds farm-months; leave-farm-out withholds whole farms, so it scores entirely unseen sites, where the polynomial's squared and cross terms explode.

Algorithm	Random K-fold	Leave-farm-out (strict)
Gradient boosting (XGBoost)	0.752	**0.686**
Random forest	0.682	0.636
Support-vector regression	0.583	0.497
Linear (OLS)	0.330	0.270
Polynomial (degree 2)	0.437	−19.9

Table I.2: Out-of-sample R^2 of the untuned gradient-boosting model under each scheme, at country versus bidding-zone resolution. Both K-fold rows are scored out-of-fold and differ only in what they withhold: random farm-months (leaky, for comparability with Jacquet's untuned 0.82 [107]) versus whole farms (strict).

Validation scheme	Country features	Zone features
Random K-fold (comparability)	0.733	0.752
Leave-farm-out (headline)	0.598	**0.686**
Leave-farm-out, excluding 2022	0.607	0.692
North Sea → Baltic transfer	0.360	0.379
Time split (train ≤2022, test 2023–24)	−0.093	−0.032

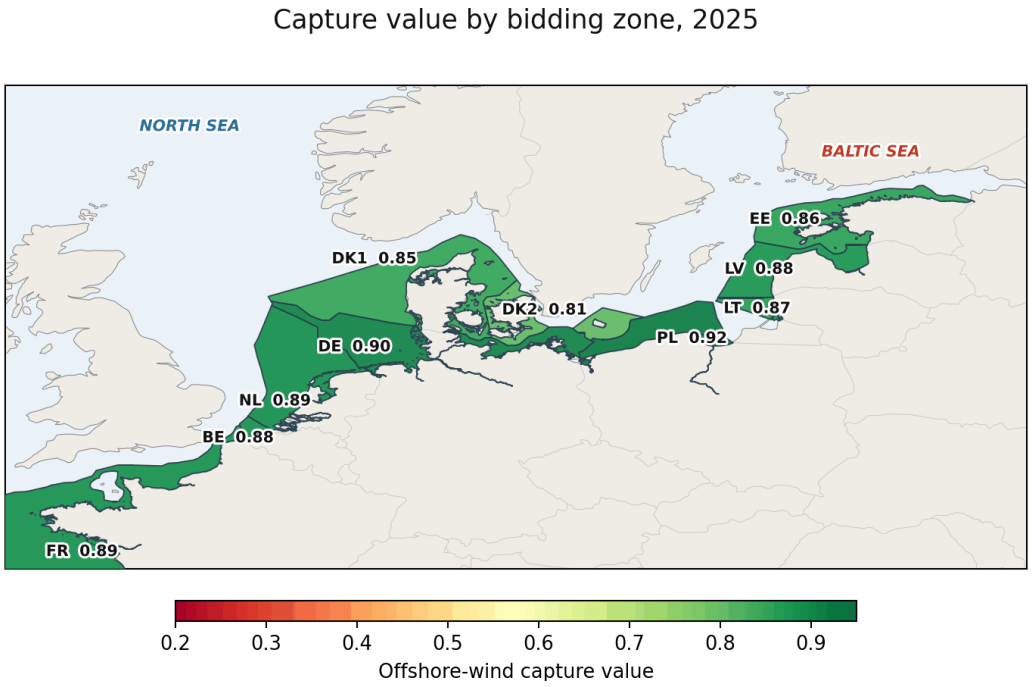


Figure I.1: Offshore capture value across both shelves, 2025. Values are high and nearly uniform: the under-built Baltic sits level with the North Sea, and the theoretical Baltic premium is not visible.

J Model Comparison Against Observed Prices: Full Detail

This appendix carries the per-zone-year detail behind Sections 4.2 and 4.3; the aggregates are quoted in the body.

K Reproducibility

The entire project is a numbered pipeline: scripts 1.01 to 1.20 (system model), 2.01 to 2.13 (regression model), and 3.01 (the comparison with observation), each reading its predecessors' outputs from disk and writing its own next to them. A single refresh chain re-solves everything from the prepared networks to the final tables in about three hours on a workstation (Apple Silicon, Gurobi academic licence); individual

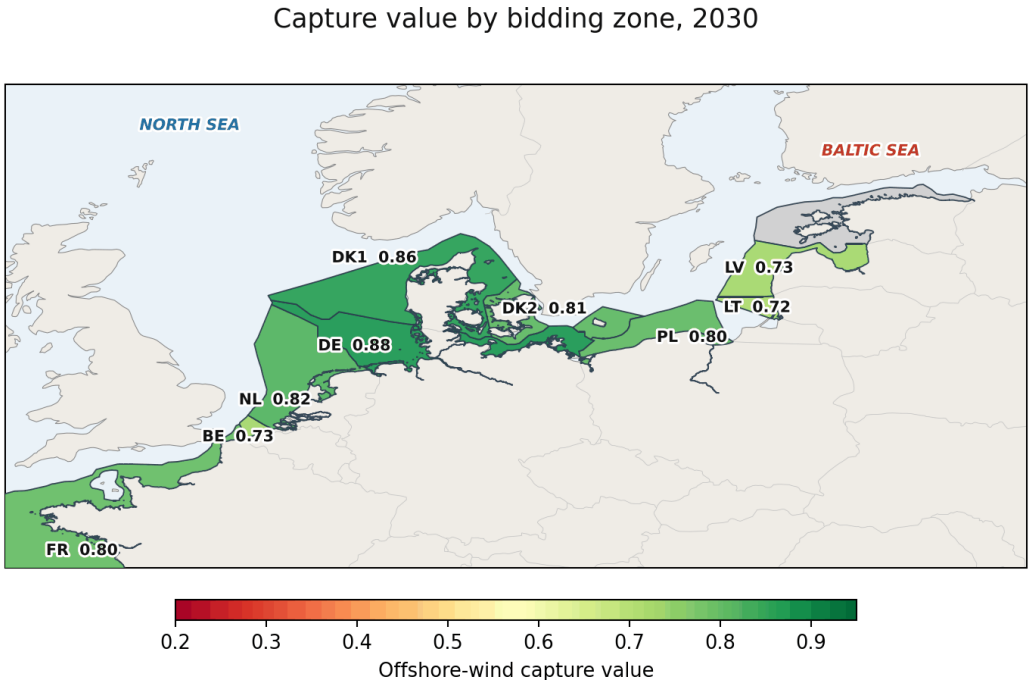


Figure I.2: The same field in 2030. The first tranche of the buildout is in place. Values have fallen slightly and remain high in both basins, with the Baltic beginning to fall further as its own correlated output crowds its price.

Table J.1: The three-way comparison in numbers: observed capture value against each model's estimate, with signed errors, for all fifteen zone-years. The regression column is out of fold, from a model that never trained on the zone's farms.

Zone	Year	Observed	Regression	System model	Reg. error	Sys. error
DE	2019	0.923	0.948	0.984	+0.025	+0.061
NL	2019	0.977	0.967	0.986	−0.010	+0.009
BE	2019	0.919	0.925	0.983	+0.006	+0.064
DK1	2019	0.924	0.929	0.986	+0.005	+0.062
DK2	2019	0.933	0.936	0.987	+0.003	+0.054
DE	2023	0.909	0.913	0.908	+0.004	−0.001
NL	2023	0.881	0.887	0.883	+0.006	+0.002
BE	2023	0.869	0.879	0.896	+0.010	+0.027
DK1	2023	0.854	0.861	0.917	+0.007	+0.063
DK2	2023	0.784	0.798	0.927	+0.014	+0.143
DE	2024	0.910	0.896	0.941	−0.014	+0.031
NL	2024	0.900	0.878	0.919	−0.022	+0.019
BE	2024	0.893	0.869	0.928	−0.024	+0.035
DK1	2024	0.843	0.833	0.947	−0.010	+0.104
DK2	2024	0.808	0.802	0.949	−0.006	+0.141

scenario solves take about 90 seconds at solver parity. Every headline number regenerates from public data: ENTSO-E Transparency, ERA5 via Open-Meteo and atlite, TYNDP 2024 workbooks, and open

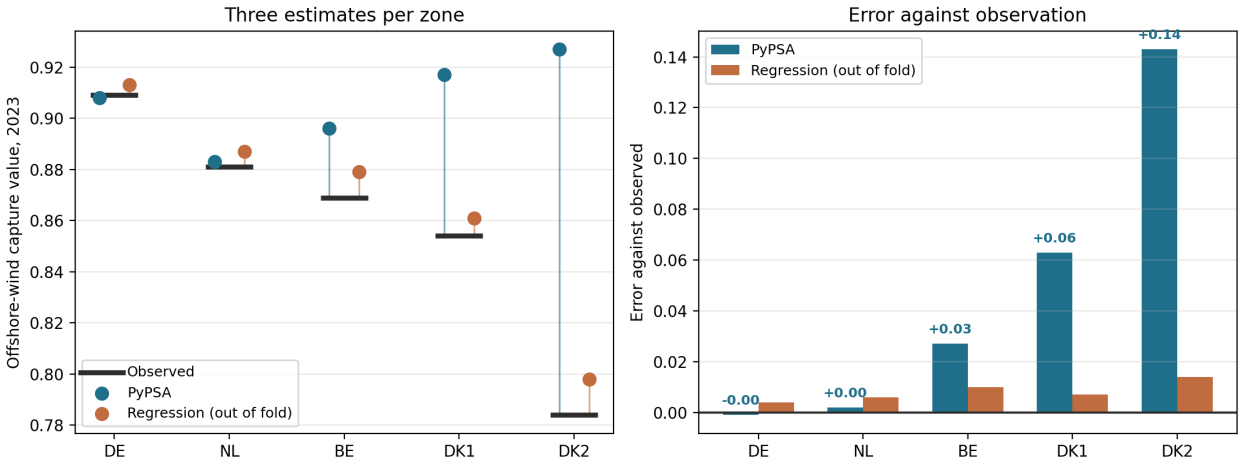


Figure J.1: The three-way comparison, 2023. Left: the three estimates per zone, observed as the black tick. Right: each model's error. The regression model stays within about 0.01 everywhere; the system model's error grows with congestion.

Table J.2: The gap between the two models, decomposed. Channels sum exactly to the total in every row. Positive means the channel moves the system model's capture value above the regression model's. Weights are offshore energy.

Zone	Regression model	System model	Total gap	Constr. residual	Gener-ation	Zero floor	Price shape
DE	0.913	0.888	−0.025	−0.014	−0.012	+0.002	+0.002
NL	0.887	0.877	−0.010	0.000	+0.009	+0.005	−0.023
BE	0.879	0.880	+0.001	+0.002	+0.020	+0.002	−0.025
DK1	0.861	0.917	+0.056	0.000	+0.007	+0.002	−0.003
Weighted mean	0.890	0.885	−0.005	−0.003	+0.004	+0.003	−0.014

The network-representation channel is −0.003 (DE), −0.001 (NL), +0.003 (BE), +0.050 (DK1) and +0.004 weighted; it is reported separately because it is the one channel already corrected in Section 4.2.3.

cost data. Configuration files pin every assumption of Appendix B; staleness guards (input modification times recorded next to fitted outputs) make outdated results detectable. A line-by-line audit log of the one major repair cycle, with before and after values, is kept in the project repository.

Table K.2 pins the software environment behind every solve and fit.

L Spatial Atlas

The main text carries one map per argument. This appendix carries the full spatial series behind them, so that every field is available at all three horizons rather than only at the year the argument needed. Each family is built on the same footing: the high-resolution capacity-factor field of Section 3.7.1, water depth and distance from GEBCO, and per-zone capture values read from the solved networks of Section 3.5.4. All maps share a fixed colour scale within their family so that the three horizons are directly comparable; a field that changes colour between 2025 and 2040 has changed in value.

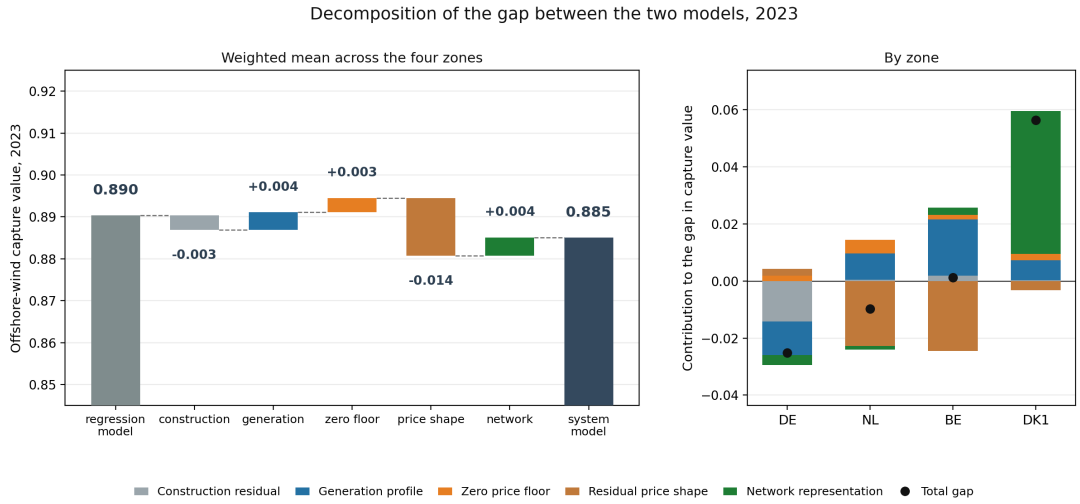


Figure J.2: Decomposition of the gap between the two models, 2023. Left: the bridge from the regression model's estimate to the system model's, energy-weighted across the four zones. Right: the same channels by zone. Positive values move the system model above the regression model.

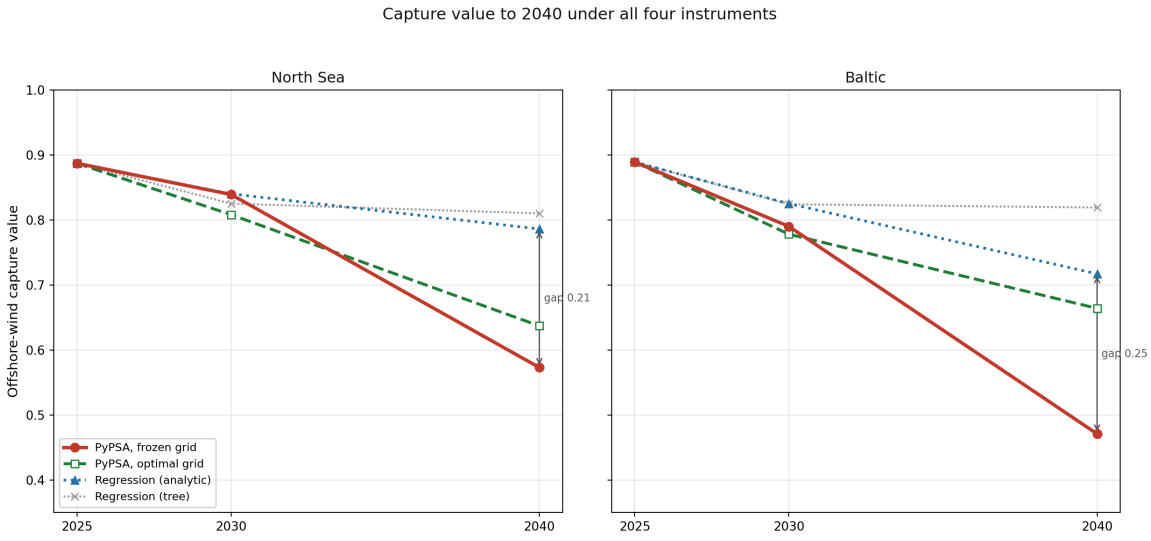


Figure J.3: Every model on one axis, 2025 to 2040. The models agree to 2030; at 2040 the regression model returns a constant, the analytical form falls gently, and the system model finds the collapse the others cannot.

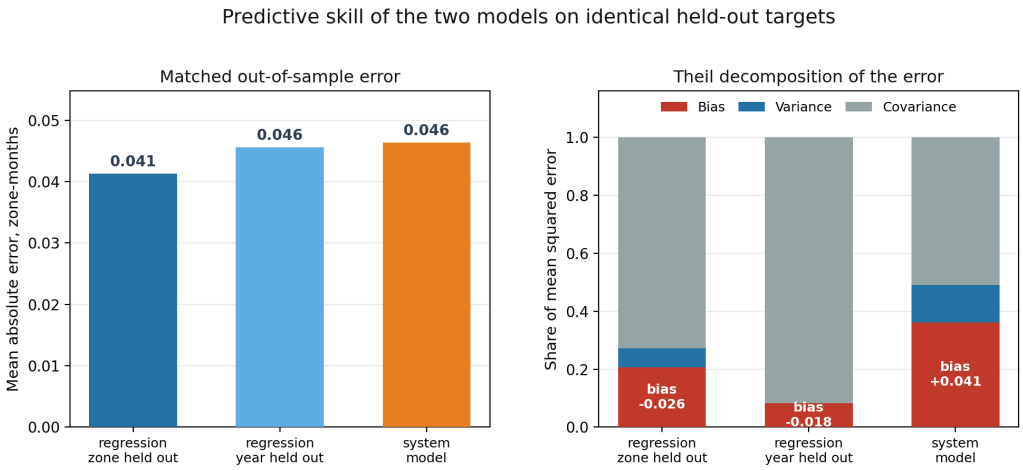


Figure J.4: Predictive skill on identical held-out targets. Left: average error, indistinguishable across the three schemes. Right: the decomposition of squared error into a systematic component, a component from failing to match the spread of the observations, and a component from imperfect correlation. The system model's error is a third systematic; the regression model's is almost entirely imperfect correlation.

Table K.1: Inventory of system-model runs. All solves use the identical model version, solver configuration, and corrections described in the main text; only the stated dimension varies.

Run set	**Solves**	**Purpose**
Historical baselines	3	2019, 2023, and 2024 calibration years
Canonical scenarios	6	2030/2040, central and high Baltic buildouts, two 2040 demand pathways
Gas-price grid	15	three buildouts × five gas prices, 15 to 120 €/MWh
Grid bracket	6	the six scenarios with optimisable transmission
Surrogate training	55	50 Latin-hypercube points plus 5 scenario anchors, three-hourly, harvested alongside the 21 full-hourly solves above
Resolution check	2	two design points re-solved at full resolution

Table K.2: The software environment. All versions are those used for the reported runs; the regression stack is pinned because untuned library defaults are version-dependent.

Python	3.13	PyPSA-Eur	v2026.02.0 (commit 23c2fac7)
pypsa	1.2.0	snakemake	9.20.0
Gurobi	13.0.2 (academic)	atlite	0.6.1
xgboost	3.2.0	scikit-learn	1.8.0
pandas	2.3.3	numpy	2.4.3

L.1 The resource field

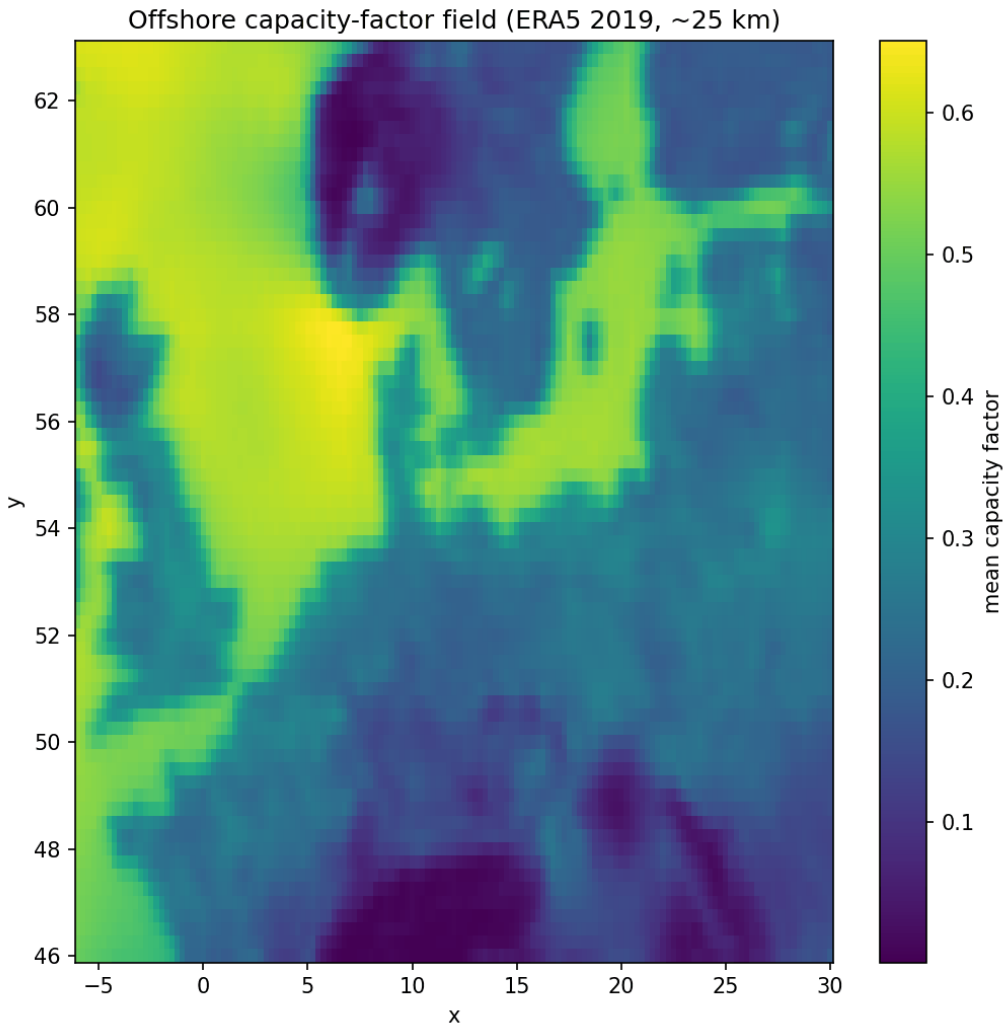


Figure L.1: The offshore capacity-factor field across both shelves, built once from the Global Wind Atlas at roughly 2 km resolution and used as the common resource layer for every economic map that follows. The gradient from the open North Sea to the eastern Baltic is the physical starting point for the whole comparison, and it is year-independent: the wind field is identical in every year, so the changes in the maps that follow come from value and cost.

L.2 Capture value under an optimal grid

Figures I.1 to 4.4 in the main text show capture value on the frozen grid. The counterfactual matters as much, since the difference between the two is the congestion component isolated in Section 4.5.4. The maps below repeat the sequence with transmission expanded to the cost optimum.

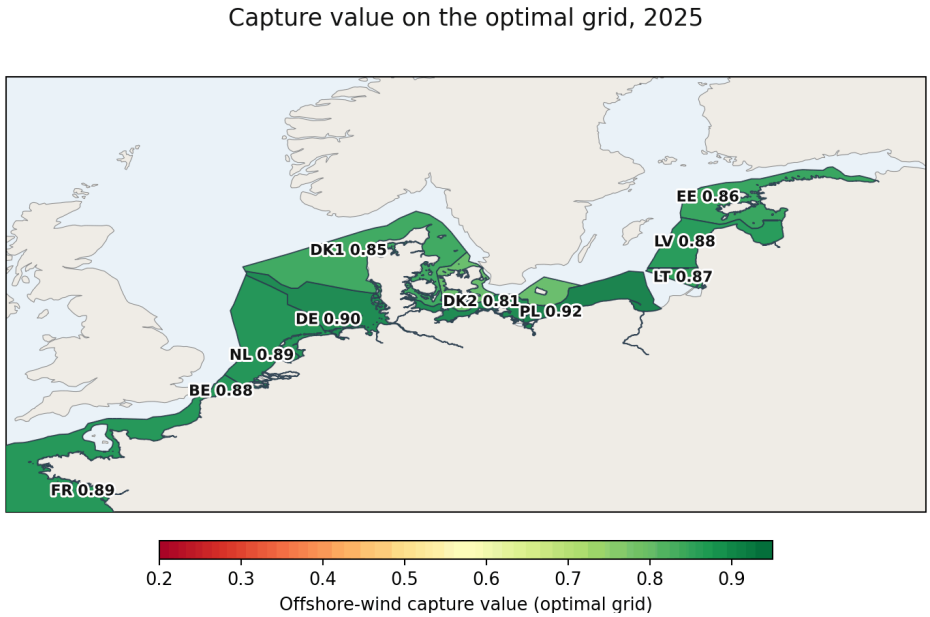


Figure L.2: Capture value on the cost-optimal grid, 2025. At today's penetration the grid assumption barely matters and the field is indistinguishable from the frozen case.

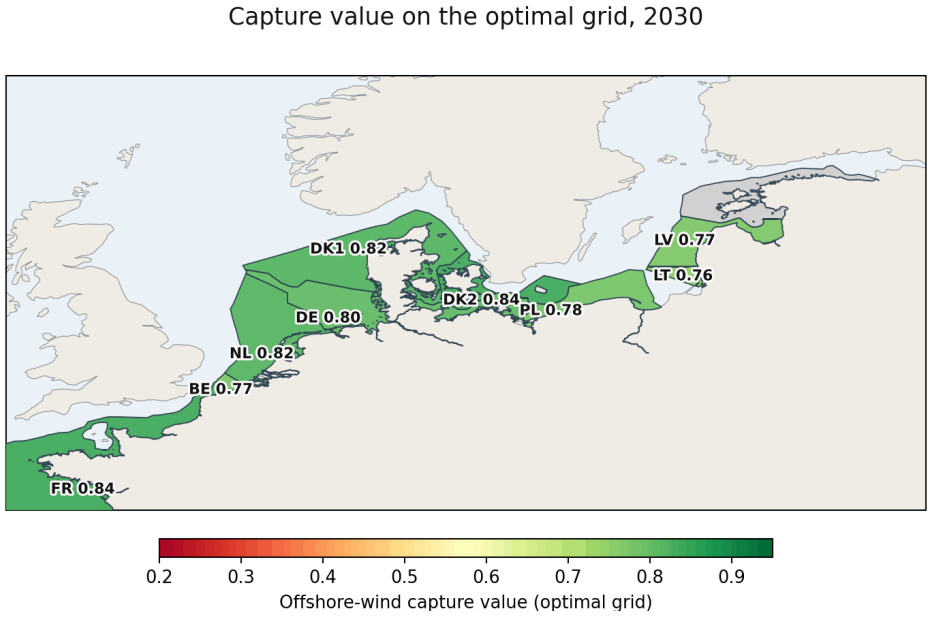


Figure L.3: Capture value on the cost-optimal grid, 2030. The bracket is still almost nil: congestion has not yet begun to bind.

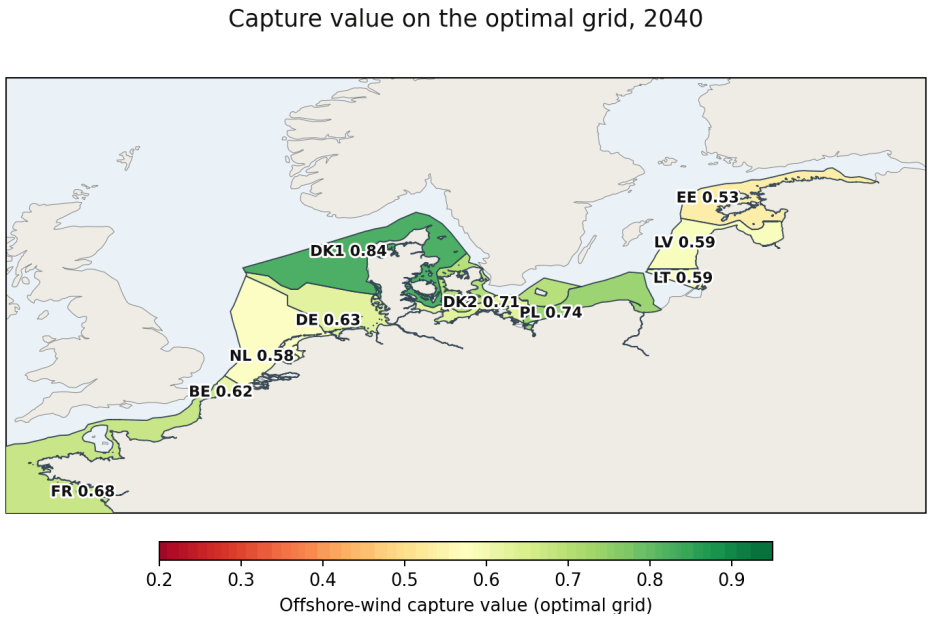


Figure L.4: Capture value on the cost-optimal grid, 2040. Compared with Figure 4.4 the eastern Baltic recovers most, which is the spatial expression of the reversal: the zones that lose the most to a frozen network are the ones an expanded network restores.

L.3 Levelised cost across the shelf

The cost field is the standard output of the spatial assessment literature [87] and the baseline against which the value-adjusted metric is compared. It falls with technology learning and is otherwise governed by depth, distance and resource.

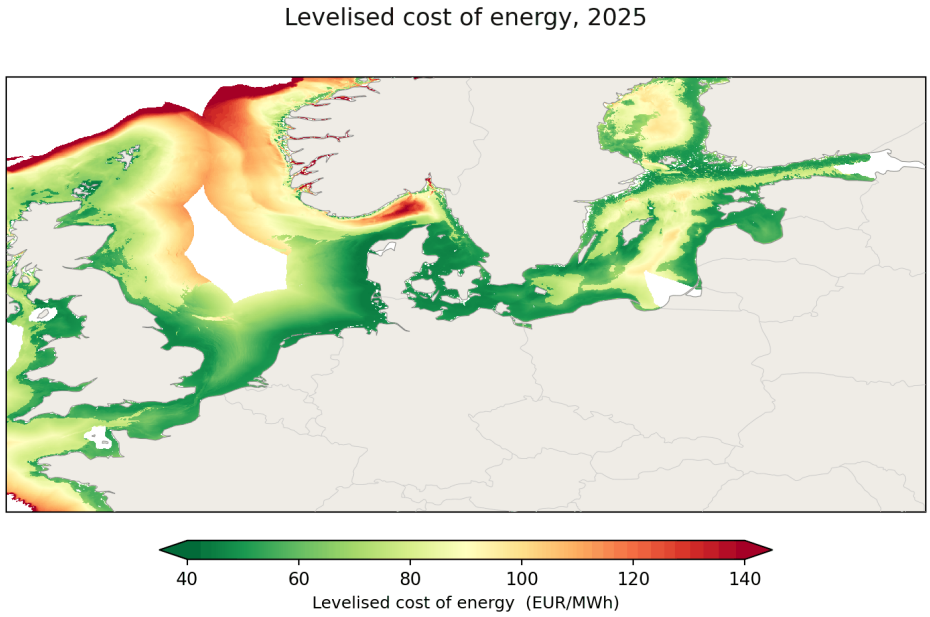


Figure L.5: Levelised cost of energy by location, 2025. Cheap sites cluster in the shallow southern North Sea and the western Baltic; the deep Norwegian trench never clears the floating-foundation premium.

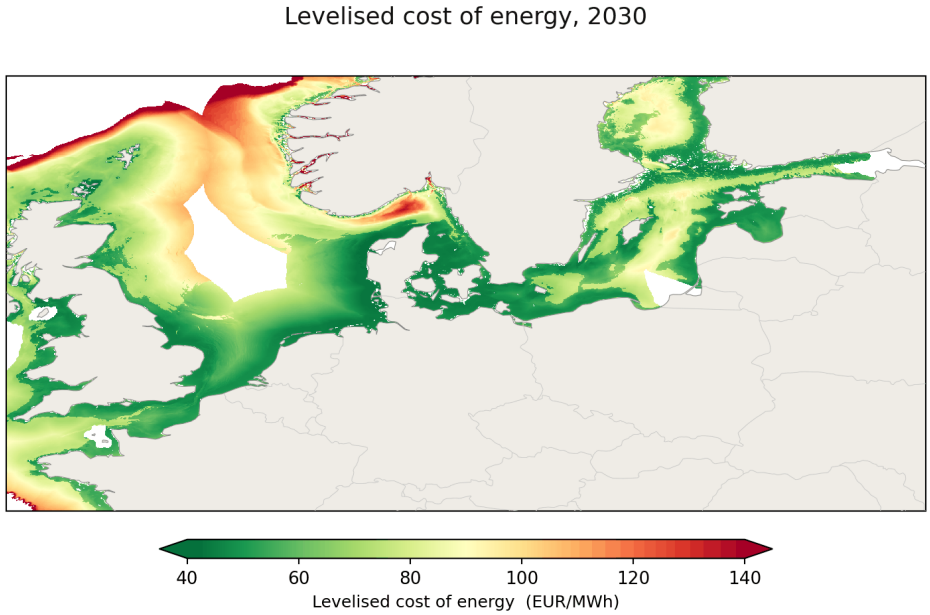


Figure L.6: Levelised cost by location, 2030. The whole field dims as capital costs fall; the spatial pattern is unchanged.

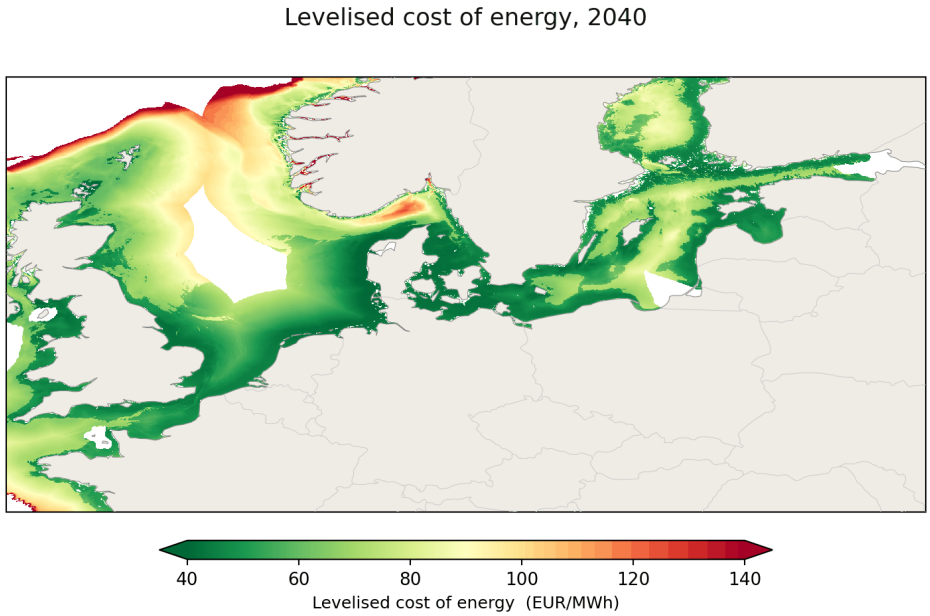


Figure L.7: Levelised cost by location, 2040. On cost alone the shelf looks steadily more attractive over time, which is precisely the impression the value-adjusted maps of Section L.4 overturn.

L.4 Value-adjusted cost

Dividing the cost field by capture value gives the wholesale price a site must earn to break even (Equation 2.3). The 2040 map appears in the main text as Figure 4.9; the earlier horizons complete the series and show the direction of travel.

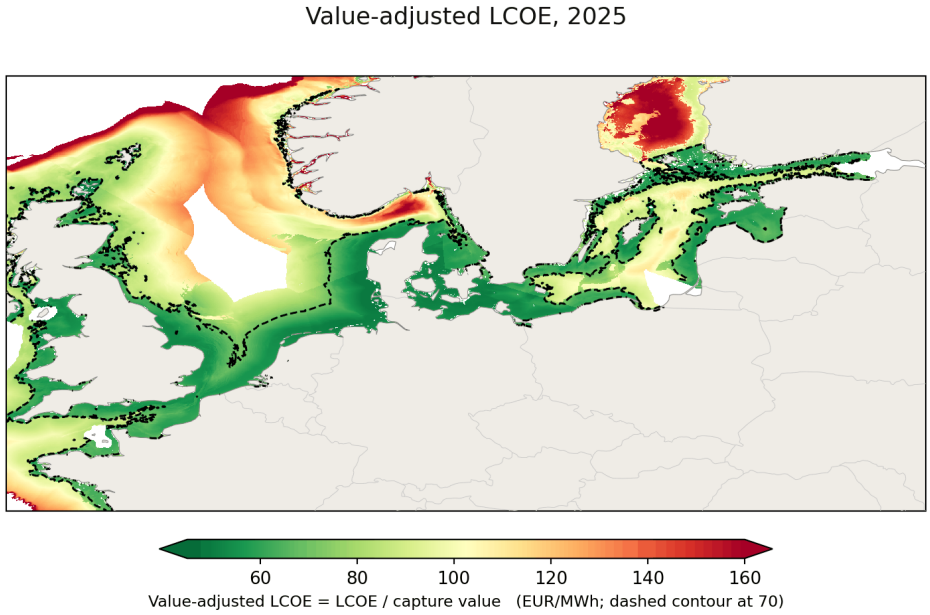


Figure L.8: Value-adjusted cost of energy, 2025. With capture values near 0.9 across both shelves the field sits close to the raw cost map.

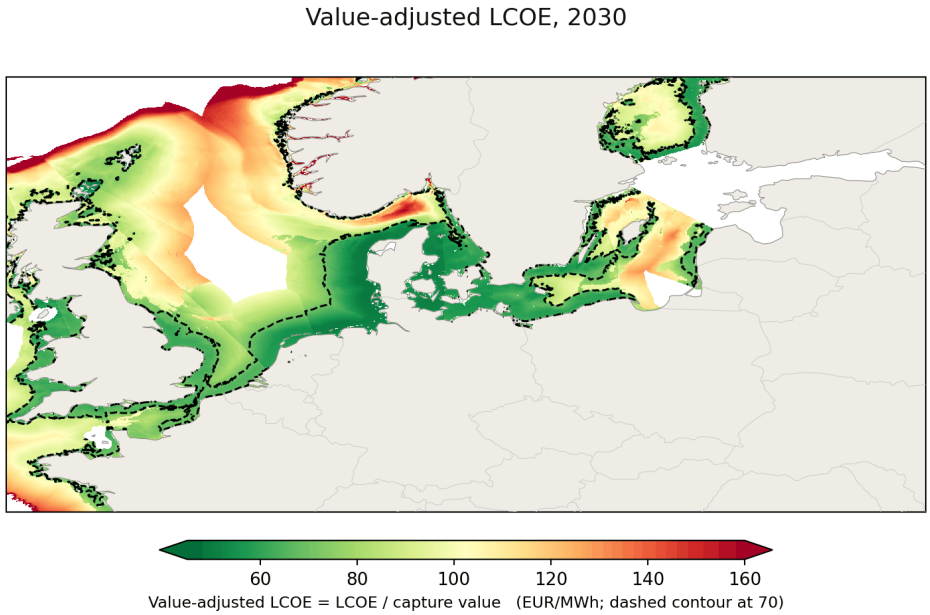


Figure L.9: Value-adjusted cost, 2030. The eastern Baltic begins to separate from its cost map as its capture value falls faster than its cost.

Read against Figures L.5 to L.7, the pair of series states the thesis result in its most compact form. The cost field improves monotonically with learning while the value-adjusted field deteriorates, and the divergence between them is cannibalisation expressed in euros per megawatt-hour rather than as a ratio.

L.5 Net value

Net value subtracts annualised cost from revenue and answers whether a site pays at all, rather than what price it needs. The series opens with the 2040 map (Figure L.10), the horizon the body's argument turns on; the earlier horizons follow.

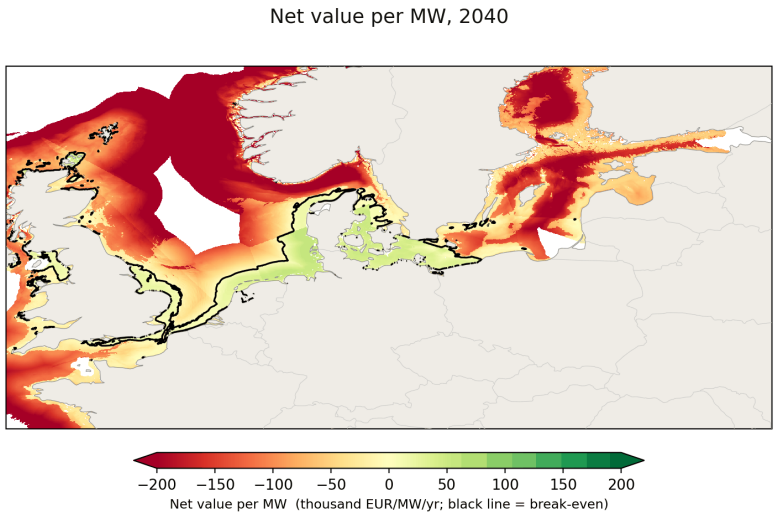


Figure L.10: Merchant net value per megawatt in 2040, captured revenue less annualised cost. Green is profitable at merchant prices; the profitable shelf shrinks from about 48% of buildable area in 2025 to 12% by 2040, as value is withdrawn while cost continues to fall.

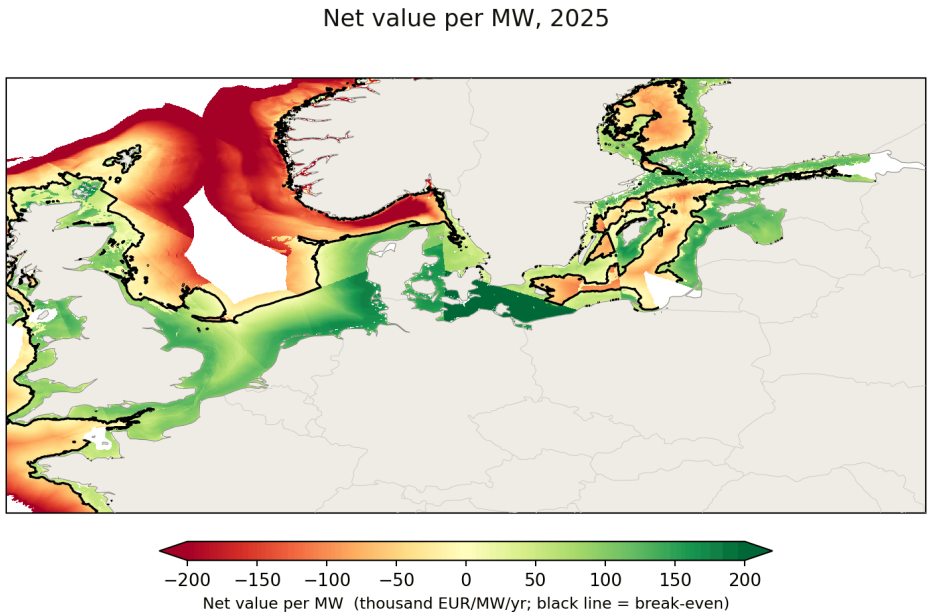


Figure L.11: Net value per megawatt, 2025. Roughly half the buildable shelf is merchant-profitable at today's capture values and prices.

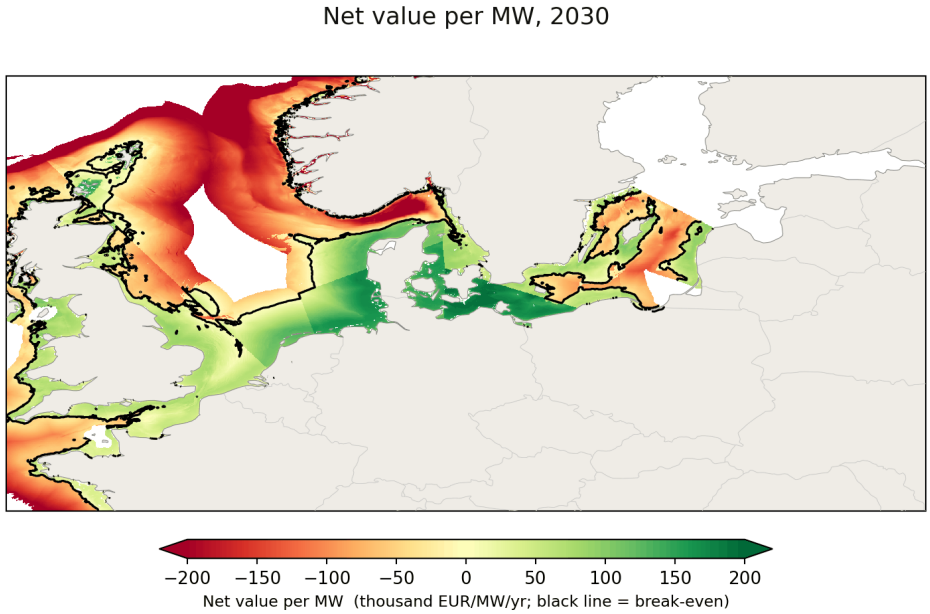


Figure L.12: Net value per megawatt, 2030. The profitable area has begun to contract toward the shallow, high-resource core.

L.6 Revenue density

The revenue-density series superimposes the resource field on the per-zone capture value and price, giving revenue per unit area rather than per megawatt. It is the most granular of the spatial products, built on the roughly 2 km Global Wind Atlas field.

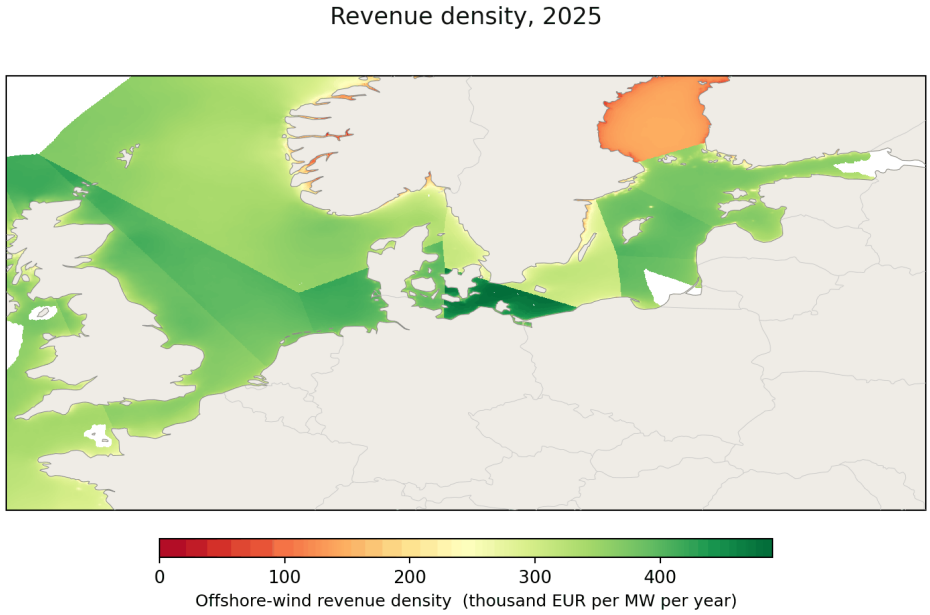


Figure L.13: Revenue density across both shelves, 2025.

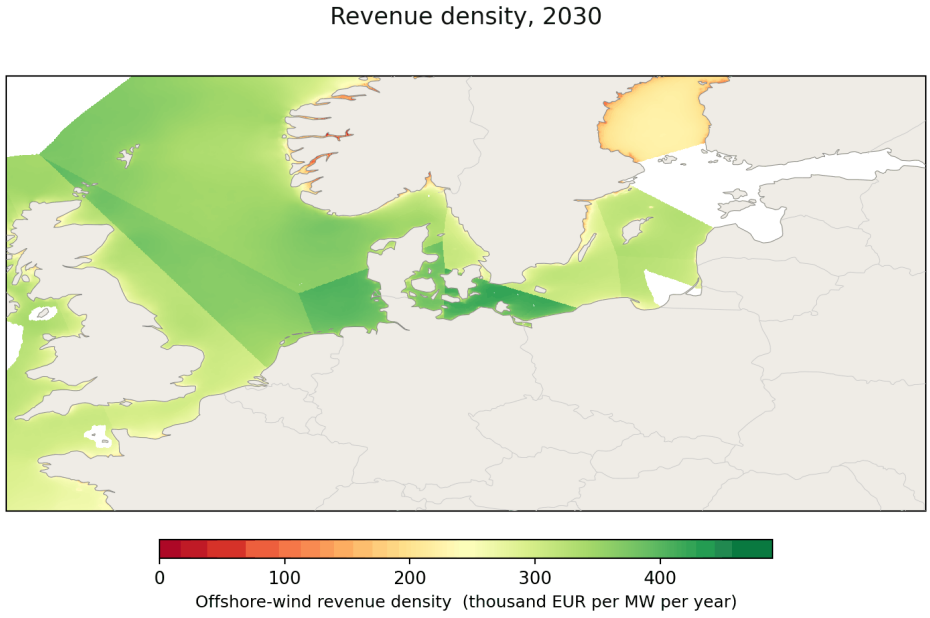


Figure L.14: Revenue density, 2030.

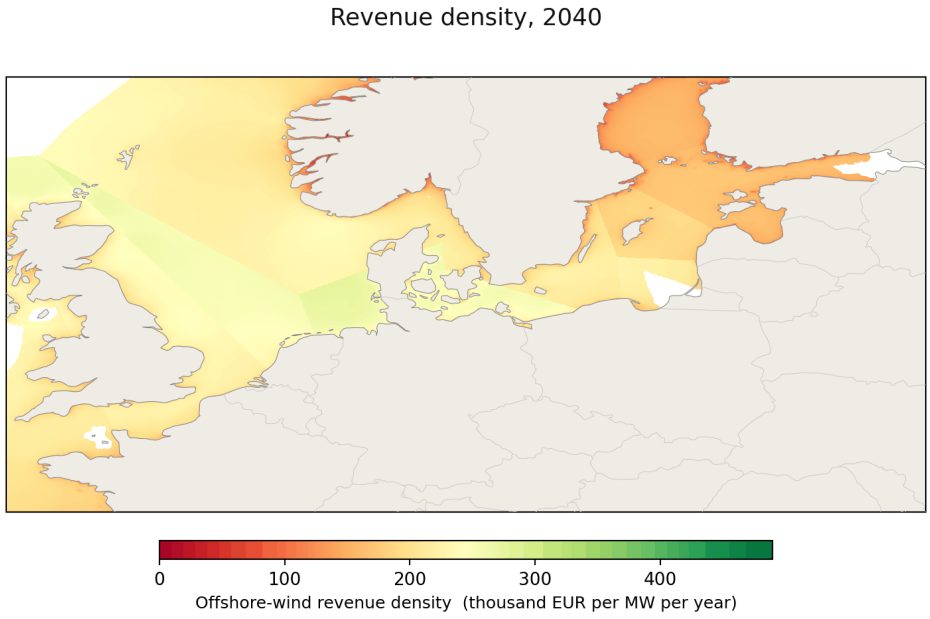


Figure L.15: Revenue density, 2040. The contraction toward the German Bight and the shallow western Baltic is the same pattern the site ranking of Section 4.6.3 produces numerically.

L.7 Candidate areas on the value grid

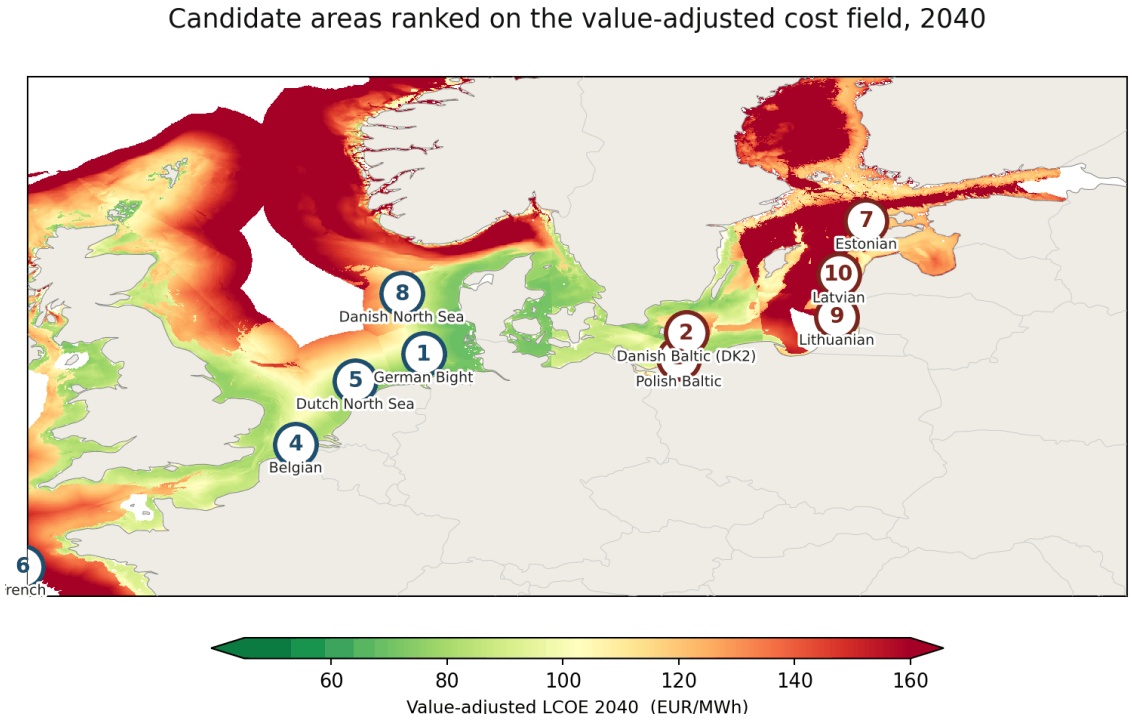


Figure L.16: The ten candidate development areas positioned on the 2040 value-adjusted field, which is the spatial backing for the ranking reported in Section 4.6.3. Circled numbers give the value-adjusted rank, 1 the best; blue marks North Sea areas, red Baltic. The areas that fall in the ranking are visibly those sitting in the reddest water.

M Supplementary Figures

This appendix holds diagnostics and intermediate results referenced in passing by the main text, grouped by the chapter they support.

M.1 Model resolution

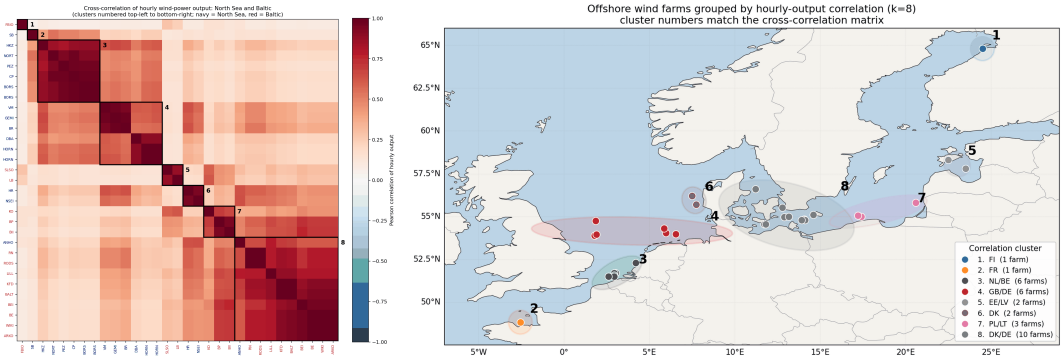


Figure M.1: The 30-node clustering of the study region, showing which physical buses are aggregated into each model node. Copper-plating within a node (the model treats transmission inside a node as unlimited) is the limitation this resolution imposes, and the reason small well-connected zones carry the largest system-model bias.

M.2 Measured capture value

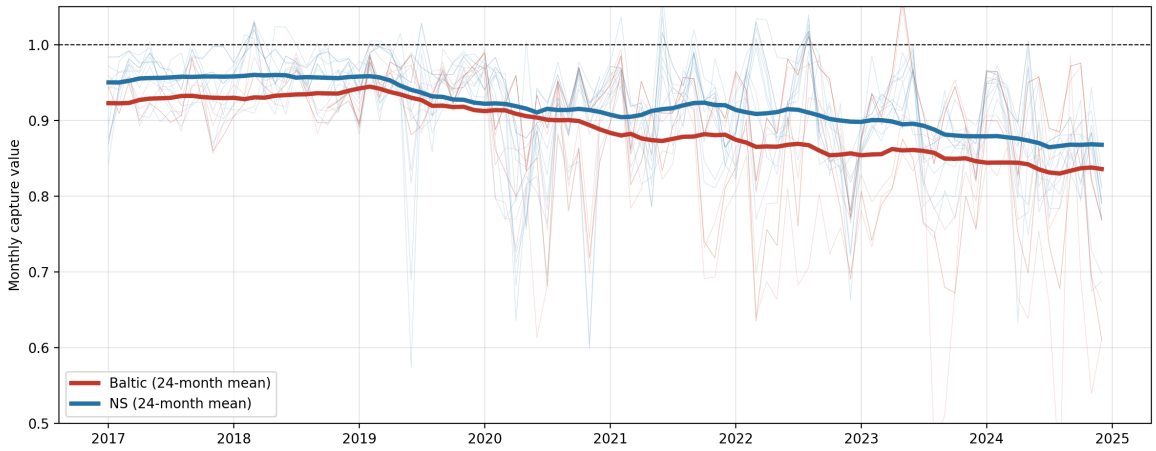


Figure M.2: Eight years of measured capture value by basin, the full history behind the summary in Section 4.1.1.

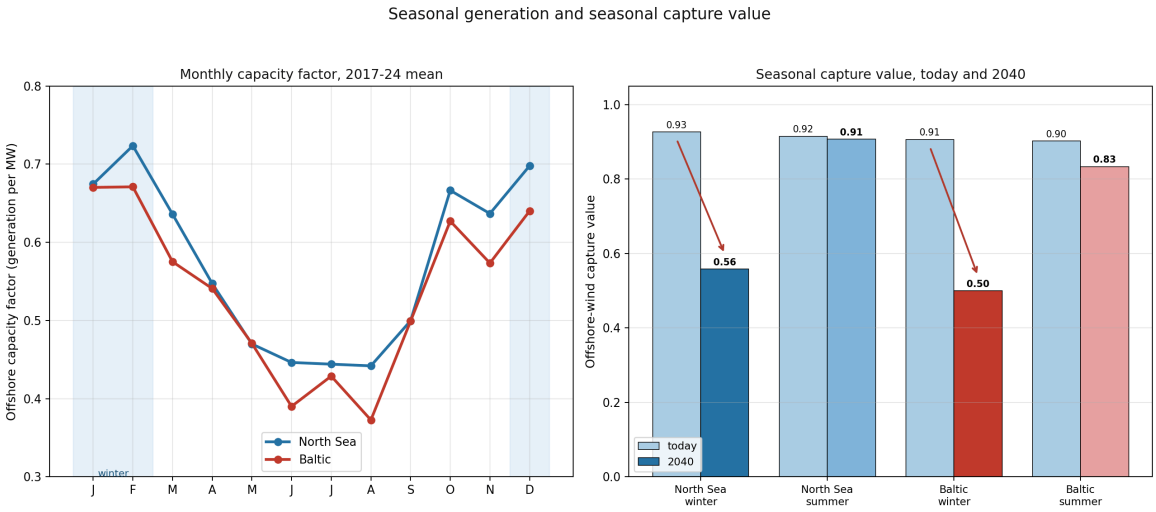


Figure M.3: The seasonal generation profile and its interaction with capture value: output peaks in winter, which is also when the basins synchronise, so the largest volumes are sold into the most depressed hours.

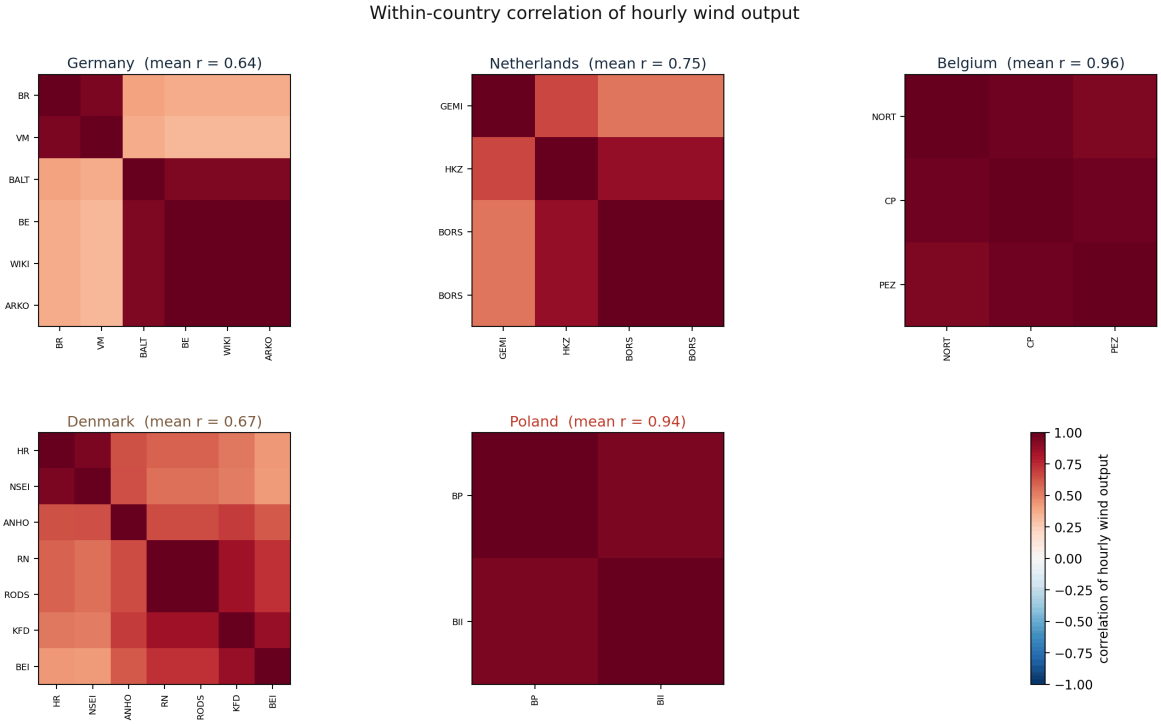


Figure M.4: Within-country wind synchronicity matrices, the granular analogue of the basin correlation figure. Compact fleets blow nearly as one (Belgium 0.96, Poland 0.94); spread fleets do not (Germany 0.64, Denmark 0.67). Germany splits into two blocks, its North Sea and Baltic farms, so the basin boundary dominates the national one.

M.3 Within-Baltic convergence

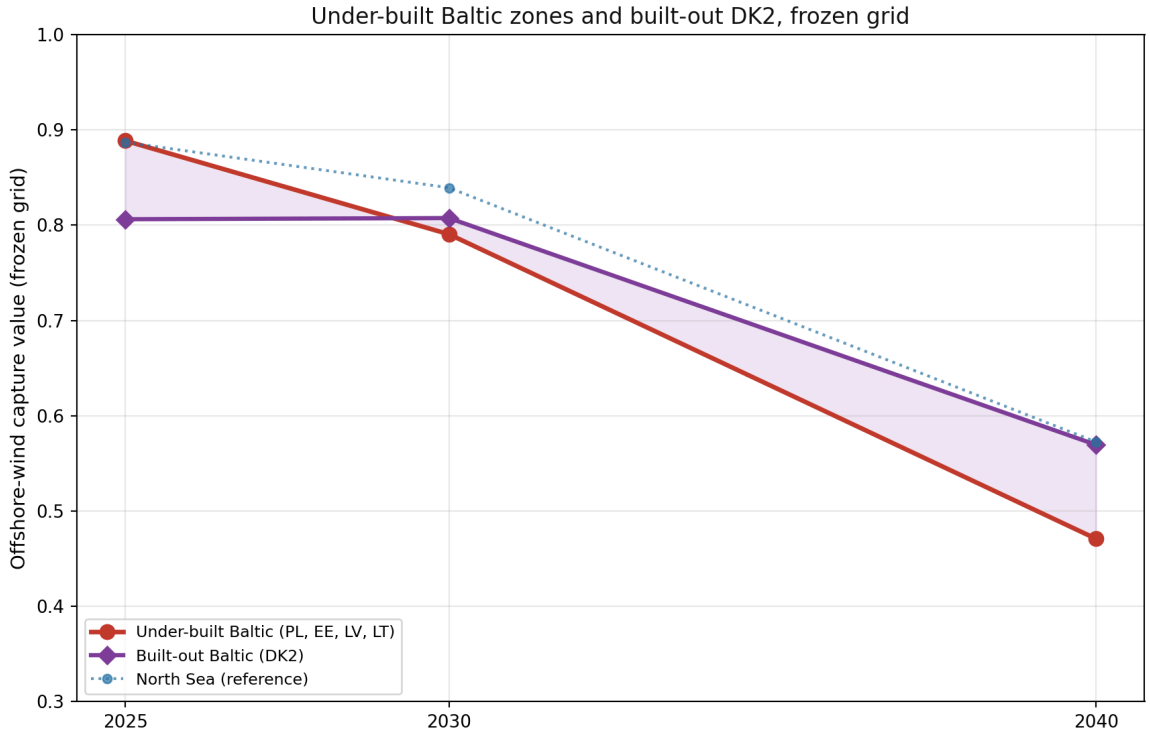


Figure M.5: Convergence of the under-built Baltic zones onto the values already reached by the built-out DK2 and SE4. The built-out zones preview where the under-built ones are heading, which is the space-for-time argument shown directly.

M.4 Model diagnostics

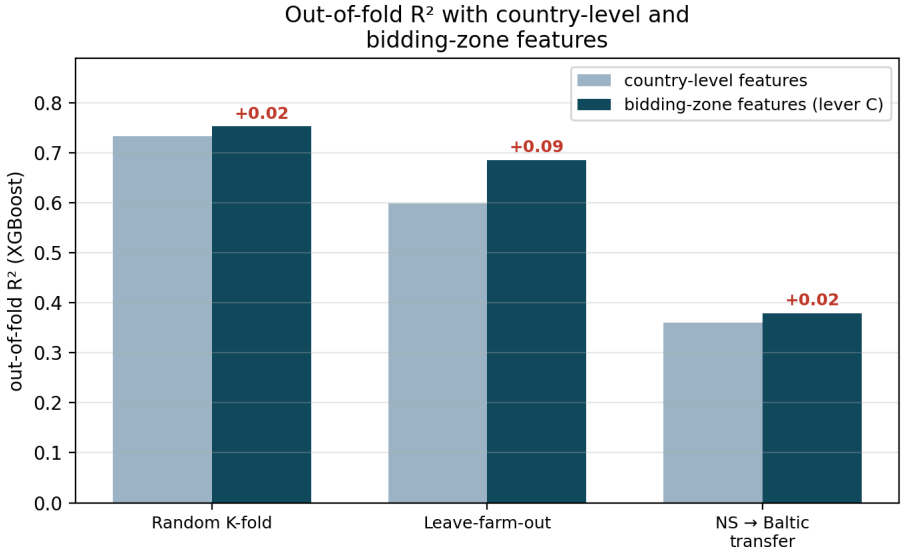


Figure M.6: Out-of-sample R^2 before and after resolving features at the bidding-zone level, the level at which prices form. The gain is what zone resolution buys the regression model.

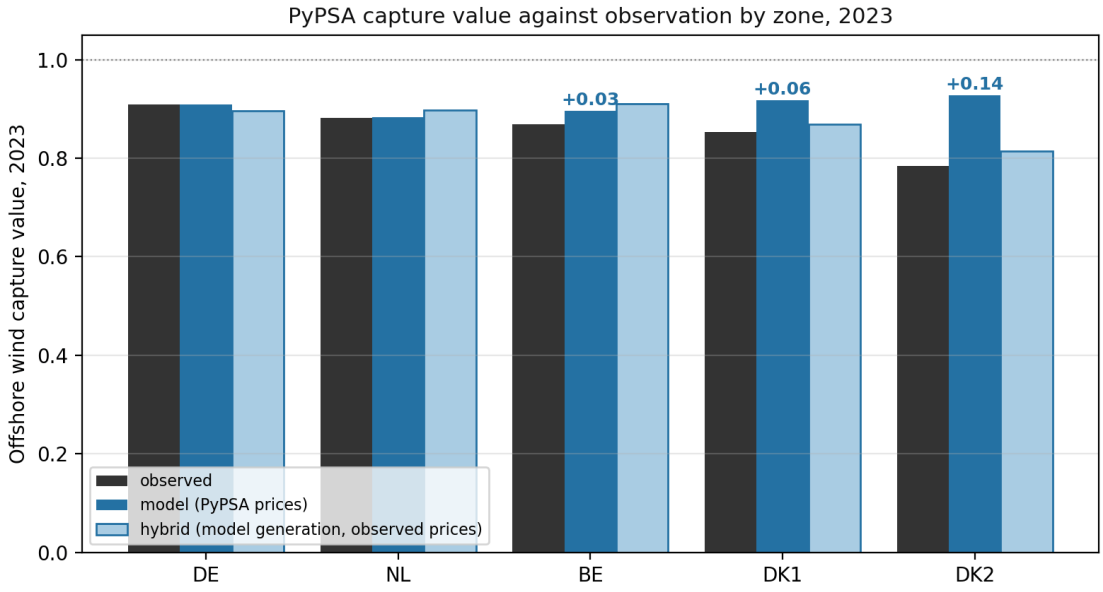


Figure M.7: The system model's capture value, validated zone by zone for 2023: observed, modelled, and the hybrid pairing modelled generation with observed prices. The model is near-exact in large zones and overstates capture value in congested zones; the hybrid locates the error in the modelled prices and shows the modelled generation to be sound.

M.5 Robustness of the projection

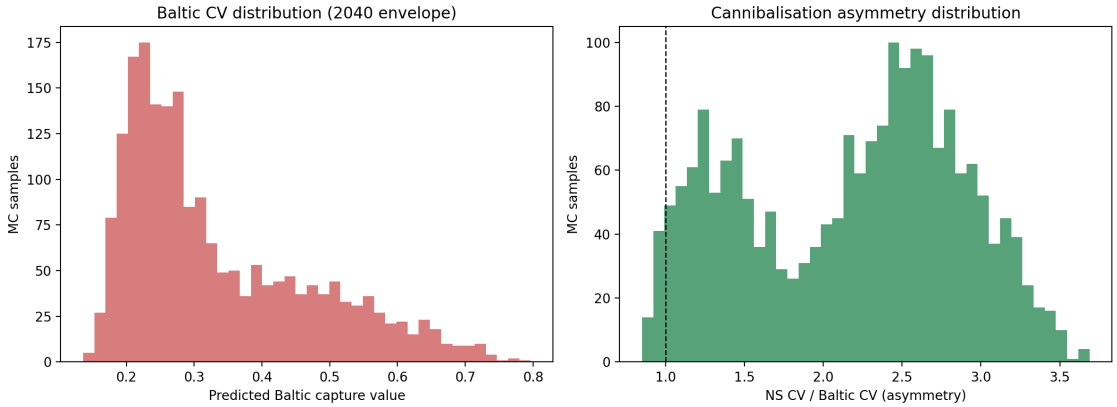


Figure M.8: Monte-Carlo fans over the full input envelope (2,000 samples, inputs beyond TYNDP by design): Baltic capture value median 0.28, 90 % interval $[0.18, 0.62]$; North Sea to Baltic asymmetry median about 2.3×. These figures describe the range of possibilities and give no single point result.

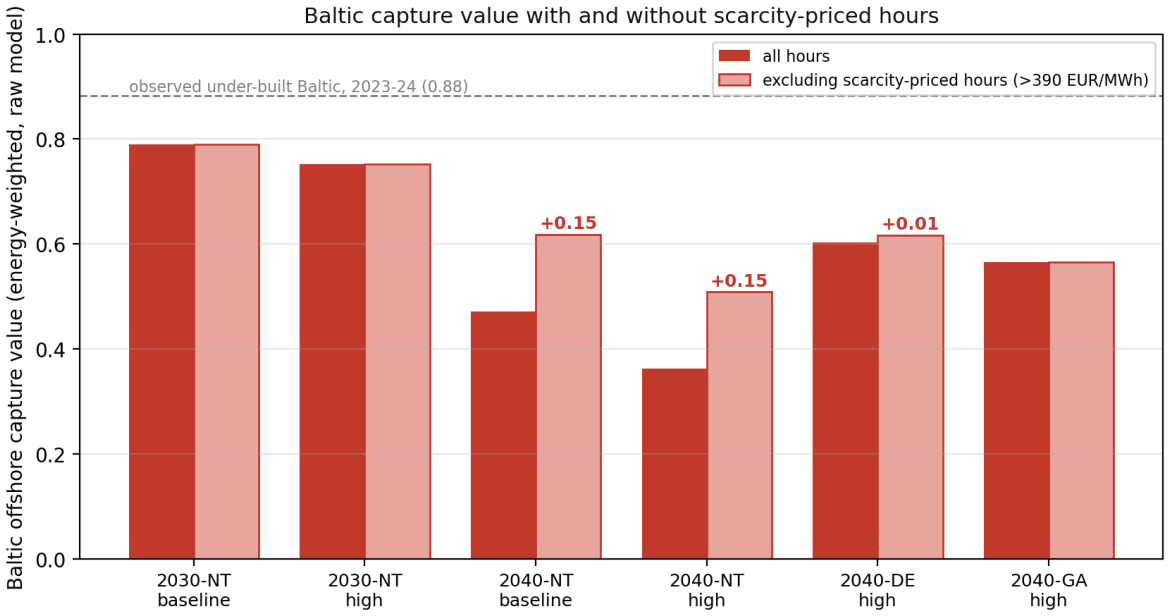


Figure M.9: Baltic capture value per scenario (uncorrected model), with and without scarcity-priced hours. The 2030 and clean 2040 scenarios are scarcity-free; the National Trends 2040 scenarios carry a Polish adequacy component; severe erosion survives the exclusion in every case.

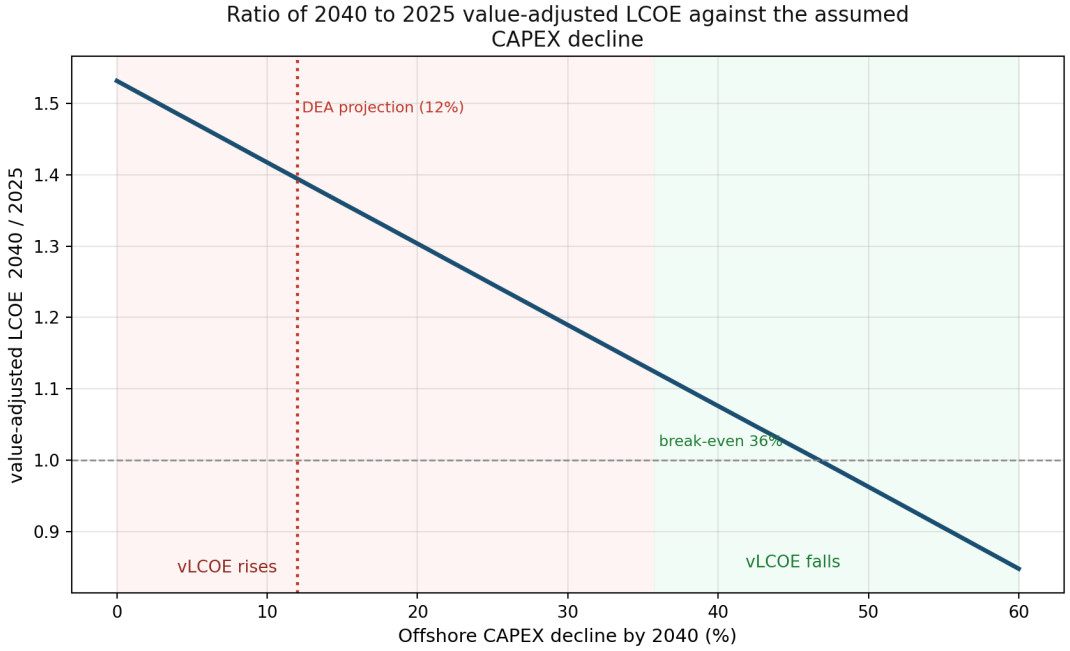


Figure M.10: Robustness of the rising value-adjusted LCOE: the 2040-to-2025 ratio against the assumed offshore capital decline. It stays above one across the plausible range, and capital would have to fall about 36%, against the database's 12%, merely to hold the metric flat.

M.6 Mechanism: supporting detail

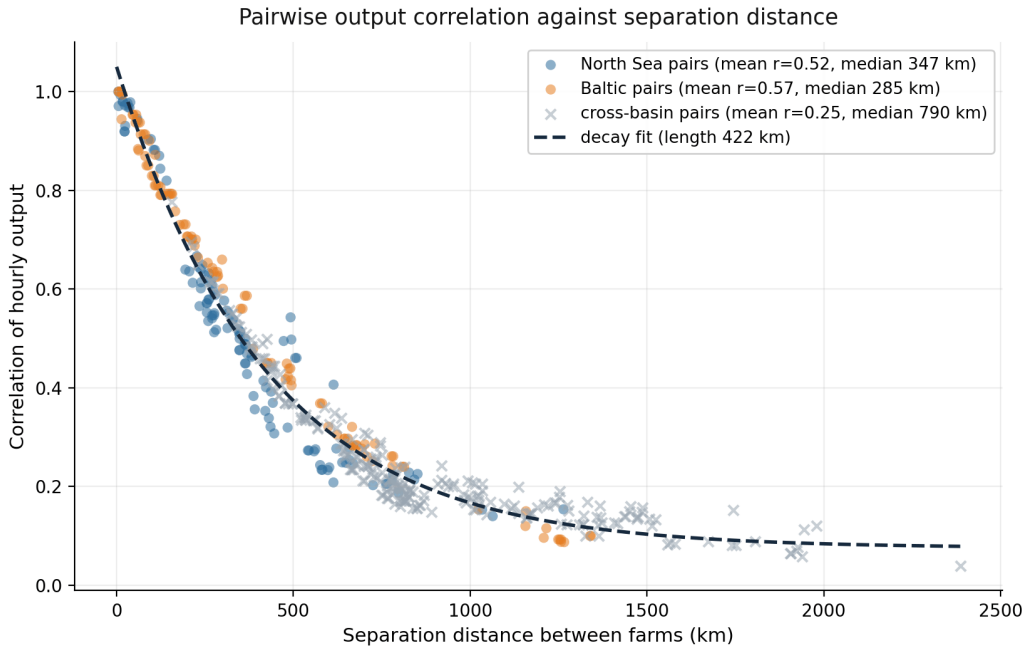


Figure M.11: Pairwise correlation of hourly wind output against separation distance, every farm pair, 2017 to 2024. All pairs follow one decay curve (fitted length 422 km); the Baltic pairs (mean 0.57) sit slightly above the North Sea pairs (mean 0.52), with a shorter typical separation (median 285 against 347 km), while cross-basin pairs (mean 0.25) lie far out along the curve.

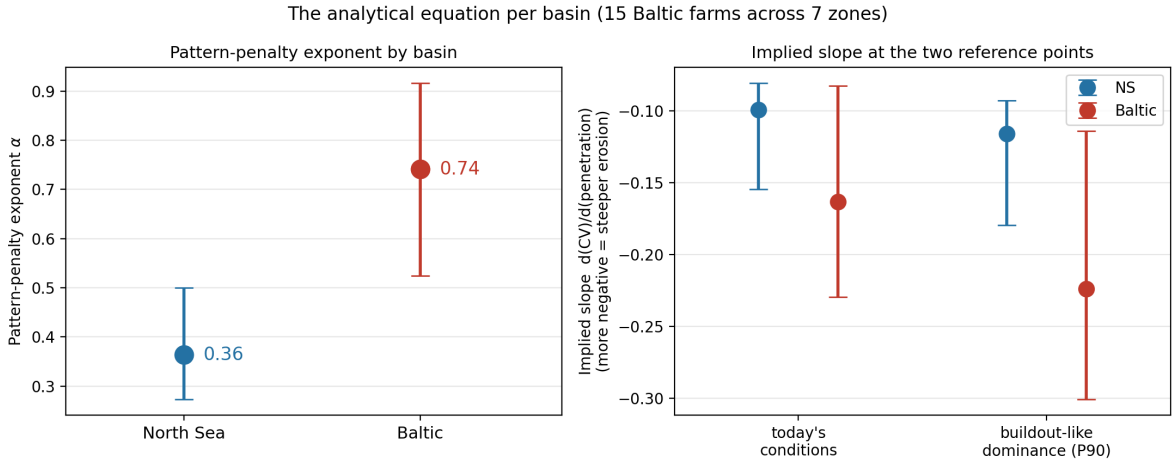


Figure M.12: The analytical equation refit per basin on the seven-zone panel (fifteen Baltic farms). Left: the pattern-penalty exponent is steeper in the Baltic (0.74 versus 0.37). Right: the implied slope of capture value against penetration is about 1.6 times steeper for the Baltic today and 1.9 times at buildout-like dominance. Whiskers are farm-block bootstrap 90 % intervals; the comparison is directional on a small panel.